

# A Cartesian Multipole Hierarchy for the Voxel

## Closed-Form Single-Site $T$ -Matrices, and the Error of Elastic Volume-Integral Scattering Isolated on a Space-Filling Layer of Cubes

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1 October 2026

### Abstract

Volume-integral (coupled-dipole) methods divide a heterogeneous elastic medium into cubic voxels, each carrying a single-site  $T$ -matrix, coupled through the background Green's tensor. Each voxel must represent the medium within it and the wavefield within it. We ask which basis each needs, and what error a given choice leaves: such schemes are built for media with no exact solution, where the error cannot be measured. We answer on a layer of voxels. Because cubes tile space,  $n$  planes of voxels are the layer itself for every  $n$ , and its exact solution isolates the discretisation error. With the wavefield expanded to Legendre degree  $p$  and the medium's contrast to degree  $r$  in each voxel, the scheme converges at order  $\min(2p + 2, 2r + 2)$ , with leading error  $c_p(kd)^{2p+2}$  in closed form ( $c_0 = 1/12$ ,  $c_1 = 1/720$ ,  $c_2 = 1/100800$ ,  $k$  the wavenumber and  $d$  the voxel side), for incident P, SV and SH waves at normal and oblique incidence. At second order in the contrast the error of the single-site  $T$ -matrix equals the relative projection error of the medium, the fraction of the squared norm of its profile that polynomials of degree  $\min(p, r)$  on the cells cannot represent: the two bases must be matched, and the error is known from the medium and the grid before anything is solved. In a plane-stratified background the orders persist, and a sphere whose contrast falls smoothly to zero converges at the order the layer predicts. The single site is the first member of a Cartesian multipole hierarchy: each voxel radiates through the Cartesian multipole moments of its source and receives the field through its gradients at its centre, the integrals over the source cube being exact at every separation. Its coefficients are moments of the Green's tensor over the cube, in closed form on four constants, and nothing in it is fitted. With third gradients it is three hundred times more accurate than the uniform-strain closure against the exact sphere, and as a voxel scheme it is fourth order. Every result is derived symbolically and checked against an independent implementation or an exact solution.

## 1 Introduction

**The problem this paper serves.** Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation

that uses one must first sample it onto a grid. A volume-integral scattering calculation divides the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site  $T$ -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green's tensor of the background, with the multiple-scattering series summed.

**The question: two bases.** A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background, which the model prescribes. The other is the *wavefield*: the displacement and strain inside the voxel, which the calculation must find. Each is expanded in a basis on the cell, and the two bases are separate choices. Here both are Legendre polynomials, the wavefield to degree  $p$  and the contrast to degree  $r$ ; the usual scheme, a uniform field in a cell of constant properties, is  $p = r = 0$ . Neither choice is new in itself. In higher-order volume integral methods for electromagnetics the material within an element is represented by polynomials whose orders are independent of those of the field [1], as it is in finite elements, and voxels that carry a linear field are in use [2]. What has been missing is the law that relates the two bases: the order a given pair attains, and the error it leaves. The answer decides the cost of everything built on it, because a voxel that represents its contents more accurately can be larger, and the number of voxels, not the size of each one's state, sets the cost of the multiple-scattering solve.

**Why the question is hard to answer.** For the media such a scheme is built for, there is no exact solution. Its error cannot be measured against the answer. Comparison with another numerical method mixes the errors of both. And a refinement study can mislead, because a voxel scheme can converge smoothly to the wrong limit. §4 shows how that happens with the most natural choice of coupling, the Green's tensor evaluated between voxel centres: it carries a local term whose size per unit volume does not shrink as the voxels do, and unless the single voxel's response is matched to cancel it, refinement converges to the wrong answer without revealing it.

**The method: a layer of voxels.** This paper therefore answers the question in the simplest configuration in which a voxel scheme runs in full (every voxel, every coupling, the whole multiple-scattering series, at normal and at oblique incidence) and the exact answer is nevertheless known: a layer filled with voxels. Because cubes tile space, a layer of identical voxels does not approximate a homogeneous layer; it *is* one. Any difference between its reflection and transmission and those of the continuous layer is therefore error of the discretisation alone, with no modelling error and no shape error behind which to hide. A sphere, the usual test object for volume-integral and discrete-dipole methods [3], cannot give this: its error mixes the discretisation with the staircase approximation of a curved surface by cubes. The layer plays the part of the patch test in finite elements: reproducing a uniform medium correctly is necessary, though not sufficient, for a scheme to be trusted in a heterogeneous one. The experiment is then made harder in three steps, each with an exact solution of its own: a randomly stratified layer, in which every cell is homogeneous but the cells differ; a layer whose properties vary smoothly within every cell, which is where the basis for the medium is tested; and a sphere whose contrast falls smoothly to

zero at its surface, which carries the layer's prediction into three dimensions without a staircase. The coupling and the single site that the layer validates are the same in any medium; a laterally heterogeneous medium remains to be tested.

**The answer.** (1) *The order is set by the weaker basis.* The scheme converges at order  $\min(2p + 2, 2r + 2)$  in the cell size (§9). A contrast held constant in each cell therefore caps every voxel at second order in a medium that varies within the cells, however rich the basis of its field; and a contrast of higher degree than the field is wasted. (2) *The leading error is in closed form, for every grid.* For a voxel of side  $d$  and a wave of wavenumber  $k$  it is  $c_p(kd)^{2p+2}$  with  $c_0 = 1/12$ ,  $c_1 = 1/720$  and  $c_2 = 1/100800$  (§8); the orders 2, 4 and 6 of  $p = 0, 1$  and 2 hold for P, SV and SH waves at normal and oblique incidence. For the collocation scheme, the uniform-field voxel in its simplest form, the error is  $+(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission (§7), and the opposite signs explain why no correction of the single site can improve both (§12). (3) *The error that remains is the single site's, and it is the projection error of the medium.* The single-site  $T$ -matrix sums the rescattering within a voxel to all orders and the coupling between voxels is exact, but the  $T$ -matrix is that of the voxel as represented, its contrast projected to degree  $r$  and its field held to degree  $p$ . The error of the term of second order in the contrast equals the *relative projection error* of the medium, the fraction of the squared norm of its profile that polynomials of degree  $\min(p, r)$  on the cells cannot represent (Eq. (43)): exactly in the long-wave limit, and to within 1.3% for all nine pairs with  $p, r \leq 2$  (§9). The wavefield basis must therefore carry what the medium imprints on the field, not only the wave, and the two bases must be matched. (4) *The error is known in advance.* The projection error is a property of the medium and the grid alone, so the cell size and the degree that a required accuracy needs are chosen before anything is solved. (5) *Where the medium is piecewise constant on the cells the projection error vanishes,* a constant suffices for the medium, and the degree of the field alone sets the order: in a randomly stratified model one first-moment voxel per model layer reaches a reflection accuracy of  $10^{-6}$ , where the uniform-field voxel needs about forty (§9), and in the uniform layer it allows voxels 58 times larger in each direction (Appendix I). (6) *In three dimensions* the smoothly graded sphere converges at second order with the mean-only voxel, as the layer predicts (§10), and at fourth order with the first-moment voxel (§11).

**What makes the experiment exact.** The answer above can be trusted only if nothing but the two bases is approximated. Three results, set against the coupled-dipole method described below, secure that. (a) *An exact coupling.* The integrated Green tensor and the filtered coupled dipoles approximate the field of a cell; for cubes the source-cell average is exact, because the cube's form factor vanishes on the reciprocal lattice (the tiling identity, §5), and at normal incidence the lattice kernel is in closed form with no lattice sum. (b) *No polarisability to choose.* The polarisability, a modelling choice in the coupled-dipole method, is here a derived object (§6): the moments of the Green's tensor over the cube, taken as distributions, yield Eshelby's  $\delta$  term without adding it by hand, the cube's departure from the sphere as a pure cubic anisotropy, and the closure's error against exact scattering by a sphere as a real  $(ka)^2$  term in closed form. The self term the single site carries is the one the averaged kernel omits, so the discrete system is a collocation of the continuum equation. (c) *The elastic state.* A voxel carries a force (through its

density contrast) and a moment (through its stiffness contrast), so its state has nine components, displacement and strain, the coupling is a  $9 \times 9$  tensor, and the single site's self-interaction is the elastic depolarisation of a cube, the discrete counterpart of the inclusion tensor of Eshelby [4]. Cubes serve all three: they tile space, so a uniform region cut into cubes is still that region, and their regular lattice makes the coupling a convolution, solved by fast transforms.

**The coupled-dipole method.** Replacing a heterogeneous body by a lattice of small scatterers that interact through the background Green's function is the multiple-scattering description of Foldy [5] and Lax [6]. In electromagnetics it became a numerical method as the coupled-dipole method, or discrete dipole approximation (DDA): each cell of a cubic grid is a polarisable point dipole driven by the incident field and by the fields of every other dipole [7, 8]. The method is a collocation of the volume integral equation for the electric field, with the cell's self-interaction absorbed into its polarisability [9], and Yurkin and Hoekstra [10] review it in that form. Its accuracy turns on two ingredients, and this paper takes up both for elasticity. The first is the polarisability. The Clausius–Mossotti value is the static response of a sphere; the lattice dispersion relation of Draine and Goodman [11] adjusts it so that an infinite lattice propagates waves as the continuum does to a given order in  $kd$ ; Rahmani et al. [12] instead derive it from the exact static solution for the whole body, so that the method is exact in the long-wavelength limit. The second is the coupling. Integrating the Green tensor over the source cell rather than sampling it at the centre improves the accuracy, most of all for strong contrasts [13], and the filtered coupled dipoles of Piller and Martin [14], revived by Yurkin et al. [15] for large refractive indices, replace the point dipole by a source of finite extent whose spectrum is band-limited to the grid. The coupling summed over a lattice extends the method to periodic structures [16]. Its error converges at second order in the dipole spacing when shape errors are absent [3], and for a cube, the one shape a cubic grid represents without a staircase, Yurkin and Kahnert [17] reach relative uncertainties of  $10^{-7}$  to  $10^{-3}$  by extrapolation in the grid. On the mathematical side, Costabel et al. [18] show that the method is a finite section of an infinite block Toeplitz matrix, stable for a large range of the contrast but not for all of it, and Costabel [19] that in a one-dimensional model it converges at second order in the interior, with a boundary layer in which the consistency error does not vanish. The layer studied here is the elastic counterpart of that cube: a body the grid represents exactly, so that the discretisation error can be isolated. What this paper adds is the closed form: for cells that tile space, the coupling that makes the lattice reproduce the continuum exactly, and the order and the constant of the error that remains, as functions of the two bases. In the terms used here the coupled-dipole method is the pair  $p = r = 0$ , and its refinements improve the polarisability and the coupling while leaving both bases constant in the cell.

**Its elastic counterparts.** The elastic problem has the same volume integral equation [20], and every cell-based solver of it is a coupled-dipole method whether or not it carries the name. Nestor [21] divided a two-and-a-half-dimensional medium into cells that interact through the background Green's tensor, each carrying a contrast renormalised by its own near field, with the multiple-scattering series summed. Kanaun and Levin [22] discretise the inclusion problem with Gaussian approximating functions whose matrix elements are analytic. Yang et al. [23] solve a

stress–velocity form of the integral equation on a grid by conjugate gradients with fast Fourier transforms, and Touhei [24] does the same for a half space with a generalised Fourier transform. Jakobsen [25] and Jakobsen et al. [26] build the seismic forward and inverse problems on the transition operator of the discretised Lippmann–Schwinger equation, the first in the acoustic approximation and the second in arbitrary anisotropic elastic media with a nine-component state vector.

**This paper.** It extends the formulation of Nestor [21] to cubic voxels, and is ordered as the experiment is built. §§2–3 set out the discrete medium and the continuous layer it must reproduce; §4 shows why the point coupling fails, and §5 why the source-averaged coupling succeeds exactly. §6 derives the single-site  $T$ -matrix as the first member of a Cartesian multipole hierarchy in closed form, and sets out what that gives (§6.7). §§7–8 take up the wavefield basis, §9 the medium basis and its projection error, §10 the sphere, and §11 the hierarchy as a voxel scheme. §12 discusses what the results imply and §13 concludes. Every result is derived symbolically in Mathematica and checked against an independent Python/Fortran implementation or an exact solution.

## 2 Setting

**The discrete medium.** Space is tiled by cubes of side  $d$  (half-side  $h = d/2$ ). Cube  $i$  carries a contrast  $\Delta\lambda, \Delta\mu, \Delta\rho$  from an isotropic background with Lamé parameters  $\lambda, \mu$ , density  $\rho$ , P and S speeds  $\alpha, \beta$  and P-wave modulus  $M_P = \lambda + 2\mu$ . Time dependence is  $e^{-i\omega t}$ ;  $k_P = \omega/\alpha$ ,  $k_S = \omega/\beta$ . Coordinates are ordered  $(z, x, y)$  with  $z$  down. The field at a voxel is the nine-component state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

displacement followed by engineering strain. A voxel responds with a force and a moment (stress dipole); the  $9 \times 9$  single-site  $T$ -matrix maps the exciting state to them, and the  $9 \times 9$  lattice Green’s tensor  $\mathbf{K}(i, j)$  maps a source voxel’s force and moment to the state at voxel  $i$ . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

with  $\psi^0$  the incident state.

**Where the kernel comes from.** Write the medium as the background plus the contrast. With time dependence  $e^{-i\omega t}$  the equation of motion is  $\partial_c \sigma_{bc} + \rho \omega^2 u_b = 0$ , with  $\sigma_{bc} = c_{bcde} \varepsilon_{de}$  and summation over repeated indices. Splitting  $\rho = \rho_0 + \Delta\rho$  and  $c = c^0 + \delta c$ , with  $\delta c_{bcde} = \Delta\lambda \delta_{bc} \delta_{de} + \Delta\mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$  the contrast stiffness, and moving the contrast terms to the right leaves the background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta\rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

a force from the density contrast, and the divergence of the contrast stress  $m$ , the moment, from the stiffness contrast. With  $\mathbf{G}$  the Green’s tensor of the background, the displacement is

$u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$ , and the moment term integrates by parts,

$$\int G_{ab}(\mathbf{x} - \mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x} - \mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x} - \mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

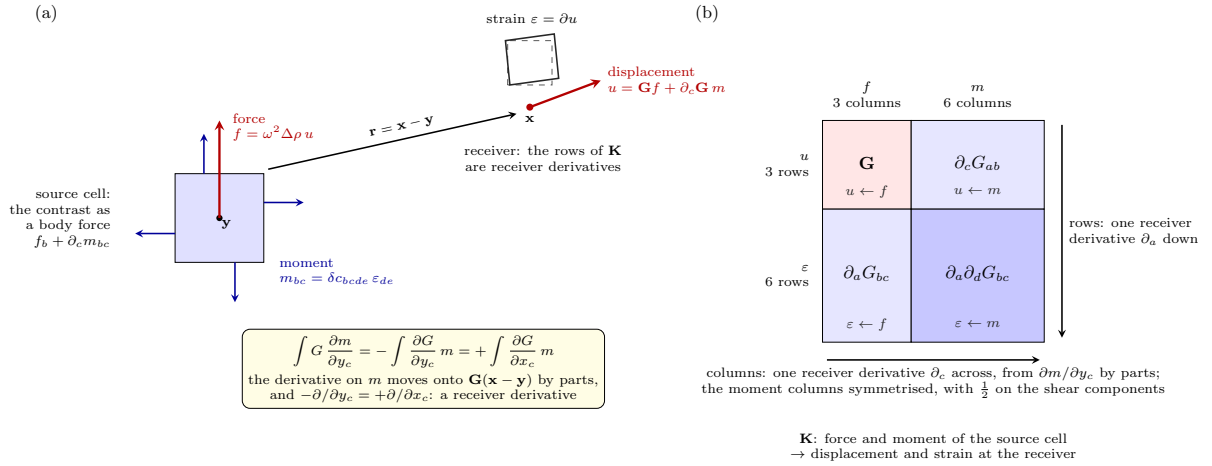
the boundary term vanishing because  $m$  is confined to the scatterers, and the last step holding because  $\mathbf{G}$  depends only on the separation. The moment therefore radiates through the *receiver* derivative of  $\mathbf{G}$ , with a plus sign, when the moment is the contrast stress itself. With the moment tensor defined instead through its equivalent body force  $-\partial_c[M_{bc} \delta(\mathbf{x} - \mathbf{y})]$ , as in Aki and Richards [27], the same field reads  $M_{bc} \partial G_{ab} / \partial y_c$ , the source derivative, with  $M = -m$ . The two are one convention written two ways; what matters is that the derivative in the kernel and the sign in the contrast operator are chosen together. The nine-component kernel follows (Fig. 1): its force columns are  $\mathbf{G}$  and its moment columns the receiver derivatives  $\partial_c G_{ab}$ , symmetrised over  $(b, c)$  with a factor  $\frac{1}{2}$  on the shear components; its displacement rows are those entries and its strain rows their receiver derivatives, symmetrised in the same way and doubled on the shear rows to give engineering strain. In that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

where  $\Delta \mathcal{C}$  is the  $6 \times 6$  Voigt form of  $\delta c$ , the contrast stiffness with  $\Delta \lambda + 2\Delta \mu$  and  $\Delta \lambda$  in the normal block and  $2\Delta \mu$  (not  $\Delta \mu$ ) on the shear diagonal.

**Averaging, and collocation.** Three operations in this paper are averages over a cell, and they are distinct. The *source-cell average* integrates the kernel over the cell that radiates:  $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) \, d\mathbf{y}$  is the field at  $\mathbf{r}$  of a uniform polarisation of the source cell. Every scheme in this paper uses it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation is imposed: at the receiving cell's centre, with the internal field of each cell uniform, so that the unknown is the field at the centre. The *receiver-cell average* imposes the equation in the mean over the receiving cell instead; with a uniform internal field that is the Galerkin scheme of degree zero, called the mean-only voxel below. The two are both *uniform-field* voxels, a term that refers to the internal field and not to the test, and they differ only in the test function (a delta at the centre against the cell's indicator) and in the constant of their  $O((kd)^2)$  error (Table 7). The *cell-mean contrast*, finally, is an average of the material, not of the field: where the medium varies within a cell, the contrast is projected onto the cell's basis, its mean for degree  $r = 0$ , rather than sampled at the centre (§§9, 10). In the language of the coupled-dipole method, the point dipole is collocation with the point kernel, the integrated Green tensor adds the source-cell average, and the filtered coupled dipoles replace the point source by one of finite extent.

**The laterally uniform layer.** A layer of thickness  $D$  is filled with  $n$  planes of voxels,  $d = D/n$ , and illuminated at normal incidence. The field is then laterally uniform, only the zero-lateral-wavenumber (specular) component of the lattice coupling acts, and Eq. (2) becomes exactly a



**Figure 1.** Where the kernel comes from. (a) A source cell at  $\mathbf{y}$  acts on the background through the body force of Eq. (3): a force  $\mathbf{f}$  from its density contrast, and the divergence of its contrast stress  $\mathbf{m}$ , the moment, from its stiffness contrast (arrows on the faces). At the receiver  $\mathbf{x}$  the state is the displacement  $\mathbf{u}$  and the strain  $\boldsymbol{\varepsilon}$ . The moment's derivative moves onto  $\mathbf{G}(\mathbf{x} - \mathbf{y})$  by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the  $9 \times 9$  kernel  $\mathbf{K}$  that results: the force columns are  $\mathbf{G}$ , the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

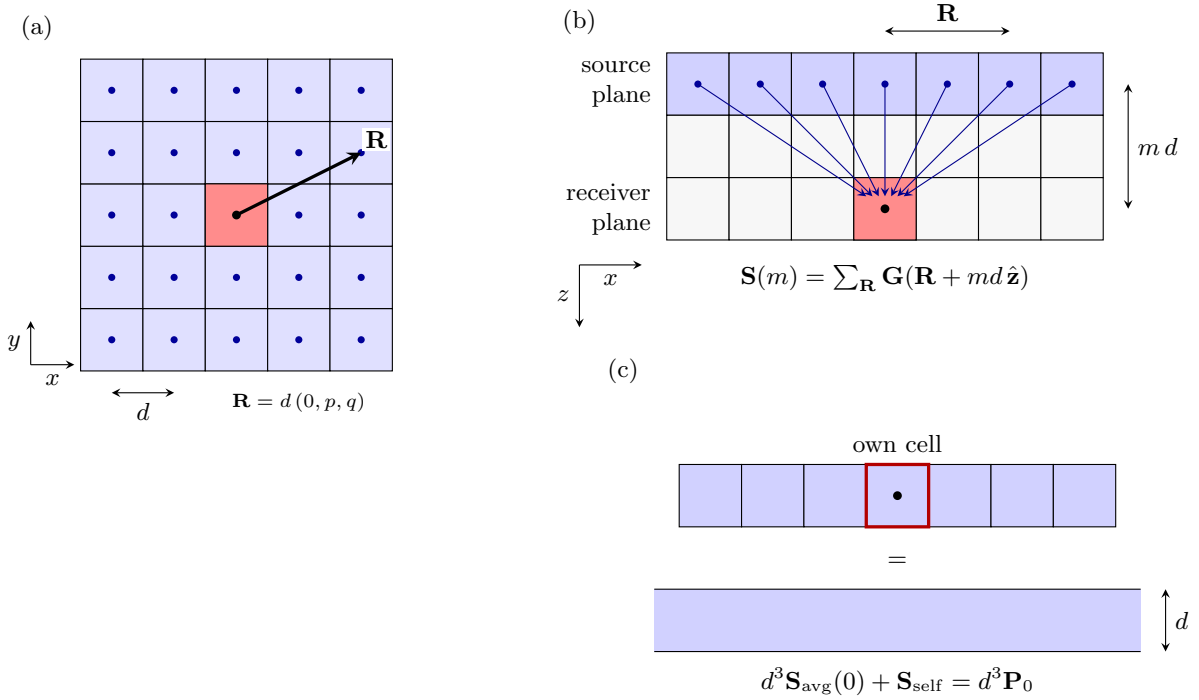
*Alt text:* Two panels. On the left, a shaded square cell with a dot at its centre marked  $\mathbf{y}$ ; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled  $\mathbf{r} = \mathbf{x} - \mathbf{y}$  runs from the cell to a red dot marked  $\mathbf{x}$  at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of  $\mathbf{G}$  times the  $y$ -derivative of  $\mathbf{m}$  equals minus the integral of the  $y$ -derivative of  $\mathbf{G}$  times  $\mathbf{m}$ , equals plus the integral of the  $x$ -derivative of  $\mathbf{G}$  times  $\mathbf{m}$ . On the right, a two-by-two grid of shaded blocks with rows labelled  $\mathbf{u}$ , three rows, and strain, six rows, and columns labelled  $\mathbf{f}$ , three columns, and  $\mathbf{m}$ , six columns; the blocks read  $\mathbf{G}$ , the first derivative of  $\mathbf{G}$ , the first derivative of  $\mathbf{G}$ , and the second derivative of  $\mathbf{G}$ ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

chain of  $n$  planes coupled by the specular lattice sums

$$\mathbf{S}(\mathbf{m}) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + \mathbf{m}d\hat{\mathbf{z}}), \quad (6)$$

where the sum runs over the lattice vectors  $\mathbf{R} = d(0, p, q)$ ,  $p, q$  integers, that join the centres of the voxels in one plane (a square lattice of pitch  $d$ ), and  $\mathbf{m}$  is the separation of the two planes in units of  $d$  (Fig. 2a,b). The question is whether, and how fast, this chain converges to the continuous layer as  $n \rightarrow \infty$  at fixed  $D$ .

**Implementations.** Every result is derived symbolically in Mathematica and compared with an independent numerical implementation or an exact solution. “The package” below is the Python/Fortran implementation of the voxel scheme (`cubic_scattering`): its cube  $T$ -matrix, its lattice kernels and its stratified propagator, which serve both as the object under test and, where validated, as a second implementation.



**Figure 2.** The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch  $d$ , and a lattice vector  $\mathbf{R} = d(0, p, q)$  joins the centre of one voxel (shaded darker) to another. (b) Cross-section ( $z$  down), between planes: the voxel of a receiver plane (shaded darker) is coupled to every voxel of a source plane  $m$  planes above it, one for each lattice vector  $\mathbf{R}$ ; the sum over  $\mathbf{R}$  is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver’s own cell (outlined) included, is the field of a uniform plate of thickness  $d$ , the tiling identity of Eq. (13).

*Alt text:* Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled  $\mathbf{R}$  runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked  $d$ . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation  $m d$  marked. Below it, a single row of voxels with one outlined as the receiver’s own cell is shown equal to a continuous shaded band of the same thickness  $d$ , a uniform plate.

### 3 The continuous layer and its thin-layer expansion

At normal incidence a plane force drives a one-dimensional problem per wave type, with modulus  $M$  ( $M_p$  for P,  $\mu$  for S), wavenumber  $k$  and plane Green’s function  $g(z) = \frac{i}{2Mk} e^{ik|z|}$ . The layer’s reflection and transmission follow from continuity of displacement and traction at its faces; they agree with the stratified propagator used as the reference in this work to  $2.5 \times 10^{-8}$ , P and S (ContinuumLimit\_Reference.wl).

For a layer of thickness  $D$  with modulus  $M_1$  and density  $\rho_1$  in a background  $M_0, \rho_0$ , write  $Z_j = \sqrt{M_j \rho_j}$  for the impedances,  $\zeta = Z_1/Z_0$  for their ratio and  $k_1 = \omega \sqrt{\rho_1/M_1}$  for the layer’s wavenumber. The reflection coefficient expands as

$$\begin{aligned}
 R = & \frac{ik_1 D}{2} (\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4} (\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24} (\zeta - \zeta^{-1})(3\zeta^{-2} + 2 + 3\zeta^2) \\
 & + \frac{(k_1 D)^4}{48} (\zeta^2 - \zeta^{-2})(3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5),
 \end{aligned} \tag{7}$$

exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each coefficient symbolically and the truncated series against the exact layer). The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[ \frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left( \frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

to first order a thin layer is an interface across which the traction jumps by the excess inertia and the displacement by the excess compliance. At oblique incidence the same statement holds with the first-order system  $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$  for the P-SV state  $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$ : the layer acts to  $O(D)$  as the jump  $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$ , which reproduces the layer's  $O(D)$  reflection coefficients  $R_{pp}, R_{ps}$  to  $10^{-12}$ ; the series truncated at  $O(D^k)$  converges at exactly order  $k$ ; and a soft thin layer with shear modulus  $\mu_1 = D/\eta$  at fixed compliance  $\eta$  tends to the linear-slip interface as  $D \rightarrow 0$ . These expansions are the target a single plane of voxels must reproduce order by order. A plane of voxels that carry Legendre moments of their internal field to degree  $p$  reproduces Eq. (7) term by term through  $O(D^{2p+2})$  and first differs at  $D^{2p+3}$  (§8): the mean-only voxel reproduces the equivalent interface and the  $D^2$  term, the first-moment voxel all four terms shown.

## 4 The specular sums of the point Green's tensor

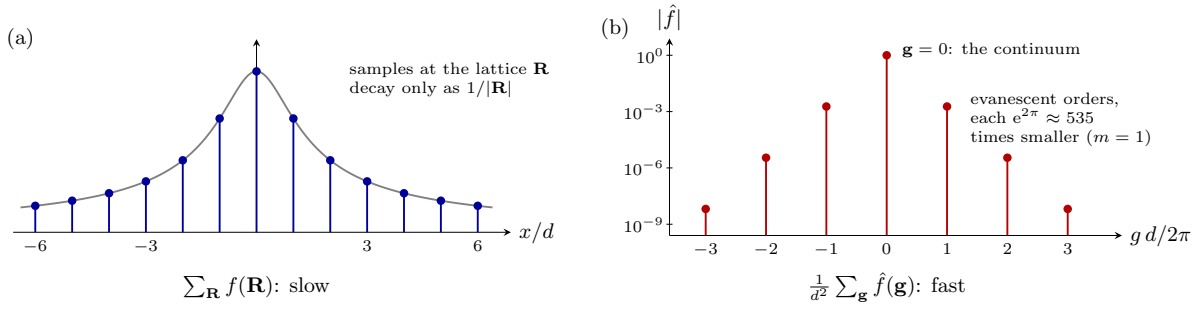
**Why transform the sum.** The specular sum (6) adds, at a receiver, the field of every voxel of a plane  $m$  planes away. In that form it is a poor object to study. The Green's tensor decays slowly with distance, so the sum converges slowly, and it hides the one question that matters here: which part of it is the continuum, and which part is the discretisation. The Poisson summation formula [28] answers both at once, and is the standard tool for lattice sums of wave fields [29]. For a function  $f$  on the plane and the square lattice of pitch  $d$  it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q), \quad (9)$$

with  $p, q$  integers: a sum of samples of  $f$  on the lattice equals a sum of samples of its Fourier transform on the reciprocal lattice, whose pitch is  $2\pi/d$  (Fig. 3). Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (10)$$

with  $\hat{\mathbf{G}}(\mathbf{g}; z)$  the two-dimensional (lateral) transform of the Green's tensor between planes a distance  $z$  apart. Each term has a physical meaning. Transformed laterally, the field of a plane of sources is a superposition of plane waves, one for each lateral wavenumber, and the lattice selects the wavenumbers  $\mathbf{g}$ . The order  $\mathbf{g} = 0$  is the laterally uniform plane wave that a continuous layer radiates: it is the continuum's term. Every other order has  $|\mathbf{g}| \geq 2\pi/d$ , far beyond the wavenumber of the field when the voxels are small compared with the wavelength, so it is evanescent and decays away from the plane as  $e^{-|\mathbf{g}|md} = e^{-2\pi m \sqrt{p^2 + q^2}}$ , independently of  $d$ . The transform therefore splits the lattice sum exactly into the continuum and a discretisation correction, and the correction converges after a few orders.



**Figure 3.** The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as  $1/|\mathbf{R}|$ , so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders  $\mathbf{g} = 2\pi\mathbf{p}/d$ . The order  $\mathbf{g} = 0$  is the continuum’s plane wave; every other order is evanescent and, at  $m = 1$ , smaller than its neighbour by  $e^{2\pi} \approx 535$ . Magnitudes are schematic: the static decay  $e^{-|\mathbf{g}|md}$  is shown.

*Alt text:* Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

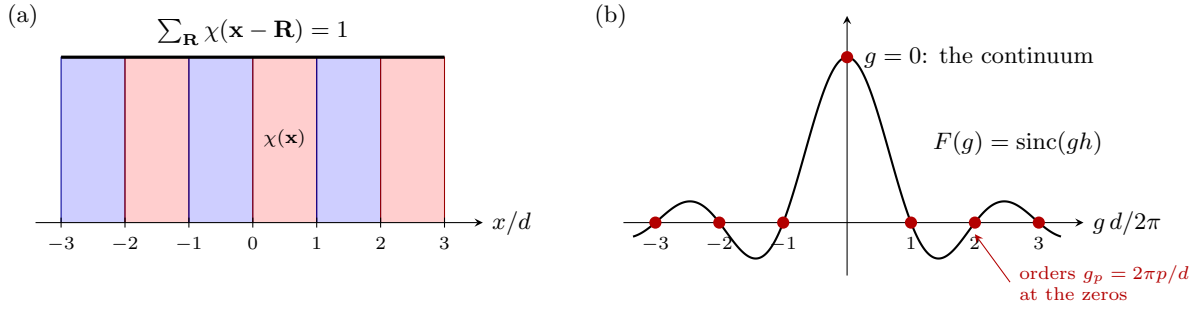
**The local term that survives.** In the strain-from-moment block the evanescent orders carry two powers of  $|\mathbf{g}| \propto 1/d$ , so their contribution per voxel volume does *not* vanish as  $d \rightarrow 0$ : at  $m = 1$  it is eleven times the continuum term (ContinuumLimit\_SpecularSums.wl, matched to the package’s Ewald lattice sum to  $5 \times 10^{-13}$ ). This surviving term is static. Summed over all planes in slab order, the static lattice sum of the point kernel is exactly the depolarisation of a uniform plate ( $-1/M_p$  on  $\varepsilon_{zz} \leftarrow M_{zz}$  and  $-1/(2\mu)$  on the normal shears, per unit volume) plus a cubic-symmetric remainder (ContinuumLimit\_StaticLocal.wl, to  $2 \times 10^{-10}$ ). The point kernel therefore reproduces the continuum only if that remainder is cancelled by the voxel’s own self term.

## 5 The tiling identity

Replace the point coupling by the *source-cell average*  $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$ , the field at a voxel centre of a uniform polarisation of the source voxel. Each Poisson order in Eq. (10) is then multiplied by the cube’s form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

which vanishes at every  $\mathbf{g} \neq 0$  of the reciprocal lattice since  $\text{sinc}(p\pi) = 0$  for integer  $p \neq 0$ . This is the Fourier form of the fact that cubes tile the plane,  $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$  for the cube’s indicator  $\chi$ : a partition of unity (Fig. 4). Two exact statements follow.



**Figure 4.** Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions  $\chi$  of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube’s lateral form factor  $F(g) = \text{sinc}(gh)$  (shown along one axis) vanishes at every reciprocal-lattice order  $g_p = 2\pi p/d$  with  $p \neq 0$ , so of all the orders in the lattice sum (10) only  $g = 0$ , the continuum’s plane wave, survives the source-cell average.

*Alt text:* Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of  $2\pi/d$ ; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

**Between planes.** For  $m \neq 0$  every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

the continuum’s plane-wave term times the vertical form factor. The package’s cell-averaged kernel agrees in all 81 entries to  $2.8 \times 10^{-16}$  (`ContinuumLimit_AveragedSums.wl`).

**Within a plane.** The sum over *all* cells of the plane, the voxel’s own included, is the field at the mid-plane of a uniform plate of thickness  $d$  (Fig. 2c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

where  $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$  is elementary: in the one-dimensional transform each entry is  $c_0 + A/(Mk_z^2 - \rho\omega^2)$ , so it averages to  $c_0/d + A(e^{ikh} - 1)/(dMk^2)$ . Its only nonzero entries are  $u \leftarrow f$  on the diagonal and the three normal strain-from-moment entries, whose local weights  $-1/M_p$  and  $-1/(2\mu)$  are the plate depolarisation of §4. Statically the identity holds for all 21 independent components of the strain-from-moment tensor to  $4 \times 10^{-9}$ , with every cell integral evaluated analytically (below).

**Consequence: an exact kernel with no lattice sum.** At normal incidence the same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

with no Ewald splitting, near shell or asymptotic tail. The route it replaces (Ewald summation with a near shell averaged by quadrature and an  $O(d^2)$  Taylor tail beyond it) carries a static strain-from-moment error of  $9.4 \times 10^{-4}$  at its default settings, falling to  $2.6 \times 10^{-5}$  only when both the shell radius and the quadrature order are raised; Eq. (14) is now the package’s kernel at normal incidence.

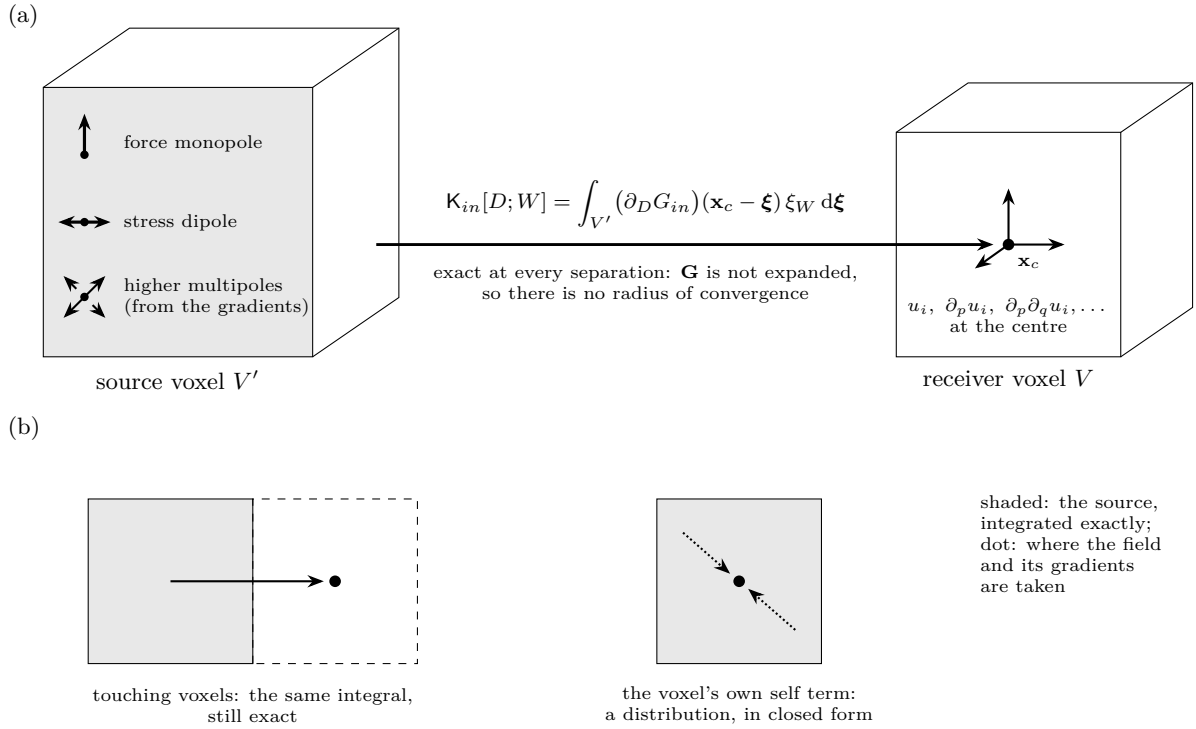
**Cell integrals without quadrature.** Write the static (Kelvin) tensor as  $G_{bc} = (c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$  with  $c_B = (1/\mu - 1/M_P)/8\pi$  and  $c_A = 1/(4\pi\mu) - c_B$ . Every cell integral of its derivatives is a cell integral of derivatives of  $1/r$  and  $r$ . One application of the divergence theorem moves it onto faces  $x_p = c_p \pm h$ , which never contain the origin, and there the two-dimensional antiderivative in the face coordinates is continuous, so the face integral is a sum over the face’s corners. The distributional term at  $r = 0$  is carried by the divergence theorem itself. (A triple antiderivative evaluated at the cube’s corners is *not* usable for repeated derivatives: its arctangent terms jump across the coordinate planes through the origin, and differentiating twice there discards exactly that term.) The self term so obtained reproduces the classical closed form of the cube’s depolarisation [30, 31] to machine precision.

## 6 The single site: the cube $T$ -matrix from moment integrals

Every error of a voxel scheme is made at the single site (§12), so the single site is derived here rather than imported. The classical route imports it. For an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a uniform load uniform, with a depolarisation tensor in closed form [4, 32, 33]; its dynamic generalisation is known for spheres and ellipsoids [34–36]. For a cube or a general polyhedron the static field of a uniform eigenstrain is also in closed form [30, 31, 37], and it has been computed for a cuboid in an anisotropic matrix [38], but the strain it produces is not uniform, so there is no exact uniform-strain inclusion to import. Closed-form Eshelby tensors for polyhedral inclusions with polynomial eigenstrains have since been given [39], the dynamic inclusion problem has been formulated for shapes other than the ellipsoid, cubes among them [40], and the Green’s tensor has been integrated analytically over a cuboid for the coupled-dipole method, to an error of  $O((kd)^4)$  [41]; these are the static and electromagnetic counterparts of individual moments below. In scattering theory the volume integral equation of an elastic inclusion [20] is closed in the same spirit by a self-consistent or effective-field approximation [42, 43]: a first-order (Born) scatterer [44] renormalised by the geometric series of its own near field, the step that turns a contrast into an effective contrast. Nestor [21] introduced the single site in this form, summing the series of the cell’s self integral, with the singular near field evaluated over an infinitesimal cell. The moments below evaluate the same self integral over the finite cube, as a distribution. The elastic  $T$ -matrix of Waterman [45] represents a scatterer in spherical multipoles, the natural basis for a sphere; a voxel in a lattice needs one compatible with the lattice, and the moments over the cube provide it. The construction below makes the approximation of a voxel explicit rather than implicit: it derives the cube’s  $T$ -matrix from the moments of the elastodynamic Green’s tensor over the cube, with the uniform-strain closure as the first truncation of a hierarchy.

**A Cartesian multipole hierarchy.** We call the construction the Cartesian multipole hierarchy (Fig. 5). The contrast acting on the voxel's field is a source: the density contrast a body force  $\omega^2 \Delta \rho \mathbf{u}$ , the stiffness contrast a stress  $\delta \mathbf{c} : \boldsymbol{\varepsilon}$ . Each voxel radiates through the Cartesian multipole moments of these sources over the cube, a force monopole and a stress dipole from the uniform field (the cell of the coupled-dipole method) and higher multipoles from the field's gradients, and receives the field through its gradients at its centre. The name needs one caution. A multipole expansion usually expands the kernel or the field about a centre and truncates it, which holds only for well-separated cells or outside a circumscribing sphere: in the fast multipole method [46], whose elastodynamic form leaves adjacent cells to direct integration [47], in the spherical-wave  $T$ -matrix of Waterman [45], and in the force-dipole description of a point defect, an expansion of the Green's function far from it [48]. Here the Green's tensor is never expanded. Only the source is resolved into Cartesian moments, and the integral over the source cube is exact at every separation, so there is no radius of convergence, and touching voxels and the self term, a distribution in closed form, are included. The source side is shared with the multipole extensions of the coupled-dipole method in optics [49, 50] and with Cartesian multipole decompositions of a source [51]. The word hierarchy has its sense in kinetic theory and cosmology [52]: a chain of coupled moment equations closed by truncation at a chosen order, whose truncation error is here in closed form.

**Its origin in the equivalent inclusion method.** In its static form the construction is the equivalent inclusion method with polynomial eigenstrains of Moschovidis and Mura [53]: the eigenstrain is expanded about the centre of the inhomogeneity, the consistency equation is expanded there in a Taylor series, and the coefficient of each monomial is equated, which amounts to imposing the Lippmann–Schwinger equation and its derivatives at the centre [54, §2.2]. Johnson et al. [55, 56] applied an expansion about the origin to a cuboidal inhomogeneity, and Wu et al. [39] expanded the eigenstrain of a polyhedron to second degree about its centroid; both report that the expansion about the centre does not follow the field near the edges, corners or vertices. Fu and Mura [34] extended the method to time-harmonic waves for an ellipsoid, with a second unknown for the density mismatch, and in the cases they report evaluated it with uniform eigenstrains. Brisard et al. [54] replaced the Taylor form by a Galerkin form of the same equation, because the Taylor form need not give an invertible system and a higher degree need not improve it. What is added here, to our knowledge, is the elastodynamic hierarchy, with density and stiffness contrasts together, to any degree and for a body that is not an ellipsoid; its moments as distributions in closed form; and the structure and the error of its truncations, tested degree by degree against the exact solution for a sphere. The moments here are integrals of the Green's tensor against monomials over the body. They are not moments in the sense of the method of moments [57], of which the hierarchy is the instance with monomials about the centre as expansion functions and the  $\delta$  function and its derivatives at the centre as testing functions; and the exact static response of an inclusion of any shape to polynomial loads is described by the polarisation and elastic moment tensors of Ammari and Kang [58].



**Figure 5.** The Cartesian multipole hierarchy. (a) The source voxel radiates through the Cartesian multipole moments of its source: a force monopole from the density contrast, a stress dipole from the stiffness contrast, and higher multipoles from the field's gradients. The receiver voxel takes the field through its value and gradients at its centre  $\mathbf{x}_c$ . The two are coupled by the integral of the derivatives of the Green's tensor over the whole source cube, Eq. (47), which is exact at every separation because the Green's tensor is not expanded. (b) The cases a multipole expansion of the Green's tensor cannot reach are therefore included: touching voxels, by the same integral, and the voxel's own self term, a distribution in closed form.

*Alt text:* Two panels. In the upper panel a shaded cube on the left, labelled source voxel, contains three small symbols: a single arrow labelled force monopole, a pair of opposed arrows labelled stress dipole, and a cross of four arrows labelled higher multipoles, from the gradients. A long arrow runs from it to the centre of an unshaded cube on the right, labelled receiver voxel, where a dot with three short arrows marks the field and its gradients at the centre. Above the arrow is the coupling integral of the derivatives of the Green's tensor over the source cube, and below it the words exact at every separation:  $\mathbf{G}$  is not expanded, so there is no radius of convergence. In the lower panel, on the left, a shaded square touches a dashed square whose centre is marked by a dot, with an arrow from the shaded square to the dot, labelled touching voxels: the same integral, still exact; on the right a single shaded square has a dot at its centre and two dotted arrows pointing to it from within, labelled the voxel's own self term: a distribution, in closed form.

## 6.1 The closed hierarchy

Let the voxel be  $V = [-h, h]^3$  with contrast  $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$  and  $\Delta\rho$ . The hierarchy is stated below for any degree; its third degree is written out first, being the lowest at which the stiffness channel converges at fourth order (§6.4). Expand the internal displacement about the centre to third degree and project the Lippmann–Schwinger equation onto  $1, x_p, x_p x_q$  and  $x_p x_q x_r$ . The unknowns are  $u_i(0)$ ,  $\partial_p u_i(0)$ ,  $\partial_p \partial_q u_i(0)$  and  $\partial_p \partial_q \partial_r u_i(0)$ , and the result is the

closed system

$$(\delta_{ij} - \omega^2 \Delta \rho G_{ij}) u_j + (N_{in,k}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho K_{ij}^{rs}) \partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$(\delta_{pr} \delta_{ij} + \omega^2 \Delta \rho N_{ij,p}^r - M_{in,pk} \delta c_{nkrj}) \partial_r u_j + (\frac{1}{6} \omega^2 \Delta \rho B_{ij,p}^{rst} - \frac{1}{2} P_{in,pk}^{st} \delta c_{nkrj}) \partial_r \partial_s \partial_t u_j = \partial_p u_i^0, \quad (16)$$

$$-\omega^2 \Delta \rho M_{ij,pq} u_j + (\delta_{qr} \delta_{ps} \delta_{ij} + Q_{in,pqk}^r \delta c_{nksj} - \frac{1}{2} \omega^2 \Delta \rho P_{ij,pq}^{rs}) \partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

$$(\omega^2 \Delta \rho Q_{ij,pqv}^r - X_{in,pqvk} \delta c_{nkrj}) \partial_r u_j + (\delta_{pr} \delta_{qs} \delta_{vt} \delta_{ij} + \frac{1}{6} \omega^2 \Delta \rho J_{ij,pqv}^{rst} - \frac{1}{2} Z_{in,pqvk}^{st} \delta c_{nkrj}) \partial_r \partial_s \partial_t u_j = \partial_p \partial_q \partial_v u_i^0, \quad (18)$$

all fields evaluated at the centre. Dropping the third gradients, that is Eq. (18) and the second line of Eq. (16), leaves the system of second degree; dropping the second gradients as well, that is Eq. (17) and the  $\partial \partial u$  terms of Eq. (15), leaves the uniform-strain closure. Equations (15) and (17) are the same at second and third degree, because every term the third degree would add to them vanishes by parity (§6.3). The coefficients are ten moments of the background Green's tensor  $\mathbf{G}$  over the voxel

$$\begin{aligned} G_{ij} &= \int_V G_{ij} dV, & K_{ij}^{rs} &= \int_V G_{ij} x_r x_s dV, & N_{in,k}^r &= \int_V \partial'_k G_{in} x_r dV, \\ M_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} dV, & P_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s dV, & Q_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r dV, \\ B_{ij,p}^{rst} &= \int_V \partial'_p G_{ij} x_r x_s x_t dV, & J_{ij,pqv}^{rst} &= \int_V \partial'_v \partial'_q \partial'_p G_{ij} x_r x_s x_t dV, & X_{in,pqvk} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} dV, \\ Z_{in,pqvk}^{rs} &= \int_V \partial'_v \partial'_q \partial'_p \partial'_k G_{in} x_r x_s dV, \end{aligned} \quad (19)$$

the first six being those of the second degree and the last four those the third gradients add. The derivatives are taken with respect to the source point,  $\partial' = -\partial$ . That sign reaches the moments with an odd number of derivatives,  $\mathbf{N}$ ,  $\mathbf{Q}$ ,  $\mathbf{B}$  and  $\mathbf{J}$ , and no symmetry, parity or scaling check can detect it, so it is fixed by convention. The moments are graded by  $(D, W)$ , the number of derivatives and the degree of the weight:  $\mathbf{G}(0, 0)$ ,  $\mathbf{K}(0, 2)$ ,  $\mathbf{N}(1, 1)$ ,  $\mathbf{M}(2, 0)$ ,  $\mathbf{P}(2, 2)$ ,  $\mathbf{Q}(3, 1)$ , and  $\mathbf{B}(1, 3)$ ,  $\mathbf{J}(3, 3)$ ,  $\mathbf{X}(4, 0)$ ,  $\mathbf{Z}(4, 2)$ .

**The hierarchy at any degree.** Nothing in the construction is special to the third degree. Let the internal displacement be expanded to degree  $\ell$ ,

$$u_j(\mathbf{x}) = \sum_{|W| \leq \ell} \frac{x_W}{|W|!} U_{j;W}, \quad U_{j;W} = \partial_W u_j(0), \quad x_W = x_{w_1} \cdots x_{w_m}, \quad (20)$$

the sum running over ordered sets of indices  $W = (w_1, \dots, w_m)$ , and let the strain be its derivative. Write the moments with any number of derivatives and any weight as

$$\mathbf{g}_{in}[D; W] = \langle \partial_D G_{in}, x_W \mathbf{1}_V \rangle, \quad (21)$$

the derivatives being those of  $G_{in}$  as a function of its argument and the pairing distributional (§6.2). The ten moments of Eq. (19) are these  $\mathbf{g}$  themselves: they are the integrals of  $\partial_D' G_{in}(\mathbf{0} - \mathbf{x})$ , and the factor  $(-1)^{|D|}$  of the source derivative cancels the parity  $(-1)^{|D|}$  of  $\partial_D G_{in}$ , so that, for example,  $\mathbf{N}_{in,k}^r = \mathbf{g}_{in}[k; r]$  and  $\mathbf{Z}_{in,pqvk}^{rs} = \mathbf{g}_{in}[pqvk; rs]$ . Beyond the third degree the moments are written as  $\mathbf{g}$  only. Imposing the Lippmann–Schwinger equation and its derivatives to order  $\ell$  at the centre gives, for every set of derivative indices  $P$  with  $|P| \leq \ell$ ,

$$U_{i;p} - (-1)^{|P|} \omega^2 \Delta \rho \sum_{|W| \leq \ell} \frac{\mathbf{g}_{ij}[P; W]}{|W|!} U_{j;W} + (-1)^{|P|} \sum_{|W| \leq \ell-1} \frac{\mathbf{g}_{in}[Pk; W]}{|W|!} \delta c_{nkrj} U_{j;rW} = \partial_P u_i^0(0). \quad (22)$$

Equations (15)–(18) are the case  $\ell = 3$ , with the moments of Eq. (19) put for the  $\mathbf{g}$ ; their truncations above are  $\ell = 2$  and the uniform-strain closure  $\ell = 1$ . With the field rather than the eigenstrain as the unknown, the moments  $\mathbf{g}[D; W]$  play the part of the polynomial Eshelby tensors of the equivalent inclusion method [33, Ch. 4], which Brisard et al. [54] call influence pseudotensors, here for the elastodynamic kernel and for the density channel as well. For an ellipsoid the static self term of a polynomial eigenstrain is itself a polynomial of the same degree, so the expansion about the centre is exact in the self term and approximate only in the interaction with other bodies [54]; for a cube it is approximate in both. The cube is centrosymmetric, so a moment vanishes unless  $|D| + |W|$  is even, and the system needs the grades

$$\begin{aligned} (D, W) &= (m, W), \quad W \leq \ell && \text{(density),} \\ (D, W) &= (m+1, W), \quad W \leq \ell-1 && \text{(stiffness),} \end{aligned} \quad m = 0, \dots, \ell, \quad D+W \text{ even.} \quad (23)$$

Table 1 lists what each degree adds. The unknowns number  $3\binom{\ell+3}{3}$ : 12, 30, 60 for  $\ell = 1, 2, 3$ .

**Table 1.** The grades  $(D, W)$  of moments,  $D$  derivatives on the Green’s tensor and a weight of degree  $W$ , that the hierarchy of degree  $\ell$  needs beyond those of degree  $\ell - 1$ .

| $\ell$ | new grades                                     | moments in all      |
|--------|------------------------------------------------|---------------------|
| 1      | (0, 0), (1, 1), (2, 0)                         | 3                   |
| 2      | (0, 2), (2, 2), (3, 1)                         | 6: G, K, N, M, P, Q |
| 3      | (1, 3), (3, 3), (4, 0), (4, 2)                 | 10: with B, J, X, Z |
| 4      | (0, 4), (2, 4), (4, 4), (5, 1), (5, 3)         | 15                  |
| 5      | (1, 5), (3, 5), (5, 5), (6, 0), (6, 2), (6, 4) | 21                  |

**Two independent systems.** In either sum of Eq. (22) the unknown  $U_{i;p}$  is coupled to  $U_{j;W}$ , or to  $U_{j;rW}$ , through a moment of grade  $(|P|, |W|)$  or  $(|P| + 1, |W|)$ , and on a centrosymmetric body that moment vanishes unless  $|P|$  and the degree of the unknown it multiplies have the same parity. The hierarchy of any degree therefore separates exactly, at every frequency and contrast, into two systems: one for the even gradients  $u$ ,  $\partial\partial u, \dots$  and one for the odd gradients  $\partial u$ ,  $\partial\partial\partial u, \dots$ . Raising the degree from odd to even adds a block to the even system alone, and from even to odd a block to the odd system alone. At  $\ell = 3$  the even system is Eqs. (15) and (17), unchanged from  $\ell = 2$ , and the odd system is Eqs. (16) and (18). Of the sixty unknowns, twenty-one belong to the even system and thirty-nine to the odd. The leading radiation of a density contrast,  $\int_V \Delta \rho u \, dV$ , is drawn from the even system, and that of a stiffness contrast,

$\int_V \delta c : \nabla u \, dV$ , from the odd one. The separation is thus the structure behind the result of §6.4, that the blocks improve the density and the stiffness channels alternately. At first degree it is implicit in the dynamic equivalent inclusion method for an ellipsoid, where the force and the strain systems uncouple at the centre [34]; here it holds at every degree, for any centrosymmetric body.

## 6.2 The moments as distributions

Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[ \delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (24)$$

[27, 59]. Expanding  $g_c$  in powers of  $r$  reduces every moment in Eq. (19) to a finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (25)$$

For  $m = -1$  and  $D \geq 2$  the classical integrand is not absolutely integrable.  $\partial^2(1/r)$  decays as  $r^{-3}$ , its angular average vanishes, and the value of the integral depends on the shape of the region excluded around the origin. This is the source of the  $\delta$ -function term of the inclusion problem, which a numerical quadrature misses. Without it the depolarisation is wrong; for a sphere it vanishes altogether (§6.3). It is not added afterwards here. The derivatives are distributional from the start, and each is moved onto the voxel's faces by pairing with the test function  $w \mathbf{1}_V$ , using  $\partial_q \mathbf{1}_V = -n_q \delta_{\partial V}$ :

$$\langle \partial_q T, w \mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (26)$$

The surface term is evaluated on the faces, a distance  $h$  from the origin, where every derivative of  $r^m$  is smooth. The recursion lowers  $D$  and  $W$  together and ends on convergent integrals. The face integrals are done in closed form by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p; ] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (27)$$

which the face route satisfies exactly and a spherical excision returns as 0. The closed forms contain logarithms of the units  $(2 \pm \sqrt{3})^n$  of the Pell equation  $x^2 - 3y^2 = 1$ . Left unreduced, these cancel catastrophically: for example,  $\log(72010600134783751 - 41575339372323900\sqrt{3})$  is a cancellation of 35 digits. They are therefore reduced to  $n \log(2 - \sqrt{3})$ , which is exact. Because every power of  $r$  is paired this way, the dynamic moments keep the even-power terms of the expansion ( $k^2 r$ ,  $r^3$ , ...) as well as the odd ones. A route that integrates only the polynomial (odd-power) terms of  $\mathbf{G}$  misses those terms, at a real  $O((kh)^2)$  cost (§6.6).

**Every grade, in closed form on four constants.** Closed forms for monomial-weighted integrals of  $1/r$  over bricks are known [60], as are closed forms for the potential and its first two derivatives over polyhedra with polynomial density, from the gravity literature [61]. The engine used here differs in taking derivatives of any order as distributions, in treating the kernel

$r$  as well as  $1/r$ , and in the even-power terms of the dynamic series. The recursion (26) does not depend on the grade, so the same engine gives every scalar moment of Table 1. Each is evaluated once and stored. The scalar moments of every grade through  $\ell = 5$ , with up to six derivatives and weights to the fifth degree, have been evaluated for the kernels  $r^{-1}$  and  $r$ : 4224 closed forms. (The second term of Eq. (24) places two further derivatives on  $r$ ; those moments are stored through  $\ell = 3$  and are evaluated numerically, by the second route below, beyond it.) Every one of them is a power of the voxel's width times

$$q_0 + q_1 \sqrt{3} + q_2 \pi + q_3 \log(2 - \sqrt{3}), \quad q_i \text{ rational.} \quad (28)$$

The integration returns many of them in other forms, with  $\log(\sqrt{3} - 1)$ ,  $\log(1 + \sqrt{3})$ , inverse hyperbolic tangents and logarithms of powers of two, which are equal in value but not canonical: two equal moments can then differ in appearance, and sums of large logarithms that cancel lose digits. Each stored form is therefore reduced to Eq. (28) by an integer-relation search at 400 digits, with  $\log 2$  and  $\log 3$  admitted to the search, and the reduction is accepted only if it agrees with the stored form to 1200 digits. All 4224 reduce, and the coefficients of  $\log 2$  and  $\log 3$  are zero in every one.

Every grade is gated by two checks whose answers do not depend on the engine, the Laplacian sum rules of Eq. (48), which test the  $\delta$  function of the inclusion problem and its derivatives where a quadrature fails, and confirmed by a second route that shares nothing with the faces of the cube, to  $1.1 \times 10^{-12}$  or better (Appendix A).

### 6.3 Parity, and the nine-component block

The cube is centrosymmetric. A moment that couples a field quantity of degree  $a$  to one of degree  $b$  therefore vanishes unless  $a + b$  is even.  $u$ ,  $\partial u$  and  $\partial\partial u$  have degrees 0, 1 and 2, so at second degree Eq. (16) contains the first gradient alone:  $\partial u$  is determined by its own block and couples to neither  $u$  nor  $\partial\partial u$ . At any degree it couples to nothing of even degree; from  $\ell = 3$  it couples to  $\partial\partial\partial u$  (the second line of Eq. (16)). The leading scattering by a stiffness contrast is the stress dipole  $\int_V \delta c : \nabla u \, dV$ , which needs only  $\partial u$ . The leading far field of the stiffness contrast is therefore fixed by Eq. (16), and *no treatment of the second-gradient block (17) can change it*. What Eq. (17) contributes reaches the far field only through the  $O((kh)^2)$  coupling in Eq. (15). That is the order, and the origin (the variation of the field across the voxel), of the collocation error of §7, which the first-moment voxel of §8 removes, as does the hierarchy itself once its third-gradient block is kept (§6.4).

With  $\Delta\rho = 0$ , Eq. (16) at second degree is the  $9 \times 9$  operator  $\delta_{pr}\delta_{ij} - M_{in,pk}\delta c_{nkrj}$ , built from  $M$  alone. Under  $O_h$  the nine gradient components decompose as  $A_{1g} \oplus E_g \oplus T_{1g} \oplus T_{2g}$ , and the

operator is diagonal in that basis. Its eigenvalues are the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (29)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (30)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (31)$$

The density channel is the displacement block of Eq. (15) with the second gradients dropped, an amplification  $(1 - \omega^2 \Delta\rho \mathbf{G})^{-1}$  with  $\mathbf{G}$  the cube integral of  $\mathbf{G}$ .

**The cube against the sphere.** For a sphere, Eshelby's tensor gives a dilatational channel  $1 + \Delta K/(\lambda + 2\mu)$ , with  $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$ , and a single deviatoric channel  $1 + 2\Delta\mu S_{\text{sph}}$ , with  $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$ . The cube's dilatational channel, Eq. (29), is the sphere's exactly: its eigenvalue involves  $\mathbf{M}$  only through the contraction  $\int_V \partial_i \partial_n G_{in} dV$ , and the static Navier equation makes  $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$ , whose integral does not depend on the shape. The cube's shear channels differ from each other. Their isotropic average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (32)$$

The  $\sqrt{3}$  terms cancel in the average. What a cube adds to a sphere at this order is therefore a pure cubic anisotropy. It splits the deviatoric channel about the sphere's value, weighted 3 : 2 by the dimensions of  $T_{2g}$  and  $E_g$ , and the diagonal shear is the more depolarised. At the background of §7 ( $\nu = 7/32$ ),  $2\mu S_{\text{off}} = 0.4314$ ,  $2\mu S_{\text{diag}} = 0.5929$  and the sphere's value is 0.4960.

**Where Eshelby enters.** Equations (29)–(31) follow from the moment integrals and the group theory alone. No ellipsoidal result is used, and no  $\delta$  term is added by hand. What corresponds to Eshelby is the *truncation*. Keeping  $\partial u$  and discarding  $\partial\partial u$  is exactly the uniform-internal-strain ansatz, and by the uniformity theorem that ansatz is exact if and only if the inclusion is an ellipsoid. For a sphere the construction is therefore exact in the static limit, and for a cube it is an approximation. The hierarchy (22) states that approximation as the omitted blocks, rather than as an unquantified shape correction. A polynomial expansion about the centre cannot represent the edge and corner concentrations of an isolated cube's internal field at any finite degree; Table 4 measures what each degree leaves. A voxel in a uniform layer has no such concentrations: its neighbours have the same properties, so its internal field is the continuum's, which is smooth. That is why a low-degree internal field suffices in a lattice (§§7–8) although it is only an approximation for an isolated cube.

**Against the exact sphere.** These statements are tested against the exact elastic Mie solution, with no Eshelby formula in between (Appendix F). Applied to a ball, the construction's dilatational and shear channels equal Mie's static concentrations to  $2 \times 10^{-10}$ , and the cube's dilatational

channel and isotropic shear average meet the same numbers; taken as a principal value the moment has no depolarisation at all, so the whole static response is the  $\delta$  term. At finite frequency the closure departs from Mie only by a real term of order  $(ka)^2$  at first order in the contrast, given in closed form in Eq. (55); the radiation damping is exact. That term is the uniform-field ansatz's own error, the order the hierarchy assigns to the omitted second gradients.

#### 6.4 What each block of the hierarchy buys

A hierarchy invites the question of what its next block is worth. The answer has a simple form, and it is not the one a count of unknowns suggests: the blocks help in pairs.

**The experiment.** The layer of §7 at normal incidence is cut into  $n$  cells, and in each the hierarchy of degree  $\ell$  is imposed: the displacement is its Taylor polynomial of degree  $\ell$  about the centre, the strain the derivative of that, and the equation and its first  $\ell$  derivatives hold at the centre, with the  $\delta$  terms kept. In one dimension the moments are elementary and the exact layer is known, so the order of each truncation can be measured for a density contrast, a modulus contrast and both. The schemes are computed symbolically to 60 digits for  $\ell = 1$  to 6 and, to  $\ell = 3$ , independently in double precision.

**The prediction.** A cell radiates the integral of its source against the kernel. If the source is held to a polynomial of degree  $d$  about the centre, the first power it omits is  $d + 1$ . When  $d + 1$  is odd, that term integrates to zero against the even part of the kernel and the error is two orders smaller; when  $d + 1$  is even it does not. The density source is the displacement, of degree  $\ell$ ; the stiffness source is the strain, of degree  $\ell - 1$ . So a density contrast gains an order pair at every even  $\ell$ , and a stiffness contrast at every odd  $\ell$ . This is the parity of §6.3, read as an order of convergence.

**Table 2.** The order of convergence of the hierarchy of degree  $\ell$  on the layer, measured from 16 to 32 cells in reflection and in transmission at  $\omega = 300 \text{ rad s}^{-1}$ , and its reflection error on 16 cells with both contrasts. Every measured order equals the predicted one to the digits shown.

|                       | $\ell = 1$           | 2                    | 3                     | 4                     | 5                     | 6                     |
|-----------------------|----------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| density contrast only | 2                    | 4                    | 4                     | 6                     | 6                     | 8                     |
| modulus contrast only | 2                    | 2                    | 4                     | 4                     | 6                     | 6                     |
| both                  | 2                    | 2                    | 4                     | 4                     | 6                     | 6                     |
| error, both, $n = 16$ | $5.4 \times 10^{-6}$ | $1.4 \times 10^{-6}$ | $5.8 \times 10^{-12}$ | $9.5 \times 10^{-13}$ | $2.6 \times 10^{-18}$ | $3.1 \times 10^{-19}$ |

**The result.** Table 2: all thirty-six orders are as predicted. The second gradients raise the density channel to fourth order and leave the stiffness channel at second, with a constant three to four times smaller in reflection; the third gradients bring the stiffness channel to fourth order; and so on in pairs. With both contrasts, fourth order therefore needs  $\ell = 3$  and sixth order  $\ell = 5$ . The truncation of Eqs. (15)–(18) at second gradients is one block short of the next order for a stiffness contrast, which is why it improves the uniform-strain closure by a constant and not by an order.

**The error in closed form.** At first order in the contrast a cell whose source is held to its Taylor polynomial of degree  $d$  returns the exact term times  $1 + E_d$ , where, with  $h$  the half-width,

$$E_d = \begin{cases} (-1)^{(d-1)/2} \frac{(kh)^{d+1}}{(d+2)!} & d \text{ odd, } R \text{ and } T, \\ (-1)^{d/2} \frac{(kh)^{d+2}}{(d+2)!} & d \text{ even, } R, \\ (-1)^{d/2+1} \frac{(d+1)(kh)^{d+2}}{(d+3)!} & d \text{ even, } T, \end{cases} \quad (33)$$

$R$  and  $T$  standing for reflection and transmission, with  $d = \ell$  for the density term and  $d = \ell - 1$  for the stiffness term. It follows from the series of  $\int_{-1}^1 P_d(ikhx) e^{\mp ikhx} dx$  with  $P_d$  the Taylor polynomial of  $e^{ikhx}$ , and it is confirmed symbolically for  $d = 0$  to 6 and against the scheme on one cell, at a contrast of  $10^{-20}$  and  $kh = 0.03$ , to 0.5% in all twelve cases. For  $d = 0$  it is  $(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission: the constants of the collocation scheme, Eq. (36), which are thus the first rung of a ladder (`ContinuumLimit_GradientHierarchy.wl`, `scripts/measure_layer_taylor_hierarchy.py`).

## 6.5 The single site against a sphere and a cube

The hierarchy is a statement about one scatterer, so it is tested first on one scatterer. The assembly of Eq. (22), with the full tensor moments in their dynamic form (the series of  $e^{ikr}/r$  taken to  $r^{11}$ ), is the same for any centrosymmetric body; only the scalar moments change.

**A sphere.** For a ball the scalar moments are elementary, by the surface route of Eq. (26) on  $|\mathbf{r}| = a$ , and the exact answer is known. A homogeneous sphere of radius  $a = 10$  m with the contrast of §7 is illuminated by a P plane wave; the hierarchy's polynomial source is radiated to nine scattering angles between  $11^\circ$  and  $169^\circ$ , and the error is the largest component of the difference from the exact displacement [62, 63] over the largest exact component. The prediction, from the pairs of §6.4 with  $kh$  replaced by  $ka$ , is an error of  $O((ka)^2)$  for  $\ell = 1$  and 2 and of  $O((ka)^4)$  for  $\ell = 3$ , and  $O((ka)^4)$  already at  $\ell = 2$  for a density contrast alone. Table 3 bears

**Table 3.** One homogeneous sphere by the hierarchy of degree  $\ell$ : relative far-field error against exact scattering, and the order between the two lowest frequencies. All nine orders are as predicted.

| contrast     | $\ell$ (unknowns) | $k_S a = 0.4$        | 0.2                  | 0.1                  | order | predicted |
|--------------|-------------------|----------------------|----------------------|----------------------|-------|-----------|
| density only | 1 (12)            | $6.4 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | $3.9 \times 10^{-4}$ | 2.0   | 2         |
|              | 2 (30)            | $4.8 \times 10^{-5}$ | $2.6 \times 10^{-6}$ | $1.8 \times 10^{-7}$ | 3.9   | 4         |
|              | 3 (60)            | $1.5 \times 10^{-5}$ | $9.5 \times 10^{-7}$ | $6.9 \times 10^{-8}$ | 3.8   | 4         |
| moduli only  | 1                 | $1.9 \times 10^{-2}$ | $5.0 \times 10^{-3}$ | $1.1 \times 10^{-3}$ | 2.3   | 2         |
|              | 2                 | $5.4 \times 10^{-3}$ | $1.4 \times 10^{-3}$ | $3.4 \times 10^{-4}$ | 2.0   | 2         |
|              | 3                 | $6.2 \times 10^{-5}$ | $4.1 \times 10^{-6}$ | $2.0 \times 10^{-7}$ | 4.3   | 4         |
| both         | 1                 | $1.4 \times 10^{-2}$ | $3.1 \times 10^{-3}$ | $7.2 \times 10^{-4}$ | 2.1   | 2         |
|              | 2                 | $3.2 \times 10^{-3}$ | $6.9 \times 10^{-4}$ | $1.8 \times 10^{-4}$ | 1.9   | 2         |
|              | 3                 | $4.6 \times 10^{-5}$ | $2.5 \times 10^{-6}$ | $1.4 \times 10^{-7}$ | 4.2   | 4         |

the prediction out. At  $k_s a = 0.4$  the error of the uniform-strain closure is 1.4%, that of the second-gradient system 0.32%, and that of the third-gradient system 0.0046%: three hundred times below the closure. The systems are well conditioned, the sixty equations of  $\ell = 3$  having a condition number of 5.9 (`scripts/measure_ball_gradient_hierarchy.py`).

**A cube.** The same assembly with the cube’s moments gives the single site of a voxel. No exact solution exists for a cube, so the reference is the same cube, of half-width  $h = 10$  m, solved by the Legendre voxels of §8 on refined grids, a cube being the one body such a grid represents exactly. Solutions with first and second moments on  $2^3$  and  $3^3$  cells differ among themselves by at most  $2 \times 10^{-5}$ , which is the accuracy of the reference. Table 4 shows what

**Table 4.** One homogeneous cube of half-width  $h$  by the hierarchy of degree  $\ell$ : relative far-field error against the converged reference.

| $\ell$ (unknowns) | $k_s h = 0.4$        | 0.2                  | 0.1                  | 0.05                 |
|-------------------|----------------------|----------------------|----------------------|----------------------|
| 1 (12)            | $2.1 \times 10^{-2}$ | $5.4 \times 10^{-3}$ | $2.8 \times 10^{-3}$ | $2.7 \times 10^{-3}$ |
| 2 (30)            | $3.2 \times 10^{-3}$ | $1.6 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $2.4 \times 10^{-3}$ |
| 3 (60)            | $3.5 \times 10^{-4}$ | $3.4 \times 10^{-4}$ | $2.9 \times 10^{-4}$ | $3.1 \times 10^{-4}$ |

distinguishes a cube from a sphere. Each truncation has a floor that does not fall with the frequency, because the internal field of an isolated cube is not a polynomial even statically: it concentrates at the edges and corners, as the static expansions about the centre of Johnson et al. [56] and Wu et al. [39] found for the cuboid and for polyhedra. The floor is  $2.7 \times 10^{-3}$  for the uniform-strain closure and nearly the same with second gradients, as §6.3 requires, the static stress dipole being fixed by the first-gradient block and coupled next to the third. The third-gradient block lowers it ninefold, to  $3 \times 10^{-4}$ . Above the floor the error falls with the frequency as on the sphere: at  $k_s h = 0.4$  the third-gradient system is sixty times more accurate than the closure. The dip of the second-gradient row at  $k_s h = 0.2$  is a cancellation between its static and dynamic terms, which have opposite signs. The condition numbers lie between 11 and 13 (`scripts/measure_cube_gradient_hierarchy.py`). These results come from one implementation, checked against the exact sphere and against the converged cube; the moments it uses are those confirmed by two routes in §6.2.

## 6.6 The collocation closure

The collocation closure takes the internal field of a voxel as uniform, the truncation of §6.3 for the density and stiffness channels together. The voxel’s own field at its centre is then  $\mathbf{S}_{\text{self}} \Delta \mathbf{e}$  for internal state  $\mathbf{e}$ , where  $\mathbf{S}_{\text{self}}$  is the nine-component form of the moments  $\mathbf{G}$  and  $\mathbf{M}$ . By parity it is block-diagonal. So  $\mathbf{e} = \boldsymbol{\psi} + \mathbf{S}_{\text{self}} \Delta \mathbf{e}$  and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (34)$$

With the kernel (14) the self term cancels. Substituting  $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}}\Delta)e$  into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (35)$$

an identity checked on the matrices to  $4 \times 10^{-16}$ . Every block of  $\mathbf{K}_{\text{all}}$  is an *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)), so Eq. (35) is precisely the collocation of the continuum Lippmann–Schwinger equation with a strain piecewise constant over voxels. It does not contain  $\mathbf{S}_{\text{self}}$  at all.

The package’s cube  $T$ -matrix is this closure statically: both blocks agree with Eq. (34) to  $2 \times 10^{-5}$  at  $k_s h = 0.01$ , the difference falling as  $(kh)^2$  (`ContinuumLimit_CubeT.wl`). It departs dynamically by about  $0.25(kh)^2$  through its far-field form-factor corrections. Its self integrals are exact in the static and radiation parts but omit the real even-power terms of the Green’s tensor’s expansion ( $k^2 r, r^3, \dots$ ), an error of  $0.4\text{--}0.6(kh)^2$  against the analytic self term, whose series agrees with the closed-form elastodynamic tensor to  $10^{-14}$ . By Eq. (35) that error cannot enter the closure scheme.

## 6.7 What a closed-form single-site $T$ -matrix gives

The Cartesian multipole hierarchy costs more to set up than the alternatives: moments of the Green’s tensor to fourth derivatives, evaluated as distributions and reduced to closed form. What that buys follows from the Lippmann–Schwinger equation of one body rather than from a model fitted to it. *One derivation.* The static depolarisation, its dynamic corrections, the radiation damping and every higher block come from that one equation and one engine (§6.2), so they cannot be inconsistent with one another, and the static, spherical and low-frequency limits are results of the construction rather than conditions imposed on it. *Nothing is fitted.* Every coefficient of Eq. (22) is a moment of the background Green’s tensor over the body: there is no polarisability to choose and no depolarisation factor to calibrate, where other routes import the single site from the ellipsoid [4, 33] or close the volume integral equation with an effective field [42, 43]. As in the equivalent inclusion method [34, 53], the uniform-strain closure is the first member (§6.3); what the hierarchy adds is the closed form at finite frequency for a body that is not an ellipsoid, with the error of each truncation known. *The singular self-interaction is exact:* the  $\delta$ -function content that a quadrature misses is part of the moments from the start, and the sum rule (27) and the traces (48) hold at every grade (§6.2). *The exact limits are built in:* the cube’s dilatational channel and its isotropic shear average equal the sphere’s, confirmed against the exact Mie solution to  $2 \times 10^{-10}$  and  $10^{-8}$  (§6.3), and at finite frequency the closure departs from Mie only through a real term of order  $(ka)^2$ , the radiation damping agreeing exactly in all three channels (Eq. (55)). *The error is stated before anything is solved:* a truncation is a statement of which blocks are omitted; the orders come in the predicted parity pairs, all thirty-six on the layer and all nine on the sphere (Tables 2 and 3), and the leading error factor is in closed form, Eq. (33). *It improves block by block, on fixed constants:* raising the degree adds moments, confirmed by a second route that shares nothing with the first (§6.2), and no quadrature; against the exact sphere at  $k_s a = 0.4$  the error falls from 1.4% for the uniform-strain closure to 0.0046%

with third gradients, from a system of sixty equations with a condition number of 5.9 (Table 3). That each degree improves on the one below is shown here for one sphere and one cube, not in general: in the Taylor form of the equivalent inclusion method, for interacting inhomogeneities, a higher degree need not improve the solution and the system need not be invertible [54, 64]. *It says what a body's shape does:* at the order of the uniform-strain closure a cube is a sphere plus a pure cubic anisotropy, which splits the deviatoric channel about the sphere's value in the ratio 3 : 2 (Eq. (32)). *It describes a scatterer, not only a cell:* the assembly is the same for every centrosymmetric body, only the scalar moments changing (§6.5), and the same moments are the self block of a voxel scheme (§11).

**What it does not give.** A polynomial about the centre cannot follow the edge and corner concentrations of an isolated cube's internal field, so each truncation there has a floor that does not fall with the frequency ( $3 \times 10^{-4}$  with third gradients; Table 4). And in a lattice the Legendre voxel of §8 is the more economical in unknowns: both are fourth order on a smoothly graded sphere, the hierarchy with sixty unknowns a voxel and the first-moment voxel with thirty-six (§11). The hierarchy's case is not the number of unknowns; it is that it yields the single site itself, in closed form, with its error known.

## 7 Convergence, and the closed-form error

**The experiment.** Every convergence test of this paper is the same experiment, varied in one respect at a time, and it is set out here once.

1. *The medium.* The background has  $\alpha = 5000$  and  $\beta = 3000 \text{ m s}^{-1}$  and  $\rho = 2500 \text{ kg m}^{-3}$ . The layer occupies  $0 < z < D$  with  $D = 2 \text{ m}$ , and differs from the background by  $\Delta\lambda = 2 \text{ GPa}$ ,  $\Delta\mu = 1 \text{ GPa}$  and  $\Delta\rho = 100 \text{ kg m}^{-3}$ : 6.4% in the P modulus, 4.4% in the shear modulus and 4% in the density.
2. *The source.* A plane force at  $z_s = -12 \text{ m}$ , above the layer, radiates a plane wave towards it. The incident field in the layer is then known in closed form, and it is smooth there because the source is outside.
3. *What is measured.* The scattered displacement, total less incident, at two observers outside the layer: at  $z = -1 \text{ m}$  above it, where it is the reflected wave, and at  $z = 4 \text{ m}$  below it, where it is the change the layer makes to the transmitted wave. Outside the layer the scattered field is a single outgoing wave, so one observer on each side holds everything; the two are reported separately, as reflection / transmission.
4. *The exact answer.* The reflection and transmission of the layer from continuity of displacement and traction at its two faces (§3); for a layer whose properties vary with depth, from the differential equation integrated to 30 digits.
5. *The error.*  $e = |u_{\text{scheme}} - u_{\text{exact}}| / |u_{\text{exact}}|$  at each observer: relative to the scattered field, not to the total, so that an error of  $10^{-6}$  is a millionth of what the layer scatters.

6. *The refinement.* The layer is cut into  $n$  planes of voxels of side  $d = D/n$ , with  $n$  from 1 to 32, at  $\omega = 60, 300$  and  $600 \text{ rad s}^{-1}$ , for which  $k_P D = 0.024, 0.12$  and  $0.24$  and  $k_S D = 0.04, 0.2$  and  $0.4$ . Three frequencies show that an order is not an accident of one.
7. *The arithmetic.* The errors of the higher-degree voxels fall below the round-off of double precision, so the symbolic implementation computes them to 40 digits; the independent numerical implementation, in double precision, agrees with it down to its own floor of about  $10^{-13}$ .

The later sections change one item each: the incident wave to SV or SH and its angle (§8), the medium to a stratified or a graded one and the background to a stratified one (§9), and the body to a sphere (§10).

**Second order.** For a layer  $D = 2 \text{ m}$  with the contrast  $\Delta\lambda = 2 \text{ GPa}$ ,  $\Delta\mu = 1 \text{ GPa}$ ,  $\Delta\rho = 100 \text{ kg m}^{-3}$  in a background  $\alpha = 5000$ ,  $\beta = 3000 \text{ m s}^{-1}$ ,  $\rho = 2500 \text{ kg m}^{-3}$ , illuminated by a P plane force, the reflected displacement of the chain converges to the continuous layer's at order 2.000 for  $n = 4 \rightarrow 16$  at  $\omega = 60, 300$  and  $600 \text{ rad s}^{-1}$ , with the collocation (35) and with the package *T-matrix* alike (`ContinuumLimit_Chain.wl`). The error depends on  $k_S h$  alone. Throughout, an order quoted for a pair of grids is the apparent order  $\ln(e_1/e_2)/\ln(n_2/n_1)$  between them, with  $e$  the relative error against the exact solution and  $n$  the refinement; the orders in the legend of Fig. 7 are least-squares slopes of  $\ln e$  against  $\ln n$  over the grids marked as fitted, with their standard errors. The two agree where a sequence is asymptotic and differ where it is not, which the apparent orders show.

**The leading error in closed form.** In Eq. (35) the kernel is integrated exactly; the only approximation is the internal field's variation across a voxel. At first order in the contrast the discrete scattered field is therefore a midpoint rule applied to the continuum's Born integral, with the kernel's cell integral exact. For reflection the integrand carries the two-way phase  $e^{2ikz}$ ; for transmission it is flat. With  $k$  the background wavenumber of the wave concerned ( $k_P$  for an incident P wave,  $k_S$  for an incident S wave),  $R$  the reflection coefficient and  $T^{\text{sc}}$  the scattered part of the transmitted field, for *every*  $n$

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (36)$$

confirmed for  $n = 1$  to 16 to  $10^{-4}$  of the error itself (`ContinuumLimit_ErrorConstant.wl`); for an incident S wave, with  $k = k_S$ , the same forms and those of Eq. (39) hold to  $4 \times 10^{-4}$  (`ContinuumLimit_IncidentS.wl`, §8). At the full contrast the reflection constant is unchanged (0.124 against  $1/8$ ); the transmission constant grows to  $-0.092$ .

**Why a uniform-field single site stops at second order.** The two constants have opposite signs, so no scalar correction of  $\mathbf{T}$ , no single  $O((kh)^2)$  factor, can cancel both. The package *T-matrix* illustrates it: its form-factor corrections reduce the reflection error by a factor 4.45 and *increase* the transmission error by 1.7, at every  $n$ . The missing term in Eq. (36) is the first

moment of the internal field over each voxel, which radiates through the derivative of the kernel. A fourth-order scheme must carry it: a single site with a gradient channel, and a kernel whose cell integrals include the kernel's derivative.

## 8 Fourth order and beyond: Legendre moments of the voxel field

Let each voxel carry the mean *and* the first moment of its internal field. On voxel  $j$ , with local coordinate  $s \in [-h, h]$  and Legendre functions  $\phi_b(s) = P_b(s/h)$  ( $\phi_0 = 1$ ,  $\phi_1 = s/h$ ,  $\phi_2 = (3s^2/h^2 - 1)/2$ , ...), the field is  $w = \sum_b c_{b,j} \phi_b$ ,  $w = (u, \varepsilon)$  the displacement and strain of the P channel, and the polarisation is  $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$  with  $\Delta M = \Delta \lambda + 2\Delta \mu$ . Testing the continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z - z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_p, \quad (37)$$

with the same functions on every voxel gives

$$\left( \int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[ \iint_j \phi_a(z) \mathbf{K}(z - z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (38)$$

$\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$ . Between voxels the double integral is separable, a product of Legendre moments of  $e^{\pm i k z}$ ; within a voxel it is split at  $z' = z$  and the local term contributes  $-\int_i \phi_a \phi_b dz / M_p$ . Every entry is closed-form. Keeping only  $\phi_0$  recovers a second-order scheme; keeping  $\phi_0$  and  $\phi_1$  converges at fourth order in both reflection and transmission (`ContinuumLimit_FourthOrder.wl`; Table 5). A single plane of first-moment voxels represents the 2 m layer 6500 times more accurately

**Table 5.** Relative error of the scattered displacement of the layer, reflection / transmission, at  $\omega = 300 \text{ rad s}^{-1}$ , against the number of voxel planes  $n$ : collocation, the mean-only voxel and the first-moment voxel of Eq. (38). The order is 4.00 at  $\omega = 60$  and  $600 \text{ rad s}^{-1}$  too.

| $n$   | collocation                               | mean only                                 | mean and first moment                       |
|-------|-------------------------------------------|-------------------------------------------|---------------------------------------------|
| 1     | $1.8 \times 10^{-3} / 1.3 \times 10^{-3}$ | $1.2 \times 10^{-3} / 1.8 \times 10^{-3}$ | $2.8 \times 10^{-7} / 2.8 \times 10^{-7}$   |
| 4     | $1.1 \times 10^{-4} / 8.3 \times 10^{-5}$ | $7.5 \times 10^{-5} / 1.1 \times 10^{-4}$ | $1.1 \times 10^{-9} / 1.1 \times 10^{-9}$   |
| 16    | $7.0 \times 10^{-6} / 5.2 \times 10^{-6}$ | $4.7 \times 10^{-6} / 6.9 \times 10^{-6}$ | $4.2 \times 10^{-12} / 4.2 \times 10^{-12}$ |
| order | 2 / 2                                     | 2 / 2                                     | 4 / 4                                       |

than a single plane of uniform ones. The collocation column is the scheme of §6 and reproduces the chain of the full  $9 \times 9$  system to all printed digits.

**Higher moments.** Carrying the Legendre moments to degree  $p$  changes nothing in the construction: the moments and the same-cell double integrals of  $P_b(s/h)$  against exponentials are again closed-form, and for  $p \leq 1$  the general-degree solver reproduces the chain above exactly. The second-moment voxel ( $p = 2$ ) converges at sixth order and the third-moment voxel ( $p = 3$ ) at eighth, in reflection and transmission (Table 6; `ContinuumLimit_SecondMoment.wl`); the

second-moment voxel's order is 6.00 at  $\omega = 60, 300$  and  $600 \text{ rad s}^{-1}$ . The errors fall to  $10^{-22}$ , so they are computed in 40-digit arithmetic.

**Table 6.** Relative error of the scattered displacement of the layer, reflection / transmission, at  $\omega = 300 \text{ rad s}^{-1}$ , for voxels carrying Legendre moments to degree  $p$ . The order is the mean over the halvings  $n = 2 \rightarrow 4 \rightarrow 8$ .

| $n$   | first moment ( $p = 1$ )                    | second moment ( $p = 2$ )                   | third moment ( $p = 3$ )                    |
|-------|---------------------------------------------|---------------------------------------------|---------------------------------------------|
| 1     | $2.8 \times 10^{-7} / 2.8 \times 10^{-7}$   | $2.8 \times 10^{-11} / 4.2 \times 10^{-11}$ | $1.5 \times 10^{-15} / 1.5 \times 10^{-15}$ |
| 2     | $1.7 \times 10^{-8} / 1.7 \times 10^{-8}$   | $4.4 \times 10^{-13} / 6.5 \times 10^{-13}$ | $6.0 \times 10^{-18} / 6.0 \times 10^{-18}$ |
| 4     | $1.1 \times 10^{-9} / 1.1 \times 10^{-9}$   | $6.9 \times 10^{-15} / 1.0 \times 10^{-14}$ | $2.4 \times 10^{-20} / 2.4 \times 10^{-20}$ |
| 8     | $6.7 \times 10^{-11} / 6.7 \times 10^{-11}$ | $1.1 \times 10^{-16} / 1.6 \times 10^{-16}$ | $9.2 \times 10^{-23} / 9.2 \times 10^{-23}$ |
| order | 4 / 4                                       | 6 / 6                                       | 8 / 8                                       |

**The leading error in closed form.** Without scattering, the Galerkin solution is the  $L^2$  projection  $\Pi$  of the incident field onto each voxel's basis, and the kernel is integrated over each voxel exactly; at first order in the contrast the error is therefore the kernel's cell integral applied to  $\Pi w^0 - w^0$ . The field varies as  $e^{iks}$  across a voxel and the kernel toward an outside observer as  $e^{iqs}$ ,  $q = +k$  for reflection and  $q = -k$  for transmission, so for every scheme and every  $n$

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (39)$$

with  $\Pi$  the centre value for collocation, the cell mean for the mean-only voxel and the projection onto  $\{P_0, \dots, P_p\}$  for the voxel of degree  $p$  (`ContinuumLimit_FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results are in Table 7. The chain of Eq. (38) repro-

**Table 7.** The leading error of the discrete layer in closed form, Eq. (39): discrete over exact scattered field at first order in the contrast, for every number of voxel planes  $n$ .

|               | reflection                        | transmission                      |
|---------------|-----------------------------------|-----------------------------------|
| collocation   | $1 + (kd)^2/8 + O(d^4)$           | $1 - (kd)^2/24 + O(d^4)$          |
| mean only     | $1 + (kd)^2/12 + O(d^4)$          | $1 - (kd)^2/12 + O(d^4)$          |
| first moment  | $1 - (kd)^4/720 + O(d^6)$         | $1 - (kd)^4/720 + O(d^6)$         |
| second moment | $1 + (kd)^6/100800 + O(d^8)$      | $1 - (kd)^6/100800 + O(d^8)$      |
| third moment  | $1 - (kd)^8/25401600 + O(d^{10})$ | $1 - (kd)^8/25401600 + O(d^{10})$ |
| degree $p$    | $1 + (-1)^p c_p (kd)^{2p+2}$      | $1 - c_p (kd)^{2p+2}$             |

duces the forms for collocation and degrees 0 and 1 at  $n = 1$  to 8 to  $10^{-4}$  of the error, the remainder being the second order in the contrast, and those for degrees 2 and 3 to  $4 \times 10^{-7}$  at a contrast reduced by  $10^{-6}$ . The constant for degree  $p$  follows from the Legendre expansion  $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$ . The projection drops the terms  $j > p$ , led by  $j = p+1$ , whose coefficient is  $O((kh)^{p+1})$ ; its moment against  $e^{iqs}$  is  $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$ . With

$j_n(x) \approx x^n/(2n+1)!!$  and  $h = d/2$ ,

$$c_p = \frac{2p+3}{4^{p+1}[(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)!(2p+3)!}, \quad c_0 = \frac{1}{12}, \quad c_1 = \frac{1}{720}, \quad c_2 = \frac{1}{100800}, \quad (40)$$

and the leading error carries  $(q/k)^{p+1}$ : the same sign in reflection and transmission for odd  $p$ , opposite signs for even  $p$ . The first-moment voxel's leading error therefore does not depend on the direction in which the field is observed. Eq. (40) is checked symbolically for  $p = 0$  to 5 (every lower power of  $kd$  vanishes) in Mathematica and, independently, in sympy.

**One plane of voxels and the thin-layer series.** Refining at fixed thickness is one limit; thinning a single plane is the other. For one plane of voxels ( $n = 1$ ,  $d = D$ ) the series in  $D$  of the reflection coefficient, computed exactly at the full contrast, coincides with that of the exact layer, Eq. (7), term by term through  $D^{2p+2}$  and first differs at  $D^{2p+3}$ , for  $p = 0$  to 3 and for a layer both stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment.wl`). Each further Legendre moment adds two terms of the thin-layer series.

**In three dimensions.** The three-dimensional first-moment voxel carries the nine-component state with Legendre weights  $\{1, x/h, y/h, z/h\}$ , 36 unknowns in all, tested with the same functions and coupled by cell-to-cell double integrals of the Green's tensor. For a laterally uniform layer at normal incidence it reduces *exactly* to Eq. (38) (`ContinuumLimit_Reduction3D.wl`). The lateral first moments are never excited: the lattice and the incident field are symmetric under reflection about every voxel's centre planes, and  $x/h, y/h$  are odd there. The mean and the z-moment carry weights that are constant laterally, so their lateral form factor  $\text{sinc}(g_x h) \text{sinc}(g_y h)$  vanishes on the reciprocal lattice and the plane sum of the double integrals is the plate-to-plate integral in  $z$  alone. Two facts mark where genuinely three-dimensional structure begins. The lateral first moment's form factor,  $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$ , does not vanish on the reciprocal lattice ( $F_1 = i(-1)^p/p\pi$ ), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift,  $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$ , so the tiling identity holds only to  $O(k_{\parallel} h)$ .

**Other angles, other waves, and degree two.** The orders and their constants hold beyond normal incidence and incident P. At oblique incidence, where the tiling identity holds only to  $O(k_{\parallel} h)$  and the gradient channels couple through the evanescent orders, the mean-only and first-moment voxels remain second and fourth order in the reflected P and the converted S wave; for incident SV and SH waves the orders, and with  $k = k_s$  the constants, are those of incident P; and the voxel of degree two converges at sixth order for P, SV and SH waves, its error on four planes at normal incidence being the  $c_2(kd)^6$  of Eq. (40) to 4%. Appendix B gives the spectral construction, Eq. (49), the tests, and the second implementations that check them; Figure 7(a,b) shows the convergence.

## 9 A heterogeneous layer

A heterogeneous medium is not a layer of identical cells. The simplest heterogeneous case with an exact answer is a stratified model built from layers of constant properties, with real interfaces between them: the form of many layered crustal velocity models. The last test takes such a model. A layer  $D = 4$  m thick is divided into eight model layers of 0.5 m, and in each one  $\Delta\lambda$ ,  $\Delta\mu$  and  $\Delta\rho$  are drawn uniformly within  $\pm 15\%$  of the background (the values are listed in `ContinuumLimit_Heterogeneous.wl`). The result is a random stratification with physical contrasts. The voxels refine the model's layering, with  $m$  planes of voxels in each model layer, so no voxel straddles a model interface: the natural discretisation when the model's interfaces are known. At  $\omega = 1500 \text{ rad s}^{-1}$  the layer is  $k_P D = 1.2$  and  $k_S D = 2.0$  thick, so the field reverberates between the model interfaces. The schemes are those of §§7–8, with the contrast now changing from plane to plane; nothing else in them changes. In the terms of the experiment of §7, the test is this.

1. *The medium.* The eight sets of contrasts are drawn once, from a fixed seed, and the same eight are used in every run, so that every grid and every scheme sees the same model.
2. *Normal incidence.* The source is the plane force at  $z_s = -12$  m, and the scattered displacement is taken at  $z = -1$  m, where it is the reflected wave, and at  $z = D + 2 = 6$  m, where it is the change in the transmitted wave. The P problem has the modulus contrast  $\Delta\lambda + 2\Delta\mu$  and the wave speed  $\alpha$ ; the S problem is the same one-dimensional problem with  $\Delta\mu$  and  $\beta$ .
3. *Oblique incidence.* A P plane wave is incident at  $20^\circ$  to the vertical. What is measured is the pair of displacement amplitude vectors of the reflected P and the reflected S plane waves,  $R_{PP}$  and  $R_{PS}$ , in the specular order. The lateral sums of the coupling (§8) run over the reciprocal-lattice orders  $|p|, |q| \leq 8$ .
4. *The exact answer.* The state of displacement and traction, two components at normal incidence and four for P–SV at  $20^\circ$ , is carried across each model layer by the matrix exponential of that layer's system matrix times its thickness, the eight matrices are multiplied in order, and the radiation conditions of the two half-spaces are imposed on the product. The arithmetic carries 30 to 40 digits. The reference shares nothing with the voxel scheme but the model. It is checked against a third algorithm, the reflection–transmission recursion of Kennett [65] over the nine interfaces: the same-type coefficients,  $R_{PP}$  and  $R_{SS}$  at normal incidence and  $R_{PP}$  at  $20^\circ$ , agree to  $10^{-14}$  (`scripts/gate_heterogeneous_reference_vs_kennett.py`). Only same-type coefficients are compared, because they do not depend on how the up- and downgoing waves are normalised; a converted coefficient does. One sign differs, by convention: at normal incidence the one-dimensional model's  $R_{PP}$  is a ratio of vertical displacements, the recursion's a ratio of amplitudes along each wave's own direction, and the reflected P wave travels upwards.
5. *The error.* As in §7, the difference from the exact scattered quantity relative to its size: for the amplitude vectors at  $20^\circ$ ,  $|\mathbf{R}_{\text{scheme}} - \mathbf{R}_{\text{exact}}|/|\mathbf{R}_{\text{exact}}|$ .

6. *The refinement.* Each model layer is cut into  $m$  planes of voxels, so that the layer holds  $n = 8m$ :  $m = 1, 2, 4, 8, 16$  at normal incidence and  $m = 1, 2, 4$  at  $20^\circ$ , where the lateral sums make each run dearer. The orders are apparent orders, from  $m = 4$  to 16 and from  $m = 2$  to 4.

The errors are in Table 8 and Fig. 7(c).

**Table 8.** Relative error against the exact stratified layer, for  $m$  planes of voxels per model layer (§9). At normal incidence the order is measured from  $m = 4$  to 16, where it is also 2.00 for collocation and 4.00 in transmission; at  $20^\circ$  it is measured from  $m = 2$  to 4.

|                          |               | $m = 1$              | $m = 2$              | $m = 4$              | order |
|--------------------------|---------------|----------------------|----------------------|----------------------|-------|
| P, normal, R             | mean only     | $1.9 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $1.2 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $8.5 \times 10^{-7}$ | $5.3 \times 10^{-8}$ | $3.3 \times 10^{-9}$ | 4.00  |
| S, normal, R             | mean only     | $5.3 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $3.3 \times 10^{-4}$ | 2.00  |
|                          | first moment  | $6.2 \times 10^{-6}$ | $3.9 \times 10^{-7}$ | $2.4 \times 10^{-8}$ | 4.00  |
| P, $20^\circ$ , $R_{PP}$ | mean only     | $1.4 \times 10^{-3}$ | $3.6 \times 10^{-4}$ | $9.0 \times 10^{-5}$ | 2.00  |
|                          | first moments | $3.3 \times 10^{-7}$ | $2.1 \times 10^{-8}$ | $1.3 \times 10^{-9}$ | 4.00  |
| P, $20^\circ$ , $R_{PS}$ | mean only     | $2.8 \times 10^{-3}$ | $6.9 \times 10^{-4}$ | $1.7 \times 10^{-4}$ | 2.00  |
|                          | first moments | $1.4 \times 10^{-6}$ | $8.8 \times 10^{-8}$ | $5.5 \times 10^{-9}$ | 4.00  |

**The interfaces cost nothing.** Every order is the one measured for the uniform layer. At a model interface the strain of the true field jumps. A voxel basis that is discontinuous from voxel to voxel represents that jump exactly when the interface lies on a voxel boundary, so the interface adds no error of its own. The only requirement is that voxel boundaries coincide with the model's interfaces. A voxel that straddled an interface would carry a contrast that is not uniform within it, a case not examined here.

**What fourth order buys.** For a reflection error of  $10^{-6}$  at normal incidence (P), the mean-only voxel needs 347 planes of voxels, about 43 per model layer. That figure is not a run: it is the second-order law carried from the finest grid computed, 128 planes, whose error is  $7.3 \times 10^{-6}$ , to the target,  $n = 128 (7.3 \times 10^{-6} / 10^{-6})^{1/2}$ . The first-moment voxel needs 8, one per model layer, and that figure is a run: the model's own layering already gives  $8.5 \times 10^{-7}$ . Counting unknowns (9 per voxel against 36), that is 3123 against 288 per vertical column of voxels, a factor of eleven. In three dimensions, with the lateral grid refined as the vertical one is, the number of voxels falls by  $(347/8)^3 \approx 8 \times 10^4$  and the number of unknowns by about  $2 \times 10^4$ . The same model has been solved independently in Python, with a contrast for each plane: the raw reflected displacements agree with the Mathematica scheme to  $1.1 \times 10^{-15}$  at normal incidence and at  $20^\circ$ , with and without the first moments. Lateral heterogeneity needs the full three-dimensional solver; it is the first extension discussed below.

**A medium that varies within a cell.** The stratified model is exact on its own grid: every cell is homogeneous. A smoothly varying medium is not, and sampling it as cells of constant properties is an approximation of its own. Let the contrast of cell  $j$  be its Legendre projection

to degree  $r$ ,  $\Delta(z) = \Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$ , with the voxel's field of degree  $p$  as in §8. The product of contrast and field is again a Legendre sum,  $P_c P_b = \sum_e A_{cbe} P_e$ , with  $A_{cbe} = (2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$  the square of a Wigner  $3j$  symbol, so the coupling of Eq. (38) becomes

$$\iint_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[ \iint_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : (41)$$

the same kernel moments, now to degree  $p+r$ , and nothing else new. Against the exact graded layer, a high-precision solution of  $(M(z)u)' + \omega^2 \rho(z)u = 0$ , the order of convergence is

$$\min(2p+2, 2r+2), \quad (42)$$

or  $2p+2$  when degree  $r$  represents the profile exactly (Table 9; `ContinuumLimit_GradedContrast.wl`). A contrast that is constant in each cell therefore caps every voxel at second order, however rich its field basis, and for the smooth profile the voxel of degree 2 with a linear contrast has the error of the voxel of degree 1 to within 0.06%. At first order in the contrast this is what one expects: the contrast's projection error is orthogonal to the polynomials of degree  $r$ , so against the smooth product of the incident and adjoint fields it contributes at  $O(h^{2r+2})$ . To keep order  $2p+2$  the contrast must be represented to the degree of the field. The closed forms of the construction are tested

**Table 9.** A contrast that varies within the cells: the field of each voxel to Legendre degree  $p$ , its contrast to degree  $r$ . Relative reflection error at  $n = 16$  planes and the order ( $n = 4 \rightarrow 8 \rightarrow 16$ ),  $\omega = 300 \text{ rad s}^{-1}$ , for the profiles  $1 + (2z/D - 1)/2$  (linear) and  $1 + \frac{1}{2} \sin(2\pi z/D)$  (smooth) times the contrast of §7. Predicted: Eq. (42), and  $2p+2$  for the linear profile once  $r \geq 1$ .

| $p$ | $r$ | linear profile        |       | smooth profile        |       |
|-----|-----|-----------------------|-------|-----------------------|-------|
|     |     | error                 | order | error                 | order |
| 0   | 0   | $7.7 \times 10^{-5}$  | 2.00  |                       |       |
| 1   | 0   | $7.6 \times 10^{-5}$  | 2.00  | $5.5 \times 10^{-5}$  | 1.95  |
| 1   | 1   | $6.4 \times 10^{-11}$ | 4.00  | $1.4 \times 10^{-7}$  | 3.95  |
| 2   | 1   | $2.3 \times 10^{-17}$ | 6.00  | $1.4 \times 10^{-7}$  | 3.95  |
| 2   | 2   |                       |       | $1.6 \times 10^{-10}$ | 5.98  |

one by one by an independent implementation (`scripts/crosscheck_graded_contrast.py`): the Legendre moments of the kernel against  $2hi^n j_n(ch)$  and 30-digit quadrature (to  $2 \times 10^{-16}$ ), the linearisation coefficients against the  $3j$  symbols (exactly), and the same-cell double integrals against 30-digit quadrature ( $1.4 \times 10^{-16}$ ). It then assembles the scheme without linearisation, evaluating the projected contrast at the quadrature nodes, against its own 30-digit solution of the graded layer; its errors agree with those of Table 9 to  $10^{-3}$ , and at  $\omega = 1200 \text{ rad s}^{-1}$  it measures the orders of Eq. (42) again.

**Which basis sets the error.** Equation (42) gives the order. The size of the error, and which of the two bases it belongs to, appear when the scattered field is separated by its order in the contrast: the Born term  $T_1$ , and the second-order term  $T_2$  in which the field scattered once is scattered again.

Appendix G gives both terms for the exact layer in closed form (Eqs. (57)–(62)) and for the

scheme, where the first needs no solve (Eq. (63)), and shows how both are read from solutions at four scaled contrasts (Eq. (64)). The terms must be separated: at the full contrast the terms of third and higher order are 7% of the second and carry errors of their own.

With  $f(z)$  the profile and  $\Pi_q$  the  $L^2$  projection onto the Legendre polynomials of degree at most  $q$  in each cell, define the *relative projection error* of the medium at degree  $q$ ,

$$\mathcal{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 dz}{\int f^2 dz} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \quad \|f\|^2 = \int f^2 dz : \quad (43)$$

the fraction of the squared norm of the profile that polynomials of degree  $q$  on the cells cannot represent. It is a property of the medium and the grid alone; it involves neither the wave nor the solution. For the smooth profile and every one of the nine pairs with  $p, r \leq 2$ , the relative error of  $T_2$  is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathcal{E}_{\min(p,r)} \quad (44)$$

to within 1.3% at  $\omega = 300 \text{ rad s}^{-1}$  on 4 to 16 cells: the ratio of the two sides is 0.998 to 1.005 when  $\min(p, r)$  is 0 or 1, and 1.010 to 1.013 when it is 2. The error of the Born term is smaller by two to six orders of magnitude. The error of the whole is therefore that of  $T_2$ , the projection error times  $|T_2|/|T_1|$  (0.044 here): for  $p = r = 1$  on 16 cells that predicts  $1.6 \times 10^{-7}$ , and Table 9 has  $1.4 \times 10^{-7}$ .

Equation (44) is exact in the long-wave limit (Appendix G). There the strain scattered once is the profile itself, a cell holds it as  $\Pi_{\min(p,r)} f$ , and the second scattering takes place in the cell of the first: the law is a statement about the single site, whose  $T$ -matrix is exact for the cell as represented and whose error is that the cell is not the medium. Measured, the part of  $T_2$  between different cells carries at most 0.002 of the error, and the departure from the law falls as  $(kD)^2$ .

Three consequences follow. The two bases must be matched: a contrast of higher degree than the field is wasted (the pairs  $(p, r) = (0, 1)$  and  $(0, 2)$  have the errors of  $(0, 0)$  to three digits), and a field of higher degree than the contrast improves only the Born term. The wavefield basis is there to carry what the medium imprints on the field, not only the wave: at these frequencies the wave itself varies little across a cell. And the error is known from the medium, before anything is solved:  $\mathcal{E}_q$  is a property of the profile and the cells. It vanishes when the medium is piecewise constant on the cells, which is the stratified model above; there the contrast needs no more than a constant in each cell, and the degree of the field alone sets the order.

**A stratified background.** Everything above takes the background to be a whole space. The method is meant for a background that is plane stratified, with the voxels carrying the departure from it, and the results must hold there. They do, in the one case tested: normal incidence, P, with the background two half-spaces joined at  $z = z_I$  below the layer, the lower one faster and denser (displacement reflection coefficient  $-0.19$ ). The voxels then interact through the Green's function of that background,

$$g(z, z') = \frac{i}{2M_p k} \left[ e^{ik|z-z'|} + \rho_I e^{ik(2z_I - z - z')} \right], \quad z, z' < z_I, \quad (45)$$

with  $\rho_I$  the reflection coefficient of the interface: the whole-space term and a reverberation that is smooth and carries no local term. The incident field is the field of the source in the background, and the exact answer is the reflection of the whole stack less that of the background. With the interface 3 m below the layer, and again with the lowest voxel against it, the scheme converges at order 2.00, 4.00 and 6.0 for  $p = 0, 1$  and 2 on the uniform layer, and at 1.98, 3.98 and 5.98 on the smooth profile with  $r = p$ . The first-order term is predicted to round-off ( $5 \times 10^{-16}$ ): it is the sum over four paths, the incident wave travelling down or up and the scattered wave leaving up or down, of the exact first-order term of each path times the factor  $\sum_{a \leq p} (2a + 1) j_a(k_{\text{in}} h) j_a(k_{\text{out}} h) / j_0((k_{\text{in}} - k_{\text{out}}) h)$  for that path's pair of wavenumbers, whose leading terms are the constants of Eq. (40). And the relative error of  $T_2$  over  $\mathcal{E}_p$  is 1.00 to 1.02 at  $kD = 0.12$  on 4 to 16 cells for all three degrees, departing from one by  $1.1 \times 10^{-4}$  and  $3 \times 10^{-5}$  at  $kD = 0.012$  for the two positions of the interface: Eq. (44) holds as in the whole space (`scripts/measure_layer_stratified_background.py`; the exact stack is computed in 30-digit arithmetic). The reason is the one given for Eq. (65): the reverberation adds no local term, so the second scattering in the cell of the first is unchanged. This check is by one implementation against an exact solution.

**Oblique incidence, and a three-layer background.** The same holds at oblique incidence, for P, SV and SH waves, in the two half-spaces and in a three-layer background in which the layer of voxels reverberates between two interfaces (Appendix C). At  $20^\circ$  the orders are 2, 4 and 6 for  $p = 0, 1$  and 2 in all thirty-six cases (Table 15), and the errors are close to those of the whole space: the background changes the field the voxels must represent, not how well they represent it. Both terms of the error remain known in advance. The scheme's first-order term equals a closed-form prediction, built from the thirty-six pairs of incident and radiated plane waves (Eq. (52)), to  $2 \times 10^{-15}$ ; and the law of Eq. (44) holds to 3% for  $p \leq 1$  and 6% for  $p = 2$  at  $k_S D = 0.2$ , and to 0.6% at  $k_S D = 0.067$ , in every background and for every wave type (Table 16).

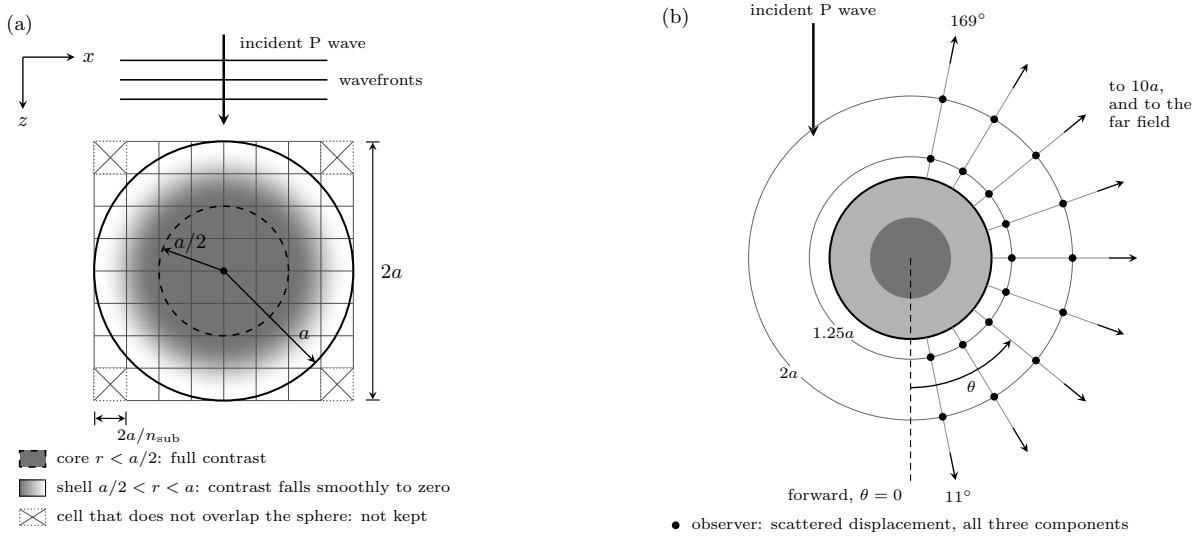
## 10 A sphere: what the layer predicts in three dimensions

The layer and the stratified layer are laterally uniform. A sphere is not, and elastic scattering by a sphere, homogeneous [62, 63] or radially varying [66], has an exact solution: it is the usual benchmark for volume-integral and discrete-dipole methods. This section solves two spheres with the full three-dimensional voxel scheme: the sphere is voxelised on an  $n_{\text{sub}}^3$  grid, the Foldy–Lax system is solved with the cell-averaged coupling by an FFT-accelerated iterative solver, and a P wave is incident. For the graded sphere the voxel is the mean-only voxel of §8 in three dimensions: each cell carries the mean of its nine-component state, tested with the same constant, and the  $L^2$  projection of its contrast, which is its cell mean; every cell that overlaps the sphere is kept, and the cells are coupled by cell-to-cell double integrals of the Green's tensor, one block per orbit of the cube group (Appendix E). For the sharp sphere the voxel is the package's cube  $T$ -matrix, which differs from the collocation closure only through its far-field corrections (§6.6) and has the same order, each cell keeping the contrast of its centre. The far field is computed from the solution for the incident plane wave itself, each voxel's displacement and strain carrying

the plane-wave phase at its centre; adding to that phase a linear variation about the scatterer's centre, as a Taylor expansion of the incident field would, counts the wave's variation twice and leaves an error of order  $(ka)^2$  that no refinement removes. In the terms of the experiment of §7, the test is this.

1. *The body.* A sphere of radius  $a = 10$  m (Fig. 6) in the background of §7, carrying the contrast of that section, at  $\omega = 150$  and  $300 \text{ rad s}^{-1}$ , for which  $k_S a = 0.5$  and  $1$  and  $k_P a = 0.3$  and  $0.6$ .
2. *The incident wave.* A P plane wave travelling along  $+z$ .
3. *The grid.* The cube of side  $2a$  that circumscribes the sphere is cut into  $n_{\text{sub}}^3$  cells of side  $2a/n_{\text{sub}}$ ; which of them are kept is stated with each voxel above.
4. *The solve.* The iteration is carried to a relative residual of  $10^{-11}$ , far below every discretisation error reported, so that the errors are the scheme's and not the solver's.
5. *The exact answer.* The partial-wave series of the sphere, summed to order 10 at  $k_S a = 0.5$  and 11 at  $k_S a = 1$ , and evaluated as the exact scattered displacement at the observers, not as its asymptotic form.
6. *What is measured.* The scattered displacement vector at nine scattering angles, equally spaced from  $11^\circ$  to  $169^\circ$  in the  $z$ - $x$  plane: forward and backward scattering and everything between, P and S together.
7. *The error.* The largest component of the difference between the scheme's far field and the exact displacement over the nine angles, divided by the largest component of the exact one. It is a maximum, not an average, so that no angle is hidden.
8. *The distance.* The solution of the scheme is the field inside the voxels, and the scattered field outside follows from it at any distance. For the far-field test it is read with the asymptotic formula, which keeps the term in  $1/r$  alone: a choice of this test, not a property of the scheme. The reference is the exact field, so the two differ by the near-field terms, of relative size  $1/(kr)$ , however good the scheme. The graded sphere is therefore observed at  $5 \times 10^8$  radii, where those terms are  $10^{-8}$  of the peak: at  $5 \times 10^4$  radii they are  $1.4 \times 10^{-4}$ , which is 4% of the discretisation error at 12 voxels across and most of it at 24, and would be read as a failure to converge. The sharp sphere, whose errors are a hundred times larger, is observed at  $5 \times 10^4$  radii. The near field is tested separately, with a readout that drops nothing (Table 17).

**A sphere without a staircase.** A voxelised sphere with a sharp surface changes its shape with every grid. A sphere whose contrast falls smoothly to zero at its surface does not: the voxels that cross the surface carry a vanishing contrast. The body here has a homogeneous core  $r < a/2$  carrying the contrast of §7, and across the shell  $a/2 < r < a$  the contrast falls to zero as the smoothstep  $s(x) = 10x^3 - 15x^4 + 6x^5$ ,  $x = (a - r)/(a - a/2)$ , which vanishes like  $(a - r)^3$  at the surface. Its exact scattering follows, order by order, from the radial equations



**Figure 6.** The sphere experiment, in the section through the centre that contains the direction of incidence. (a) The body and the grid. The contrast is full in the core  $r < a/2$  and falls smoothly to zero across the shell  $a/2 < r < a$ . The circumscribing cube of side  $2a$  is cut into  $n_{\text{sub}}^3$  cells, here  $n_{\text{sub}} = 8$ ; every cell that overlaps the sphere is kept and carries the projection of the contrast it contains. A P plane wave is incident along  $+z$ . (b) The observers. The scattered displacement is taken at nine scattering angles  $\theta$ , measured from the forward direction and equally spaced from  $11^\circ$  to  $169^\circ$ , on spheres of several radii about the centre, and compared with the exact displacement at the same points.

*Alt text:* Two panels. On the left, a disc on a square grid of eight by eight cells: the inner half of the disc is uniformly dark, and the outer half fades from dark to white at the rim, which is drawn as a solid circle; the four corner cells of the grid, which the disc does not reach, are crossed out. Horizontal lines above the grid with a downward arrow mark the incident wave. On the right, the same disc drawn smaller, with two concentric circles around it at one and a quarter and at two radii; nine dots on each circle lie on nine rays from the disc, fanning from just off the downward, forward direction to just off the upward, backward one, and each ray ends in an outward arrow.

of Takeuchi and Saito [66], integrated across the shell from the core's regular solutions and matched to outgoing waves at  $r = a$ . The equations are derived symbolically from the equations of motion, and the solution is checked against a sphere of homogeneous concentric shells in closed form, by unitarity and reciprocity, and by an independent implementation that agrees to  $2 \times 10^{-11}$  (TakeuchiSaito.wl, ContinuumLimit\_GradedSphere.wl, scripts/crosscheck\_graded\_sphere.py). Each voxel carries the mean of the contrast over its cell. Table 18 shows the error falling at second order from  $n_{\text{sub}} = 6$  to 24 at both frequencies (scripts/pilot\_graded\_voxel\_sphere.py): the apparent order between successive grids is 2.02 to 2.06 at  $k_s a = 0.5$  and 2.00 at  $k_s a = 1$ , and the least-squares orders of Fig. 7(d) are  $2.045 \pm 0.004$  and  $1.994 \pm 0.004$ . With 24 voxels across the error is  $1.8 \times 10^{-4}$  and  $8.3 \times 10^{-4}$ . At  $n_{\text{sub}} = 4$  a voxel is as wide as the graded shell and the order is not yet asymptotic. This is the order the layer predicts for the mean-only voxel, and the first test of that prediction on a body with genuine three-dimensional structure.

What makes the sequence this clean is that the voxel carries a single error. The collocation voxel of the sharp sphere applied to this body, each cell at the profile's value at its centre and only the cells whose centres lie inside the sphere kept, carries two: its own internal-field error and the error of sampling the profile, both  $O(h^2)$ . Its error falls from  $1.0 \times 10^{-1}$  at  $n_{\text{sub}} = 4$  to

$4.7 \times 10^{-3}$ ,  $2.6 \times 10^{-3}$ ,  $6.4 \times 10^{-4}$  and  $2.3 \times 10^{-4}$  at 6, 8, 12 and 16, apparent orders 7.6, 2.0, 3.5 and 3.6 at  $k_s a = 0.5$ , and at  $k_s a = 1$  the apparent orders are 1.7, 2.5 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's, but its apparent order is not that of any single term and cannot be read from them. The same collocation voxel with the cell mean of the contrast, and every overlapping cell kept, gives  $3.4 \times 10^{-3}$ ,  $1.9 \times 10^{-3}$  and  $8.2 \times 10^{-4}$  at 6, 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the second term, and with it removed the collocation and mean-only voxels differ only by their constants, as in Table 7. The cell mean, with every overlapping cell kept, represents the same body on every grid, and the order the layer predicts is then the one observed.

The first-moment voxel of §8, extended to three dimensions with a contrast linear in each cell, converges at fourth order on this same sphere (Appendix E): its apparent orders are 4.01 and 4.13 where the discretisation error is isolated at weak contrast, and at the full contrast its error with 16 voxels across is  $4.9 \times 10^{-6}$  and  $5.9 \times 10^{-6}$ , 82 and 317 times below the mean-only voxel's on the same grid.

**The near field, an impedance march, and a sharp sphere.** Three further tests are in Appendix D. Read at finite distance, down to a quarter of a radius from the surface, the graded sphere converges at the orders of the far field, second for the mean-only voxel and 3.8 to 3.95 for the first-moment voxel, while the asymptotic readout would be in error there by 48% to 85% (Table 17). Against the impedance march of Haines et al. [67], which needs far fewer unknowns, the first-moment voxel reaches the march's accuracy with six voxels across and continues at close to fourth order below it, where the march is limited by the scattering between the spheres of the periodic array it must solve (Table 19). And a sphere with a sharp surface does not converge: its error tracks the volume of its staircase, which changes with the grid (Table 20); the smooth sphere, like the layer, measures the scheme.

## 11 The Cartesian multipole hierarchy as a voxel scheme

The Legendre voxels of §8 represent the field of a voxel by its moments over the voxel. The hierarchy of §6 represents it by its gradients at the centre. Both are polynomial, and §6.4 shows that on the layer the hierarchy of degree  $\ell = 3$  is fourth order, as the first-moment voxel is. This section builds the hierarchy into a voxel scheme in three dimensions and compares the two.

**The scheme.** Each voxel  $c$ , centred at  $\mathbf{x}_c$ , carries the coefficients  $U_{j;W}^c$  of Eq. (20). The equation and its derivatives to order  $\ell$  are imposed at every centre (with  $\ell = 1$  and no density contrast this is the static numerical equivalent inclusion method on cuboidal elements, with uniform eigenstrains, equivalence at the element centres and a fast Fourier transform, of Zhou et al. [68]):

$$U_{i;p}^c - \sum_{c'} \left\{ \omega^2 \Delta \rho \sum_W \frac{K_{ij}[P;W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} U_{j;W}^{c'} + \sum_W \frac{K_{in}[Pk;W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} \delta c_{nkrj} U_{j;rW}^{c'} \right\} = \partial_p u_i^0(\mathbf{x}_c), \quad (46)$$

$$K_{in}[D;W](\Delta) = \int_V (\partial_D G_{in})(\Delta - \xi) \xi_W d\xi. \quad (47)$$

The two kinds of block are of different character. For  $c' = c$  the integrand is singular and the integral is a distribution: it is the moment of the hierarchy,  $K_{in}[D; W](0) = (-1)^{|D|} g_{in}[D; W]$ , in closed form on the cube. For  $c' \neq c$  the field point is the centre of another voxel, at least half a voxel from the source voxel, so the integrand is smooth and a tensor Gauss rule integrates it; the derivatives of the Green's tensor, to any order, come from the family  $x^a y^b z^c r^p e^{ikr}$ , which is closed under differentiation. All the singular work of the scheme is thus in the self block, where it is done once and in closed form; the Legendre voxel's self block is a double integral of the same singular kernel. The blocks depend on the offset alone, so the product of the system with a vector is a convolution over the lattice and is done exactly by the fast Fourier transform. Where the contrast varies within a voxel it is projected there on polynomials of degree  $r$ , as in §9, and the source is the product of that polynomial and the field's: the moments then carry weights to degree  $\ell + r$ , and a block is a sum, over the monomials of the contrast, of blocks with the weight shifted.

**Checks of the construction.** The derivatives of the Green's tensor to fourth order agree with the closed forms of its value, gradient and second derivative, and with central differences beyond them, to  $8 \times 10^{-7}$ . A lattice of one voxel reproduces the single site of §6.5 to  $10^{-14}$ . Only the offsets of one octant are computed, the others following from the parity of each index; a block obtained that way equals one computed directly to  $10^{-15}$ . The iterative solution equals a dense solution to  $7 \times 10^{-11}$  where both can be run.

**A body with corners.** The cube of §6.5, at  $k_s h = 0.2$ , is cut into  $n^3$  voxels. Table 10 gives the far-field error against the converged reference, with the Legendre voxels on the same body. On this body every degree of the hierarchy converges at about the same rate, an order near two

**Table 10.** A homogeneous cube cut into  $n^3$  voxels,  $k_s h = 0.2$ : relative far-field error of the hierarchy of degree  $\ell$  and of the Legendre voxels, with the unknowns of each voxel.

|                        | unknowns a voxel | $n = 1$              | 2                    | 3                    | 4                    |
|------------------------|------------------|----------------------|----------------------|----------------------|----------------------|
| hierarchy, $\ell = 1$  | 12               | $5.4 \times 10^{-3}$ | $1.7 \times 10^{-3}$ | $8.3 \times 10^{-4}$ | $5.0 \times 10^{-4}$ |
| hierarchy, $\ell = 2$  | 30               | $1.6 \times 10^{-3}$ | $6.7 \times 10^{-4}$ | $3.7 \times 10^{-4}$ | $2.4 \times 10^{-4}$ |
| hierarchy, $\ell = 3$  | 60               | $3.4 \times 10^{-4}$ | $7.0 \times 10^{-5}$ | $3.0 \times 10^{-5}$ | $1.7 \times 10^{-5}$ |
| Legendre, mean only    | 9                | $5.4 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $5.8 \times 10^{-4}$ | $3.3 \times 10^{-4}$ |
| Legendre, first moment | 36               | $4.8 \times 10^{-5}$ | $6.0 \times 10^{-6}$ |                      |                      |

between the two finest grids, and the degree buys a constant: thirty times from  $\ell = 1$  to  $\ell = 3$  on 64 voxels. That rate is the body's and not the scheme's, for the mean-only Legendre voxel shows it too: a homogeneous cube has edges and corners, whose fields no polynomial follows. The uniform-strain closure and the mean-only voxel agree on one cell, as they must, being the same approximation. And the first-moment Legendre voxel is far ahead: one cell of 36 unknowns is more accurate than eight voxels of the hierarchy of degree three with 480. A projection, which the Legendre voxel is, follows a field that is not polynomial better than an expansion about a point: the advantage of a Galerkin over a Taylor discretisation of the Lippmann–Schwinger equation that Brisard et al. [54] established for the static equivalent inclusion method.

**A smooth medium.** The comparison that bears on voxel schemes is in a medium without corners. First in one dimension: the layer of §6.4 with a contrast that varies in depth, projected in each cell to the degree of the field, against the exact graded layer. With the profile  $\sin^2(\pi z/D)$  the orders from 16 to 32 cells are 1.99, 1.98 and 3.99 for  $\ell = 1, 2$  and 3: a varying contrast leaves the pairs of §6.4 as they are. With the smoothstep of §10 as the profile they are 1.97, 1.99 and 3.84, and on coarser grids they are erratic, the third-gradient scheme being worse than the second on four cells. The orders appear only when about four cells cross the graded part of the profile (`scripts/measure_layer_taylor_hierarchy_graded.py`). That fixes the conditions of the three-dimensional test. The graded sphere of §10 has a shell 5 m wide, and a solver that reaches fourteen voxels across puts fewer than four voxels across it; so the core is reduced here from  $a/2$  to  $a/10$ , which widens the shell to 9 m, and a second profile,  $\sin^2(\pi x/2)$  in the fraction  $x$  of the shell's width measured from the surface, is run beside the smoothstep. The exact reference is the radial solution of §10 for each body. The contrast is linear in each voxel for  $\ell = 3$ , which suffices for fourth order (§9), and of degree  $\ell$  for  $\ell = 1$  and 2.

**Table 11.** The graded sphere with a shell 9 m wide,  $k_s a = 0.5$ : relative far-field error against the exact solution for  $n_{\text{sub}}$  voxels across, with the apparent order between the two finest grids of each row. The lower block is the profile  $\sin^2$ .

|                        | $n_{\text{sub}} = 4$ | 6                    | 8                    | 10                   | 12                   | 14                   | order |
|------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|-------|
| hierarchy, $\ell = 1$  | $1.5 \times 10^{-3}$ | $8.4 \times 10^{-4}$ | $4.9 \times 10^{-4}$ | $3.2 \times 10^{-4}$ |                      |                      | 1.94  |
| hierarchy, $\ell = 2$  | $3.3 \times 10^{-4}$ | $2.1 \times 10^{-4}$ | $1.5 \times 10^{-4}$ | $1.0 \times 10^{-4}$ |                      |                      | 1.56  |
| hierarchy, $\ell = 3$  | $2.6 \times 10^{-4}$ | $5.0 \times 10^{-5}$ | $1.7 \times 10^{-5}$ | $7.2 \times 10^{-6}$ | $3.6 \times 10^{-6}$ | $2.0 \times 10^{-6}$ | 3.80  |
| Legendre, mean only    | $7.3 \times 10^{-3}$ | $3.2 \times 10^{-3}$ | $1.8 \times 10^{-3}$ | $1.1 \times 10^{-3}$ |                      |                      | 2.02  |
| Legendre, first moment | $2.9 \times 10^{-4}$ | $6.4 \times 10^{-5}$ | $2.2 \times 10^{-5}$ | $9.4 \times 10^{-6}$ | $4.6 \times 10^{-6}$ | $2.5 \times 10^{-6}$ | 3.92  |
| hierarchy, $\ell = 3$  | $1.9 \times 10^{-4}$ | $3.8 \times 10^{-5}$ | $1.3 \times 10^{-5}$ | $5.2 \times 10^{-6}$ | $2.7 \times 10^{-6}$ | $1.5 \times 10^{-6}$ | 3.84  |
| Legendre, first moment | $2.2 \times 10^{-4}$ | $4.8 \times 10^{-5}$ | $1.6 \times 10^{-5}$ | $6.8 \times 10^{-6}$ | $3.4 \times 10^{-6}$ | $1.8 \times 10^{-6}$ | 4.02  |

Table 11 gives the result. The first-gradient scheme converges at second order, and is three to four times more accurate than the mean-only Legendre voxel on every grid, for twelve unknowns a voxel against nine. The second-gradient scheme is more accurate again but gains no order, as §6.4 predicts; its apparent order is still rising towards two. The third-gradient scheme converges at an apparent order between 3.7 and 4.1 from grid to grid, 3.8 over the range from six to fourteen voxels across, for both profiles; the first-moment Legendre voxel does the same, between 3.7 and 4.0 and 3.85 over the range. Neither has settled on four at fourteen voxels across, and the second profile, gentler than the first, changes neither: the two fourth-order schemes behave alike. On every grid from six voxels across the hierarchy's error is the smaller, by a fifth to a quarter. It carries sixty unknowns a voxel against thirty-six, so that for a given number of unknowns the Legendre voxel remains somewhat the more accurate; the wide margin of Table 10 was the corners.

**What the hierarchy is for.** The two schemes reach the same order by different closures of the same equation, and they differ in what they cost and in what they give besides. The Legendre voxel is the more economical in unknowns. The hierarchy's blocks are simpler: its self block is known in closed form on four constants (§6.2), and its coupling is a single integral of a smooth

function where the Legendre voxel's is a double integral. The iteration converged in five steps at every size; the time goes into the tables of blocks, which are computed once for a grid and a frequency. Formed on the same footing, one block for each orbit of offsets under the 48 symmetries of the cube, the others following exactly, and every point value of the Green's tensor or its derivatives from the same compiled radial ladder, the hierarchy's tables are the cheaper: 0.6, 1.9 and 4.5 s for 6, 10 and 14 voxels across, against 4.0, 9.3 and 23.3 s for the first-moment Legendre voxel, with eight times fewer kernel evaluations. Both have a one-off cost besides, the hierarchy's the smaller (0.3 s against 2.2 s): the moments of its self block depend on neither the frequency nor the cell size, and are computed once and stored (`scripts/build_cube_unit_moments.py`; `scripts/measure_hierarchy_table_cost.py`; one machine, five threads). And the hierarchy is, in addition, a single-site  $T$ -matrix in closed form, whose accuracy improves block by block (§6.5), which no voxel defined by moments over the cell provides. The lattice results of this section come from one implementation, checked against exact solutions and, where it can be, against a dense solve; they are at one frequency and on one family of bodies (`scripts/gradient_voxel_lattice.py`, `scripts/measure_lattice_gradient_hierarchy.py`, `scripts/measure_graded_sphere_gradient_hierarchy.py`, `scripts/measure_graded_sphere_gradient_hierarchy_fft.py`).

## 12 Discussion

Figure 7 collects the convergence of every test, with the order fitted to each by least squares (`scripts/plot_convergence_orders.py`).

**What each basis is for.** A voxel holds two functions, and the results give each basis its part. The basis of the wavefield sets how well the voxel carries the field across itself: in a uniform medium the scheme converges at order  $2p + 2$  with the constant  $c_p$  of Eq. (40), for P and S waves and at normal and oblique incidence alike. The basis of the medium sets how well the voxel *is* the medium: with degree  $r$  the order is capped at  $2r + 2$  (Eq. (42)). Neither can make up for the other, and the second-order term shows why. The field scattered once is a copy of the medium's profile, so the wavefield basis must carry the profile as well as the wave, and the error of that term is the relative projection error of the profile at the lower of the two degrees (Eq. (44)). At the frequencies of interest the wave varies little across a cell and the medium may vary much, so it is the medium, not the wavelength, that decides the degree and the size of the cells. Two cases bracket the choice. Where the medium is piecewise constant on the cells, as in a layered model with voxel faces on its interfaces, a constant is exact for the medium, the projection error vanishes, and the degree of the field is spent entirely on the wave. Where the medium varies smoothly within the cells, the two degrees should be equal: a higher degree for the contrast than for the field is wasted, and a higher degree for the field improves only the Born term. That the two degrees can be chosen independently is the practice of higher-order electromagnetic volume methods [1]; what the present results add is the rule for choosing them, and the error each choice leaves.

**Where the error of a voxel scheme lives.** The results separate the two approximations a voxel scheme makes. The *coupling* between voxels makes none: averaged over the source cell it is the continuum's cell integral exactly, between planes by the form factor's zeros and within a plane by the tiling identity, and the self term it omits is exactly the one the single site carries. Every error therefore comes from the *single site*: from the internal field it assumes and, where the medium varies within the cell, from the medium it is given. Its  $T$ -matrix sums the rescattering within the voxel exactly, but for the voxel as represented, and the split of the second-order term in §9 measures the consequence: the error sits in the part whose two scatterings lie in one cell, and the part between cells carries at most 0.2% of it. For a uniform field that error is the midpoint rule's, known in closed form; for a linear field it is the next term of the same expansion. The converse is the practical warning. A kernel that is not the cell integral (the point Green's tensor, or a cell average that does not match the single site's closure) leaves a local term that does not vanish as the voxels shrink (§4): the evanescent lattice orders carry a static contribution of fixed size per voxel volume. A scheme built that way converges to the wrong limit, and refinement cannot reveal it, because the error is scale-invariant.

**What the single site is.** The Cartesian multipole hierarchy of §6 turns the uniform-field single site from a modelling choice into a derived object with a stated error (§6.7). Its one departure from exact scattering by a sphere, the real  $(ka)^2$  term of Eq. (55), and the collocation error of Eq. (36) have the same origin, the variation of the field across the scatterer, and the same remedy, a richer internal field. In a lattice the remedy is cheap: the field a voxel must represent is the smooth continuum field, not the field of an isolated cube, with its concentrations at the edges and corners.

**Relation to other methods.** The partition of the error is the coupled-dipole method's choice of polarisability seen from the other side: for space-filling cubes the tiling identity fixes the local field exactly, and the polarisability is neither needed nor a free parameter. The voxel of degree  $p$  is close kin to the spectral element of seismology; the two share a Galerkin condition on polynomial bases, and differ as a differential formulation from an integral one, and so suit different problems. Appendix H sets out both relations, and where each method is the better choice.

**The tension between the two error sources.** The coupled-dipole method has two sources of error, the single site and the coupling between sites, and each has been corrected on its own: the polarisability by the Clausius–Mossotti relation, the lattice dispersion relation or the exact static limit, and the coupling by integrating or filtering the Green tensor. The two are not independent. The point kernel's evanescent orders carry a static local term whose size per voxel volume does not shrink with the voxels (§4), and the single site must cancel exactly that term. A polarisability tuned for one kernel is therefore wrong for another: the lattice dispersion relation corrects the point-dipole lattice, and changing the kernel to an integrated or a filtered one changes the term it must cancel. Tuning either alone moves the error between the two, and refinement cannot reveal a mismatch, because the surviving term is scale-invariant. For

space-filling cubes the tension is resolved rather than balanced: the cell-averaged kernel omits exactly the self term that the closure carries (Eq. (35)), so the coupling error vanishes identically and every remaining error is the single site's, the variation of the field across the cell. That error has a second tension of its own. It is a midpoint-rule error whose constants have opposite signs in reflection and transmission (Eq. (36)), so any scalar adjustment of the single site that improves one worsens the other; the package's far-field corrections reduce the reflection error by a factor 4.45 and increase the transmission error by 1.7. No polarisability, however chosen, can remove it. What removes it is a richer internal field: the first moment lowers the error by two orders in  $kd$  in both observables at once.

**The cost of higher order, and a fast solver.** A first-moment voxel carries four times as many unknowns as a uniform one, but by the closed-form constants it reaches a reflection error of  $10^{-6}$  with voxels 58 times larger in each direction; the stratified model of §9 measures it directly, 347 planes of uniform-field voxels against 8 first-moment voxels. And because the discretisation error is known in advance and the coupling between cells is exact, a fast product need be no more accurate than that error, and can be budgeted separately from it. Appendix I sets out the cost, and what the error analysis offers a fast solver.

**Limitations and outlook.** The results concern laterally uniform media, a homogeneous layer and a randomly stratified one, for which the discrete system is exactly a chain of planes; a laterally heterogeneous medium needs the first-moment voxel in a full three-dimensional solver, where the gradient channels couple through the evanescent lattice orders (§8) and the tiling identity holds only to  $O(k_{\parallel}h)$ . The smoothly graded sphere of §10 shows the mean-only voxel converging at second order on a laterally heterogeneous body against an exact solution, and the first-moment voxel converges at fourth order there (§11). A contrast that varies within a cell, which §9 shows is needed beyond second order in a smoothly varying medium, has a further consequence in three dimensions: its linear part is odd about the cell's centre, so it couples the single site's uniform-field and first-moment blocks, which parity keeps apart for a homogeneous cube (§6.3). A graded voxel therefore needs at least the first-moment block; that voxel is built, and tested on the graded sphere, in the companion paper. The numerical evidence is at moderate, physically typical contrasts (up to 15% in density and moduli) and at  $k_{\parallel}h \leq 0.3$ , the range in which the single-site  $T$ -matrix is validated; the closed forms are first order in the contrast, and the departures from them at the full contrast are of second order. What is established is the *order* of convergence and its leading constant, not the behaviour at strong contrast, where the single site's own closure may limit the accuracy before the lattice coupling does. The single site's dynamic error is in closed form for a sphere only; for a cube, for which no exact solution exists, it is known through the layer (§§7–8) and, for the isolated cube, by comparison with a converged solution (§6.5). The hierarchy beyond the first gradient is established less widely than the rest: its orders and its error factor on the layer are derived symbolically and checked independently to the third degree, and its moments are confirmed by two routes, but its three-dimensional results, the single site against the sphere and the cube and the voxel scheme of §11, come from one implementation, at one frequency for the voxel scheme, checked against exact solutions. There the fourth order is approached and not attained on the grids reached (3.8 at fourteen

voxels across, as for the first-moment voxel), the fifth-gradient system, whose moments are in hand, has not been assembled in three dimensions, and the two fourth-order schemes have been compared in the cost of forming their tables only (§11). Two extensions follow directly. The first is the three-dimensional solver itself. The second is a stratified *background*, a layered reference medium, as distinct from the stratified contrast of §9. For a laterally uniform layer of voxels the orders, the first-order factor and the projection-error law all survive it, at normal and at oblique incidence, for P, SV and SH waves, in a background of two half-spaces and in one of three layers (§9). What remains is a body of finite lateral extent in such a background, where the image of a voxel in an interface it touches is singular.

## 13 Conclusions

A voxel of a volume-integral scheme must represent the medium within it and the wavefield within it. This paper has determined what each basis must be and what error a given choice leaves, by an experiment in which nothing else is approximated: a layer of cubes is the layer itself, so its exact solution isolates the discretisation error.

With the wavefield expanded to Legendre degree  $p$  and the contrast to degree  $r$  in each voxel, the scheme converges at order  $\min(2p+2, 2r+2)$ , with the leading error  $c_p(kd)^{2p+2}$  in closed form (Eq. (40)) for P, SV and SH waves at normal and oblique incidence, in a whole space and in a plane-stratified background. A contrast held constant in each cell caps every voxel at second order. The error of the term of second order in the contrast is the relative projection error of the medium at degree  $\min(p, r)$ : it is the single site's, whose  $T$ -matrix sums the rescattering within a voxel exactly for the voxel as represented, so the two bases must be matched, and where the medium is piecewise constant on the cells it vanishes. A sphere whose contrast falls smoothly to zero at its surface converges at the orders the layer predicts.

The single site is derived rather than assumed. Three properties of the cube make the experiment exact: the source-cell-averaged coupling is the continuum's cell integral exactly, the moments of the Green's tensor over the cube give the single site with no polarisability to choose, and its self term cancels the kernel's, so the simplest scheme is a collocation of the continuum equation. The uniform-strain closure is the first member of a Cartesian multipole hierarchy, the elastodynamic extension of the Taylor form of the equivalent inclusion method [53] to bodies that are not ellipsoids, whose moments are in closed form on four constants. Its blocks gain accuracy in pairs, with the error of each truncation in closed form: as a single site the third-gradient system is three hundred times more accurate than the uniform-strain closure against the exact sphere, and as a voxel scheme it is fourth order, as the first-moment voxel is, on a smoothly graded sphere, where in a lattice the Legendre voxel is the more economical. The results and the evidence for each are collected in Appendix J.

**The error is known in advance.** For every discretisation studied here the error is not something to be discovered by refining until the answer stops changing. Its order and its leading constant are known before anything is computed:  $+(kd)^2/8$  in reflection and  $-(kd)^2/24$  in transmission for the collocation voxel (Eq. (36)),  $(kd)^2/12$ ,  $(kd)^4/720$  and  $(kd)^6/100800$  for the

voxels of degree 0, 1 and 2 (Eq. (40)), and the order  $\min(2p + 2, 2r + 2)$  where the medium varies within a cell (Eq. (42)), with the size of the error there given by the relative projection error of the medium (Eq. (44)), which is computed from the model and the grid alone. That is worth more than the order alone, for three reasons. First, the cell size and the degree of the voxel that a required accuracy needs follow from the wavenumber and the medium, before the computation and without a refinement study. Secondly, it turns a refinement study from a hope into a test. A scheme of this kind can converge smoothly to the wrong limit (§4), and smooth convergence alone cannot reveal it; but the difference between two grids must have the known order and the known constant, and a computation whose differences do not is in error from some cause other than the discretisation. Thirdly, it shows where effort pays. The error is that of representing the field and the medium within a cell, so raising the degree of the cell does what multiplying the cells cannot: in the stratified model one first-moment voxel per layer reaches  $10^{-6}$  where the uniform-field voxel needs about forty. Whether the error of a body with three-dimensional structure can be known in advance in the same way is the question taken up in a following paper [69], where it decides where the cells of an adaptive grid are placed. That paper continues the paper's first strand, the two bases and their error, with the Legendre voxels. The second strand, the Cartesian multipole hierarchy, whose coefficients are integrals of the Green's tensor over the cube, is taken further in a paper on the coupling integrals of polynomial cubes in closed form, of which it is the worked application.

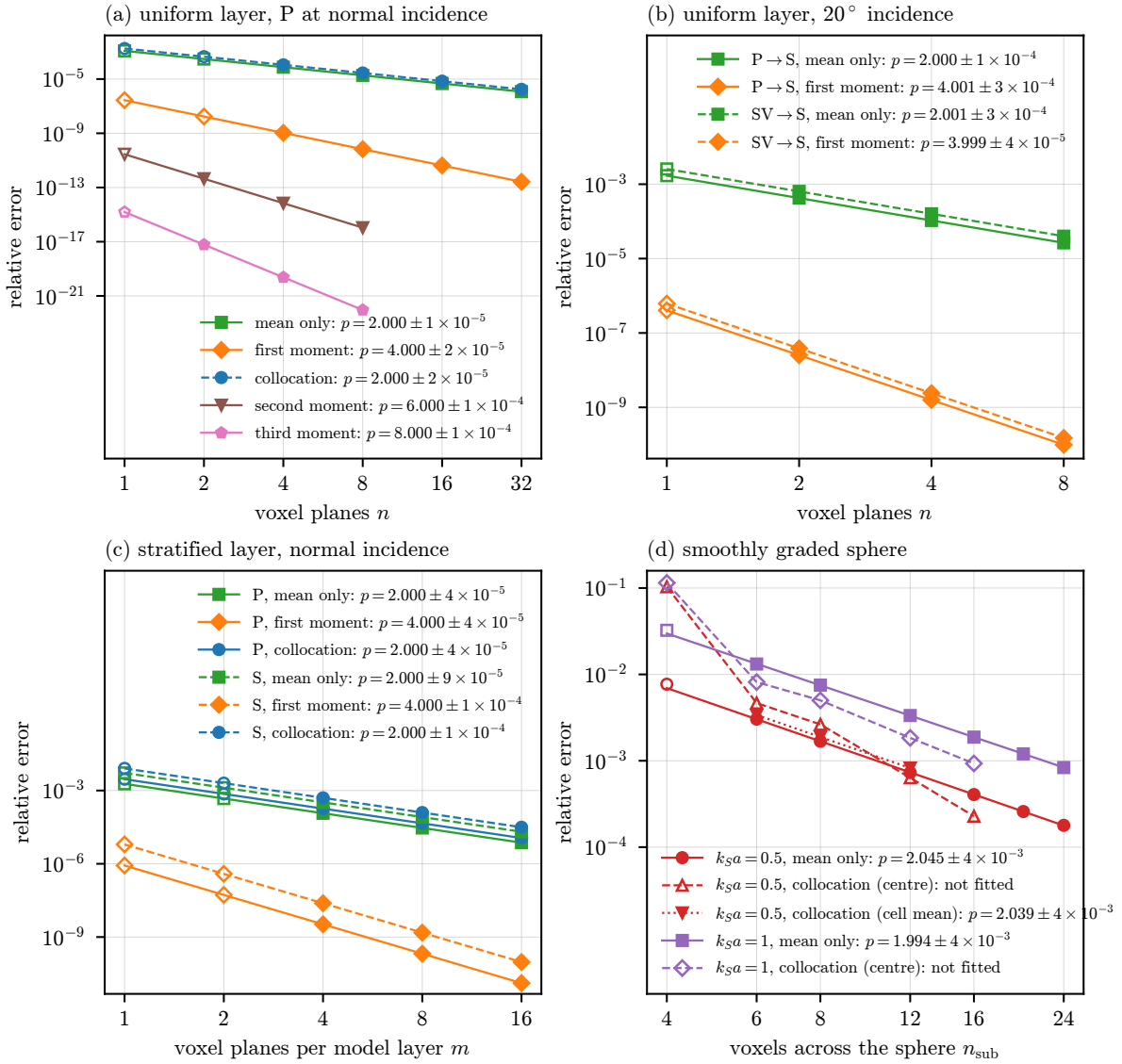
## A The moment engine: gates and a second route

Two checks, whose answers do not depend on the engine, gate every grade. Because  $\nabla^2 r^m = m(m+1)r^{m-2}$  and  $\nabla^2(1/r) = -4\pi\delta$ ,

$$\sum_p E[m; ppR; W] = m(m+1) E[m-2; R; W], \quad \sum_p E[-1; ppR; W] = -4\pi(-1)^{|R|} (\partial_R x_W)(0), \quad (48)$$

for any further derivatives  $R$  and any weight  $W$ . The second is the sum rule (27) at every grade: it tests the  $\delta$  function of the inclusion problem and its derivatives, which is where a quadrature fails. Both hold for every grade of Table 1 with two or more derivatives, 910 traces of each kind; the grades with fewer, for which the integrand is integrable, agree with numerical integration over the cube.

The closed forms are also confirmed by a route that shares nothing with the faces of the cube. The cube is its inscribed ball plus the remainder. On the ball the derivatives are moved onto the sphere, where the surface integrals of monomials are Gamma functions, and all the distributional content is there; on the remainder the integrand is an ordinary function, integrated radially in closed form and over directions through the six faces by Gauss–Legendre quadrature. This evaluation reproduces all 4224 stored values, the largest difference being  $1.1 \times 10^{-12}$ , and satisfies Eq. (48) by itself to  $5 \times 10^{-10}$  or better up to eight derivatives (CubeMomentCore.wl, CubeMomentHigherGrades.wl, CubeMomentStore.wl, CubeMomentCanonical.wl; scripts/crosscheck\_cube\_moments\_ball\_shell.py).



**Figure 7.** Convergence of every test, with least-squares regression lines. Each line fits  $\log e = c - p \log n$  to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order  $p$  and its standard error. The marker, and in colour the hue, marks the scheme, so that the figure reads in grey scale: collocation (circles; §6), the mean-only voxel (squares) and the voxels carrying Legendre moments to degree  $p = 1$  (diamonds), 2 (triangles) and 3 (pentagons) (Eqs. (38) and (49)). (a) The uniform layer, P reflection at normal incidence ( $D = 2$  m,  $\omega = 300$  rad s $^{-1}$ , the contrast of §7), fitted from  $n = 4$ ; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor 3/2 of Table 7; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from  $n = 2$ . (b) The uniform layer at  $20^\circ$ , fitted from  $n = 2$ : the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number  $m$  of voxel planes per model layer, fitted from  $m = 4$ . (d) The smoothly graded sphere of §10 against its exact solution, for  $n_{\text{sub}}$  voxels across: the mean-only voxel, fitted from  $n_{\text{sub}} = 6$  (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and not fitted, since its two  $O(h^2)$  errors of opposite sign leave no single order to fit; and, at  $k_{\text{sa}} = 0.5$ , the same collocation voxel with the cell mean of the contrast, dotted, which recovers second order.

*Alt text:* Four log-log panels of relative error against refinement, each showing data points and fitted straight lines with the fitted order in the legend. (a) Uniform layer at normal incidence, 1 to 32 voxel planes: the collocation and mean-only lines have order 2.000, the first-moment line order 4.000, falling from about  $3 \times 10^{-7}$  to  $2 \times 10^{-13}$ ; the second-moment line has order 6.000 and the third-moment line order 8.000, over 1 to 8 planes, reaching about  $10^{-22}$ . (b) Uniform layer at 20 degrees, 1 to 8 planes: P-to-S and SV-to-S lines, order 2.00 for the mean-only voxel and 4.00 for the first-moment voxel. (c) Stratified layer, 1 to 16 planes per model layer: P and S reflection, order 2.000 for collocation and mean-only, 4.000 for first moment. (d) Smoothly graded sphere, 4 to 24 voxels across: two straight falling lines of order 2 for the mean-only voxel, for  $k_{\text{sa}} = 0.5$  and 1, from about  $10^{-2}$  and  $3 \times 10^{-2}$  at 4 voxels to  $2 \times 10^{-4}$  and  $8 \times 10^{-4}$  at 24; two dashed collocation curves starting near  $10^{-1}$  at 4 voxels, dropping steeply to 6 and then falling faster than the mean-only lines, crossing below them by 16 voxels; and a dotted line of order 2 for the cell-mean collocation voxel, close to the mean-only line.

## B Further tests of the Legendre voxel

**Oblique incidence.** The scheme is built spectrally. By Poisson summation the Bloch coupling between planes  $i$  and  $j$  at lateral wavenumber  $\mathbf{k}_{\parallel}$  is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z - z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (49)$$

with  $L_a$  the lateral Legendre factors. The whole-space tensor decays as  $1/k_z^2$  and the nine-component state adds at most two powers of  $k_z$ , so each entry of  $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$  is a  $\delta(z)$  of constant weight plus, per mode,  $[i\alpha/2\gamma + i\beta \operatorname{sgn}(z)/2] e^{i\gamma|z|}$ ,  $\gamma^2 = k_W^2 - \kappa^2$ : every  $z$  integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at  $|p|, |q| \leq 8$ . Against the exact layer, for a P wave incident at  $20^\circ$  and  $40^\circ$  (`ContinuumLimit_Oblique.wl`; Table 12). The mean-only voxel is second order and the first-moment voxel fourth order in both

**Table 12.** Relative error of the reflected P ( $R_{pp}$ ) and converted S ( $R_{ps}$ ) waves for a P wave incident at  $20^\circ$ , with the mean-only and first-moment voxels of Eq. (49). At  $40^\circ$  the orders are again 2 and 4.

| $n$   | mean only            |                      | mean and first moments |                       |
|-------|----------------------|----------------------|------------------------|-----------------------|
|       | $R_{pp}$             | $R_{ps}$             | $R_{pp}$               | $R_{ps}$              |
| 1     | $9.3 \times 10^{-4}$ | $1.7 \times 10^{-3}$ | $7.2 \times 10^{-8}$   | $4.1 \times 10^{-7}$  |
| 2     | $2.3 \times 10^{-4}$ | $4.2 \times 10^{-4}$ | $4.5 \times 10^{-9}$   | $2.5 \times 10^{-8}$  |
| 8     | $1.5 \times 10^{-5}$ | $2.7 \times 10^{-5}$ | $1.7 \times 10^{-11}$  | $9.9 \times 10^{-11}$ |
| order | 2                    | 2                    | 4                      | 4                     |

the reflected P and the converted S wave: the breakdown of the tiling identity at  $O(k_{\parallel}h)$  and the evanescent coupling of the gradient channels do not cost an order. At  $\mathbf{k}_{\parallel} = 0$  the construction reproduces Eq. (38) to all printed digits.

**Incident S.** Nothing in the construction is specific to P. For a normally incident S wave the layer is the one-dimensional problem of Eq. (38) with  $M_p \rightarrow \mu$ ,  $k_p \rightarrow k_s$  and  $\Delta M \rightarrow \Delta\mu$ ; the collocation and mean-only schemes converge at order 2.00 and the first-moment voxel at 4.00, in reflection and transmission, and the full  $9 \times 9$  construction with an incident SH wave reproduces that one-dimensional result to all printed digits. At oblique incidence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders are those of Table 13. The orders, and with  $k = k_s$

**Table 13.** Observed order of convergence ( $n = 2 \rightarrow 8$ ,  $\omega = 300 \text{ rad s}^{-1}$ ) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle  $\arcsin(\beta/\alpha) = 36.9^\circ$ .

|                                            | mean only   | mean and first moments |
|--------------------------------------------|-------------|------------------------|
| SV at $20^\circ$ : $R_{SS} / R_{SP}$       | 2.00 / 2.00 | 4.00 / 3.99            |
| SV at $30^\circ$ : $R_{SS} / R_{SP}$       | 1.98 / 2.00 | 3.97 / 4.00            |
| SH at $20^\circ$ and $40^\circ$ : $R_{SH}$ | 2.00        | 4.00                   |

the closed-form constants of Eqs. (36) and (39), are those of the incident P wave.

**Degree two at oblique incidence.** In three dimensions the voxel of degree 2 carries, on each of the nine components of its state, the ten Legendre products of total degree at most two in  $(z, x, y)$ . In the construction of Eq. (49) it converges at sixth order against the exact layer for incident P, SV and SH waves (Table 14; `scripts/measure_oblique_degree.py`). The test is run at  $\omega = 3000 \text{ rad s}^{-1}$  ( $k_p D = 1.2$ ,  $k_s D = 2$ ), ten times the frequency of the other tables, because at  $300 \text{ rad s}^{-1}$  a sixth-order error on these grids lies below the round-off of double precision. The exact reflection coefficient is that of Kennett's recursion, an independent algorithm, and the same computation with the bases of degree 0 and 1 returns orders 2 and 4 in all five cases. At normal incidence the error on four planes,  $6.95 \times 10^{-9}$ , is the  $c_2(kd)^6 = 7.2 \times 10^{-9}$  of Eq. (40) to 4%. Doubling the number of lattice orders retained in Eq. (49) changes the errors by less than 1%. Unlike the degrees 0 and 1, this degree is implemented once at oblique incidence; its check is the exact solution.

**Table 14.** The voxel of degree 2 (ten Legendre products per state component) at  $\omega = 3000 \text{ rad s}^{-1}$ : relative error of the same-type reflection coefficient against the exact layer on  $n$  planes, and the order over the last halving, with the orders of the degrees 0 and 1 from the same computation.

| incident wave               | error, $p = 2$       |                      |                      | order<br>$p = 2$ | orders<br>$p = 0 / p = 1$ |
|-----------------------------|----------------------|----------------------|----------------------|------------------|---------------------------|
|                             | $n = 1$              | $n = 2$              | $n = 4$              |                  |                           |
| P, normal: $R_{pp}$         | $3.4 \times 10^{-5}$ | $4.6 \times 10^{-7}$ | $7.0 \times 10^{-9}$ | 6.05             | 2.04 / 4.05               |
| P at $20^\circ$ : $R_{pp}$  | $7.3 \times 10^{-6}$ | $1.0 \times 10^{-7}$ | $1.5 \times 10^{-9}$ | 6.04             | 2.04 / 4.06               |
| SV at $20^\circ$ : $R_{ss}$ | $2.2 \times 10^{-4}$ | $2.3 \times 10^{-6}$ | $3.4 \times 10^{-8}$ | 6.11             | 2.10 / 4.14               |
| SH at $20^\circ$ : $R_{sh}$ | $2.3 \times 10^{-4}$ | $2.5 \times 10^{-6}$ | $3.6 \times 10^{-8}$ | 6.11             | 2.10 / 4.14               |
| SH at $40^\circ$ : $R_{sh}$ | $3.0 \times 10^{-4}$ | $3.6 \times 10^{-6}$ | $5.2 \times 10^{-8}$ | 6.09             | 2.12 / 4.05               |

The spectral construction is implemented twice. The second implementation (`scripts/crosscheck_first_moment_voxel.py`) shares only the definition of the scheme: it takes the depth dependence of the transformed kernel from an independent residue construction, obtains the local weight as a numerical large- $k_z$  limit of the inverted Christoffel matrix, evaluates every double integral by quadrature and separates P from S by sampling the upgoing field at two depths. The raw reflected displacements of the two agree to  $9 \times 10^{-16}$  in all 36 cases compared: incident P, SV and SH, normal incidence and  $20^\circ$ ,  $n = 1, 2, 4$ , with and without the first moments. The one-dimensional scheme of any degree is also implemented twice. The second implementation (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral by Gauss quadrature, split at the receiver within a voxel, and the exact layer from its own interface system, in double precision. Its errors agree with the 40-digit ones within its round-off floor ( $3 \times 10^{-14}$  in reflection,  $1.6 \times 10^{-13}$  in transmission); at  $\omega = 1200 \text{ rad s}^{-1}$  it measures orders 4.02 / 3.99 and 6.02 / 6.00; for one plane as the layer thins its differences from the exact layer fall as  $D^{3.00}$ ,  $D^{5.00}$  and  $D^{7.00}$  for  $p = 0, 1$  and 2; and it rederives  $c_p$  for  $p \leq 5$ . Figure 7(a,b) shows the convergence, with the orders fitted to it.

## C The stratified-background construction

The experiment is built in four steps.

*The kernel.* At each lateral wavenumber  $\kappa$  the whole-space kernel between two depths factorises over the plane-wave modes, three going down and three going up (P, SV, SH):  $\hat{\mathbf{G}}(\kappa; z - z') = \mathbf{D}_\downarrow \mathbf{E}(z - z') \mathbf{S}_\downarrow$  for  $z > z'$ , with  $\mathbf{S}_\downarrow$  the  $3 \times 9$  matrix of mode amplitudes radiated downwards by a nine-component source,  $\mathbf{E}(L) = \text{diag}(e^{i\gamma_m L})$  the propagation over a distance  $L$  with  $\gamma_m$  the vertical wavenumber of mode  $m$ , and  $\mathbf{D}_\downarrow$  the  $9 \times 3$  matrix of the modes' states; and likewise upwards. With the interfaces at  $z_A$  above and  $z_B$  below,  $H = z_B - z_A$ , and  $\mathbf{R}_A, \mathbf{R}_B$  the  $3 \times 3$  reflection matrices of the two interfaces seen from inside the layer, a source at  $z'$  sends  $\mathbf{s}_\downarrow = \mathbf{S}_\downarrow \mathbf{q}$  down and  $\mathbf{s}_\uparrow = \mathbf{S}_\uparrow \mathbf{q}$  up, and the reverberation it sets up is

$$\begin{aligned} \mathbf{d}_A &= (\mathbf{I} - \mathbf{R}_A \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(H))^{-1} \mathbf{R}_A [\mathbf{E}(z' - z_A) \mathbf{s}_\uparrow + \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(z_B - z') \mathbf{s}_\downarrow], \\ \mathbf{u}_B &= \mathbf{R}_B [\mathbf{E}(z_B - z') \mathbf{s}_\downarrow + \mathbf{E}(H) \mathbf{d}_A], \\ \hat{\mathbf{G}}_{\text{rev}}(\kappa; z, z') \mathbf{q} &= \mathbf{D}_\downarrow \mathbf{E}(z - z_A) \mathbf{d}_A + \mathbf{D}_\uparrow \mathbf{E}(z_B - z) \mathbf{u}_B, \end{aligned} \quad (50)$$

with  $\mathbf{d}_A$  the downgoing amplitudes leaving the upper interface and  $\mathbf{u}_B$  the upgoing ones leaving the lower. It is added to the whole-space kernel in every block of Eq. (49), at every order of the reciprocal lattice. It is smooth and separable in  $z$  and  $z'$ , so its integrals over the cells are one-dimensional moments of exponentials. For two half-spaces  $\mathbf{R}_A = \mathbf{0}$  and only the second line survives.

*The interfaces.*  $\mathbf{R}_A, \mathbf{R}_B$  and the transmission matrices into and out of the layer follow from continuity of displacement and traction at each interface: six equations in the states of the six modes on either side, solved for each lateral wavenumber, evanescent orders included.

*The incident field and the output.* A plane wave of unit amplitude arrives from the upper half-space. Its field in the layer, a downgoing part and an upgoing part with the reverberation summed as in Eq. (50), is the right-hand side of the scheme. The output is the upgoing wave of the same type as the incident one, arriving at  $z_A$  from the voxels and transmitted into the upper half-space: the change that the layer of voxels makes to the same-type reflection coefficient ( $R_{PP}, R_{SS}$  or  $R_{SH}$ ), a quantity that does not depend on how the waves are normalised.

*The reference, and a check of the construction.* The exact change is the same-type reflection coefficient of the stack with the layer, less that of the background, both from Kennett's recursion [65], an independent algorithm. Before any voxel is introduced, the background's own reflection coefficient computed from the modes and the interface matrices above agrees with Kennett's to  $5 \times 10^{-15}$  in all twelve cases.

It is computed by two independent algorithms. The first is invariant imbedding, the impedance march of Haines et al. [67]: the six-component state of displacement and traction obeys  $\text{d}\mathbf{y}/\text{d}z = \mathbf{A}(z) \mathbf{y}$ , and with the four  $3 \times 3$  blocks  $\mathbf{A}_{ij}$  of  $\mathbf{A}$  the impedance  $\mathbf{Y}$ , which gives the traction from the displacement, obeys the Riccati equation

$$\frac{\text{d}\mathbf{Y}}{\text{d}z} = \mathbf{A}_{21} + \mathbf{A}_{22} \mathbf{Y} - \mathbf{Y} \mathbf{A}_{11} - \mathbf{Y} \mathbf{A}_{12} \mathbf{Y}. \quad (51)$$

It is started below the stack with the impedance of the downgoing waves of the lower half-space, which is the radiation condition, marched upwards, exactly through the homogeneous gaps and by fourth-order Runge–Kutta with the medium sampled at the stages through the graded

layer, and at the top the reflection matrix follows from continuity of displacement and traction with the modes of the upper half-space.  $\mathbf{Y}$  stays bounded, so no growing exponential is formed. The second is the propagator: the same state is integrated across the graded layer as a  $6 \times 6$  matrix, with  $\mathbf{A}(z) = \mathbf{F}(z) \text{diag}(i\gamma_m) \mathbf{F}(z)^{-1}$  built at each depth from the plane-wave modes of the medium there,  $\mathbf{F}$  the matrix of the six modes' displacements and tractions and  $\gamma_m$  their vertical wavenumbers, and the radiation conditions of the two half-spaces are imposed on the result. For the graded layer the two agree to  $10^{-12}$  in all twelve cases with 400 steps of the march, whose error falls sixteenfold for each doubling of the steps; for a uniform layer the propagator equals Kennett's recursion to  $3 \times 10^{-14}$ .

**Oblique incidence, and a three-layer background.** At oblique incidence the test is the three-dimensional scheme of Eq. (49), with P, SV and SH waves, in two backgrounds: the two half-spaces above, and a three-layer background in which the voxels lie in a layer between a slower half-space above ( $\alpha = 4200$ ,  $\beta = 2400 \text{ ms}^{-1}$ ,  $\rho = 2300 \text{ kg m}^{-3}$ ) and the faster one below (6500, 3700, 2800), so that the layer reverberates between two interfaces. The construction is the one set out above: the reverberation added to the whole-space kernel at every order of the reciprocal lattice, Eq. (50), the interface matrices, the incident field, and the reference, Kennett's recursion [65], against which the background's own reflection coefficient is reproduced to  $5 \times 10^{-15}$  in all twelve cases before any voxel is introduced.

The results are in Table 15, at  $\omega = 3000 \text{ rad s}^{-1}$  ( $k_s D = 2$ ; at  $300 \text{ rad s}^{-1}$  the sixth-order errors lie below round-off, as in Table 14) and  $20^\circ$  in the upper half-space. In all thirty-six cases the order is that of the whole space, with the interfaces 3 m from the layer and with the voxels against them (`scripts/measure_oblique_stratified_background.py`). The errors are close

**Table 15.** The layer of voxels in a stratified background at  $20^\circ$ ,  $\omega = 3000 \text{ rad s}^{-1}$ : relative error of the change of the same-type reflection coefficient on  $n = 4$  planes, and the order over the last halving ( $n = 2 \rightarrow 4$ ), for field degrees  $p = 0, 1$  and  $2$ .

| background                                    | wave | $p = 0$                     | $p = 1$                     | $p = 2$                     |
|-----------------------------------------------|------|-----------------------------|-----------------------------|-----------------------------|
| two half-spaces, interface 3 m below          | P    | $6.1 \times 10^{-3}$ (2.03) | $2.7 \times 10^{-6}$ (4.02) | $2.3 \times 10^{-9}$ (6.04) |
|                                               | SV   | $1.5 \times 10^{-2}$ (2.11) | $2.6 \times 10^{-5}$ (4.13) | $3.7 \times 10^{-8}$ (6.09) |
|                                               | SH   | $1.6 \times 10^{-2}$ (2.11) | $2.7 \times 10^{-5}$ (4.13) | $4.1 \times 10^{-8}$ (6.09) |
| two half-spaces, interface against the voxels | P    | $5.1 \times 10^{-3}$ (2.04) | $3.8 \times 10^{-6}$ (4.04) | $2.1 \times 10^{-9}$ (6.00) |
|                                               | SV   | $1.6 \times 10^{-2}$ (2.10) | $2.5 \times 10^{-5}$ (4.14) | $3.3 \times 10^{-8}$ (6.11) |
|                                               | SH   | $1.7 \times 10^{-2}$ (2.10) | $2.4 \times 10^{-5}$ (4.15) | $3.4 \times 10^{-8}$ (6.11) |
| three layers, interfaces 3 m away             | P    | $6.3 \times 10^{-3}$ (2.03) | $8.7 \times 10^{-7}$ (4.04) | $2.4 \times 10^{-9}$ (6.06) |
|                                               | SV   | $1.0 \times 10^{-2}$ (2.10) | $5.8 \times 10^{-6}$ (4.08) | $5.2 \times 10^{-8}$ (6.11) |
|                                               | SH   | $1.3 \times 10^{-2}$ (2.10) | $3.0 \times 10^{-6}$ (3.95) | $9.3 \times 10^{-8}$ (6.10) |
| three layers, interfaces against the voxels   | P    | $4.8 \times 10^{-3}$ (2.04) | $2.4 \times 10^{-6}$ (4.04) | $4.1 \times 10^{-9}$ (6.03) |
|                                               | SV   | $8.5 \times 10^{-3}$ (2.09) | $7.1 \times 10^{-6}$ (4.01) | $5.0 \times 10^{-8}$ (6.09) |
|                                               | SH   | $1.4 \times 10^{-2}$ (2.10) | $5.8 \times 10^{-6}$ (3.92) | $8.5 \times 10^{-8}$ (6.12) |

to those of the whole space at the same frequency (Table 14): the background changes the field the voxels must represent, not how well they represent it.

**The two terms of the error at oblique incidence.** The orders say that the scheme converges; the two-term analysis says what its error is. Both terms are tested at  $20^\circ$ , in the whole space and in the stratified backgrounds, for P, SV and SH waves.

*The first-order term.* The scheme is written as  $(\mathbf{M} - \mathbf{K})\mathbf{c} = \mathbf{b}$  with the output  $\mathbf{u} = \mathbf{r}\mathbf{c}$ , as in Eq. (63), so that  $T_1^{\text{scheme}} = \mathbf{r}\mathbf{M}^{-1}\mathbf{b}$  with no differencing. The prediction is built from plane waves. The background field in the layer is a sum of six, P, SV and SH going down and going up, and the voxels radiate into six. For a pair made of an incident wave of wavevector  $\mathbf{k}_{\text{in}}$  and a radiated wave of wavevector  $\mathbf{k}_{\text{out}}$ , a cubic voxel of half-side  $h$  whose field is held to total degree  $p$  gives the exact first-order term of that pair times

$$1 + E_p(\mathbf{k}_{\text{in}}, \mathbf{k}_{\text{out}}) = \frac{\sum_{|a| \leq p} \prod_{i=1}^3 (2a_i + 1) j_{a_i}(k_{\text{in},i}h) j_{a_i}(k_{\text{out},i}h)}{\prod_{i=1}^3 j_0((k_{\text{in},i} - k_{\text{out},i})h)}, \quad (52)$$

the sum being over the Legendre products  $P_{a_1}P_{a_2}P_{a_3}$  of the voxel's basis and  $j_n$  the spherical Bessel functions: the numerator is the projection of the incident wave onto the basis paired with that of the outgoing wave, the denominator the exact pairing. Its leading departure from one is  $-d^2 \mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}/12$  for  $p = 0$ , with  $d = 2h$ , and along one axis it is the factor of Eq. (39). The predicted first-order term is the sum over the thirty-six pairs of the exact term of each pair, an integral of two exponentials over the voxel, times its factor, the radiated waves then being carried to the observer by the reverberation of Eq. (50) exactly. On one and two planes of voxels of degree 0, 1 and 2, at  $\omega = 3000 \text{ rad s}^{-1}$ , the scheme's first-order term equals this prediction to  $2 \times 10^{-15}$  in all twelve combinations of background and wave type. The first-order error is therefore known in closed form at oblique incidence, in a stratified background, for every wave type.

*The second-order term.* For the law of Eq. (44) the contrast must vary within the voxels. Each plane of voxels is given the smooth profile of §9 in depth, projected to Legendre degree  $r = p$ , and the coupling is that of Eq. (41) with the reverberation added. The second-order term of the scheme is  $T_2^{\text{scheme}} = \mathbf{r}\mathbf{M}^{-1}\mathbf{K}\mathbf{M}^{-1}\mathbf{b}$ . The exact  $T_2$  needs the exact stack with a layer whose properties vary continuously in depth, at oblique incidence, for which a recursion over homogeneous layers is not exact. It is computed by two independent algorithms, invariant imbedding [67] and the propagator of the six-component state, which agree to  $10^{-12}$  in all twelve cases (above).  $T_2$  then follows from Eq. (64) at  $\epsilon = 0.1$ , where halving  $\epsilon$  changes it by  $2 \times 10^{-8}$ . The ratio of the relative error of  $T_2$  to the relative projection error  $\mathcal{E}_p$  is in Table 16. On four planes the law holds to 3% for  $p = 0$  and 1 and to 6% for  $p = 2$  at  $k_5D = 0.2$ , and to 0.6% for all three degrees at  $k_5D = 0.067$ , in every background and for every wave type: the departure falls about ninefold as the frequency falls threefold, as  $(kD)^2$ . On two planes the ratios for  $p = 0$  and 1 lie within 0.03 of those in the table; for  $p = 2$  they are lower, 0.89 to 0.99 at  $k_5D = 0.2$  and 0.988 to 0.999 at  $k_5D = 0.067$ . The law is loosest for the highest degree at the highest frequency, because the projection error of degree 2 is so small ( $1.6 \times 10^{-5}$  on four planes) that the departure from the long-wave limit is a larger share of it. Both terms of the error are thus known in advance at oblique incidence and in a stratified background, as they are at normal

**Table 16.** The relative error of the second-order term over the relative projection error of the medium, at  $20^\circ$ , on  $n = 4$  planes with  $r = p$ ; the law of Eq. (44) predicts one in the long-wave limit.

|                                             | wave | $k_s D = 0.2$ |         |         | $k_s D = 0.067$ |         |         |
|---------------------------------------------|------|---------------|---------|---------|-----------------|---------|---------|
|                                             |      | $p = 0$       | $p = 1$ | $p = 2$ | $p = 0$         | $p = 1$ | $p = 2$ |
| background                                  |      |               |         |         |                 |         |         |
| whole space                                 | P    | 1.000         | 0.998   | 1.011   | 1.000           | 1.000   | 1.000   |
|                                             | SV   | 0.998         | 0.992   | 1.032   | 1.000           | 0.999   | 1.004   |
|                                             | SH   | 0.999         | 0.992   | 1.031   | 1.000           | 0.999   | 1.003   |
| two half-spaces, interface 3 m below        | P    | 1.010         | 1.009   | 1.019   | 1.001           | 1.001   | 1.001   |
|                                             | SV   | 1.012         | 1.008   | 1.046   | 1.001           | 1.001   | 1.005   |
|                                             | SH   | 1.027         | 1.025   | 1.059   | 1.003           | 1.003   | 1.006   |
| three layers, interfaces 3 m away           | P    | 1.004         | 1.003   | 1.013   | 1.000           | 1.000   | 1.000   |
|                                             | SV   | 0.979         | 0.973   | 1.002   | 0.997           | 0.996   | 1.000   |
|                                             | SH   | 1.007         | 1.004   | 1.035   | 1.001           | 1.001   | 1.004   |
| three layers, interfaces against the voxels | P    | 1.002         | 1.001   | 1.011   | 1.000           | 1.000   | 1.000   |
|                                             | SV   | 0.995         | 0.988   | 1.018   | 0.999           | 0.999   | 1.002   |
|                                             | SH   | 1.004         | 1.002   | 1.032   | 1.000           | 1.000   | 1.003   |

incidence in the whole space (`scripts/measure_oblique_two_term.py`).

Two limits stand. The oblique construction is implemented once, and its checks are the exact solutions. And the voxels here fill their plane, so that the image of a voxel in an interface it touches is the image of a whole plane; a body of finite lateral extent against an interface, where that image is singular, has not been run.

## D The sphere: near field, an impedance march and a sharp surface

**The near field.** A receiver is seldom at infinity, and the scheme does not require it to be. The scattered field at a point  $\mathbf{x}$  outside the voxels is the propagator applied to the source that each voxel carries, the contrast times the field found in it:

$$\psi^{\text{sc}}(\mathbf{x}) = \sum_j \int_{V_j} \mathbf{K}(\mathbf{x} - \mathbf{x}') \Delta(\mathbf{x}') e_j(\mathbf{x}') d\mathbf{x}', \quad (53)$$

with  $\mathbf{K}$  the nine-component Green's tensor of §2, the same one that couples the voxels in the solve,  $\Delta$  the contrast operator of Eq. (5) and  $e_j$  the polynomial internal state of voxel  $j$ . Its first three rows are the displacement and its last six the strain. Nothing is dropped: the terms in  $1/r^2$  and  $1/r^3$  are in  $\mathbf{K}$ . The integrand's source is a polynomial in each voxel and is integrated by a tensor Gauss rule with six nodes a side; the rule's own error is measured by repeating the readout with four nodes, and from 8 voxels across the two differ by less than  $4 \times 10^{-6}$  at  $1.25a$  and by less than  $10^{-10}$  at  $2a$ . The observers are the nine angles of Fig. 6(b) on spheres of radius  $1.25a$ ,  $2a$  and  $10a$ , where  $k_s r$  runs from 0.6 to 10; the reference is the exact partial-wave displacement at the same points; and the error is the measure of the far-field test. Table 17 gives it for the mean-only voxel and for the first-moment voxel (Appendix E). The orders of the far field hold in the near field: second for the mean-only voxel at every distance and both frequencies, and 3.8

**Table 17.** The graded sphere read in the near field: relative error of the scattered displacement at 1.25, 2 and 10 radii from the centre, against the exact displacement at the same points, and the apparent order between the two finest grids.

| $n_{\text{sub}}$           | mean only            |                      |                      | first moment         |                      |                      |
|----------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                            | 1.25a                | 2a                   | 10a                  | 1.25a                | 2a                   | 10a                  |
| $k_S a = 0.5$              |                      |                      |                      |                      |                      |                      |
| 4                          | $3.3 \times 10^{-2}$ | $1.2 \times 10^{-2}$ | $5.3 \times 10^{-3}$ | $1.4 \times 10^{-2}$ | $1.6 \times 10^{-3}$ | $6.3 \times 10^{-4}$ |
| 6                          | $1.4 \times 10^{-2}$ | $5.2 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $1.2 \times 10^{-3}$ | $3.8 \times 10^{-4}$ | $1.9 \times 10^{-4}$ |
| 8                          | $8.0 \times 10^{-3}$ | $2.9 \times 10^{-3}$ | $1.1 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $1.4 \times 10^{-4}$ | $7.3 \times 10^{-5}$ |
| 12                         | $3.5 \times 10^{-3}$ | $1.2 \times 10^{-3}$ | $4.7 \times 10^{-4}$ | $7.8 \times 10^{-5}$ | $3.1 \times 10^{-5}$ | $1.7 \times 10^{-5}$ |
| 16                         | $2.0 \times 10^{-3}$ | $6.7 \times 10^{-4}$ | $2.6 \times 10^{-4}$ | $2.5 \times 10^{-5}$ | $1.0 \times 10^{-5}$ | $5.7 \times 10^{-6}$ |
| order, 12 $\rightarrow$ 16 | 2.03                 | 2.06                 | 2.02                 | 3.95                 | 3.85                 | 3.77                 |
| $k_S a = 1.0$              |                      |                      |                      |                      |                      |                      |
| 4                          | $4.8 \times 10^{-2}$ | $3.5 \times 10^{-2}$ | $3.3 \times 10^{-2}$ | $9.9 \times 10^{-3}$ | $1.3 \times 10^{-3}$ | $6.0 \times 10^{-4}$ |
| 6                          | $2.0 \times 10^{-2}$ | $1.4 \times 10^{-2}$ | $1.4 \times 10^{-2}$ | $9.3 \times 10^{-4}$ | $3.0 \times 10^{-4}$ | $2.0 \times 10^{-4}$ |
| 8                          | $1.2 \times 10^{-2}$ | $8.1 \times 10^{-3}$ | $7.8 \times 10^{-3}$ | $3.5 \times 10^{-4}$ | $1.1 \times 10^{-4}$ | $7.4 \times 10^{-5}$ |
| 12                         | $5.2 \times 10^{-3}$ | $3.6 \times 10^{-3}$ | $3.5 \times 10^{-3}$ | $6.0 \times 10^{-5}$ | $2.5 \times 10^{-5}$ | $1.7 \times 10^{-5}$ |
| 16                         | $2.9 \times 10^{-3}$ | $2.0 \times 10^{-3}$ | $1.9 \times 10^{-3}$ | $1.9 \times 10^{-5}$ | $8.3 \times 10^{-6}$ | $5.7 \times 10^{-6}$ |
| order, 12 $\rightarrow$ 16 | 2.01                 | 2.01                 | 2.00                 | 3.95                 | 3.84                 | 3.78                 |

to 3.95 for the first-moment voxel, still rising towards four as in the far field at this contrast. A quarter of a radius from the surface, with 16 voxels across, the first-moment voxel's error is  $2.5 \times 10^{-5}$  and  $1.9 \times 10^{-5}$ , 78 and 153 times below the mean-only voxel's. The error grows towards the body, by a factor of 1.5 to 7 between  $10a$  and  $1.25a$ , because the near field carries the higher multipoles of the source, which a given grid resolves less well; the order does not change. On the coarsest grid a voxel is as wide as the graded shell and twice as wide as the gap to the nearest observer, and the sequence is not yet asymptotic. Read at these distances with the asymptotic formula, the same solutions would be in error by 48% to 85%: the far-field readout is for the far field (`scripts/measure_graded_sphere_near_field.py`; the readout is `graded_field` of the package, tested against the propagator of a point source, against the gradient of its own displacement, and against the far-field readout at great distance).

**Table 18.** The smoothly graded sphere, voxelised with the mean-only voxel on grids of  $n_{\text{sub}}$  voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

| $n_{\text{sub}}$ | unknowns | $k_S a = 0.5$         |       | $k_S a = 1.0$         |       |
|------------------|----------|-----------------------|-------|-----------------------|-------|
|                  |          | error                 | order | error                 | order |
| 4                | 576      | $7.70 \times 10^{-3}$ |       | $3.24 \times 10^{-2}$ |       |
| 6                | 1 872    | $3.04 \times 10^{-3}$ | 2.30  | $1.32 \times 10^{-2}$ | 2.22  |
| 8                | 3 672    | $1.68 \times 10^{-3}$ | 2.05  | $7.52 \times 10^{-3}$ | 1.95  |
| 12               | 11 520   | $7.30 \times 10^{-4}$ | 2.06  | $3.34 \times 10^{-3}$ | 2.00  |
| 16               | 24 984   | $4.06 \times 10^{-4}$ | 2.04  | $1.88 \times 10^{-3}$ | 2.00  |
| 20               | 47 304   | $2.58 \times 10^{-4}$ | 2.03  | $1.20 \times 10^{-3}$ | 2.00  |
| 24               | 79 344   | $1.79 \times 10^{-4}$ | 2.02  | $8.34 \times 10^{-4}$ | 2.00  |

**The voxels against an impedance march.** A second route to the same body is invariant imbedding. The impedance of §9, Eq. (51), is marched upwards through the depth range of the sphere with the lateral variation of the medium carried inside the march, on a laterally periodic grid, and a second sweep carries the field down [67]; the march returns the reflected plane waves on a plane above the body and the transmitted ones on a plane below. It is run here on the graded sphere and set beside the voxels, to ask how many voxels a given accuracy costs against an established alternative. Four things are fixed so that the comparison is of like with like.

1. *The planes.* Both methods are read on the planes  $z = -1.25a$  and  $z = +1.25a$ . On the tangent planes  $z = \pm a$  themselves the observers lie on the faces of the voxels at the poles, where the Gauss rule of the voxel readout (Eq. (53)) cannot integrate the singular kernel: its change from four to six nodes is 2% to 10% there, more than the scheme's error, and below  $10^{-4}$  a quarter of a radius away. The march's amplitudes are carried across the same homogeneous gap by each order's own phase, which is exact.
2. *The periodic array.* The march needs a laterally periodic grid, so it solves a square array of spheres. Its answer is scored against the exact solution of one sphere summed over the array, by Poisson summation, which is exact but for scattering between the spheres. That part, and the lateral resolution, are errors of the method as used on a single body, and are counted.
3. *The error.* For the march, the norm of the difference over the plane-wave orders of the grid, relative to the norm of the exact scattered orders; for the voxels, the norm of the difference of the scattered displacement over a patch of side  $5a$  on each plane, relative to the norm of the exact one. The two are the same quantity in wavenumber and in space, for the array and for the single body.
4. *The refinement.* For the march, the lateral grid, with the depth step refined until the error no longer changes; for the voxels, the grid and the degree.

**Table 19.** The graded sphere at  $k_S a = 0.5$  read on the planes  $z = \mp 1.25a$ : relative error in reflection and in transmission. The march's unknowns are those of one plane, on which it carries a dense matrix; the period is  $5a$  unless stated.

| method              | setting                        | unknowns | reflection           | transmission         |
|---------------------|--------------------------------|----------|----------------------|----------------------|
| impedance march     | $12 \times 12$ , 16 steps      | 432      | $1.2 \times 10^{-2}$ | $9.9 \times 10^{-3}$ |
|                     | $16 \times 16$ , 32 steps      | 768      | $4.3 \times 10^{-3}$ | $3.6 \times 10^{-3}$ |
|                     | $20 \times 20$ , 32 steps      | 1200     | $7.6 \times 10^{-4}$ | $7.3 \times 10^{-4}$ |
|                     | $28 \times 28$ , 48 steps      | 2352     | $2.5 \times 10^{-4}$ | $5.6 \times 10^{-4}$ |
|                     | $30 \times 30$ , period $7.5a$ | 2700     | $7.2 \times 10^{-4}$ | $4.7 \times 10^{-4}$ |
| mean-only voxel     | 8 across                       | 3672     | $5.0 \times 10^{-3}$ | $7.9 \times 10^{-3}$ |
|                     | 16 across                      | 24 984   | $1.2 \times 10^{-3}$ | $2.0 \times 10^{-3}$ |
| first-moment voxel  | 6 across                       | 7488     | $3.3 \times 10^{-4}$ | $5.2 \times 10^{-4}$ |
|                     | 8 across                       | 14 688   | $1.4 \times 10^{-4}$ | $2.2 \times 10^{-4}$ |
|                     | 16 across                      | 99 936   | $1.1 \times 10^{-5}$ | $1.8 \times 10^{-5}$ |
| second-moment voxel | 8 across                       | 36 720   | $1.4 \times 10^{-5}$ | $2.3 \times 10^{-5}$ |
|                     | 10 across                      | 69 120   | $2.6 \times 10^{-6}$ | $4.2 \times 10^{-6}$ |

Table 19 gives the result. The march converges quickly in the lateral grid, because the body is smooth, and with far fewer unknowns than any voxel scheme: at  $20 \times 20$  it is at  $7 \times 10^{-4}$  with 1200 unknowns on a plane. The mean-only voxel does not reach that with sixteen voxels across and twenty times the unknowns; on a sharp sphere, where it also carries a staircase, it fares worse. The first-moment voxel reaches the march's accuracy with six voxels across, and below about  $2 \times 10^{-4}$  it continues at close to fourth order where the march does not: in transmission the march's error at a period of  $5a$  stops near  $5 \times 10^{-4}$ , which is the scattering between the spheres of its array, since lengthening the period lowers it and refining the grid does not. The costs are of different kinds. The march's unknowns are few but its operator is a dense matrix on the plane, marched through the depth, and a finer lateral grid requires a smaller depth step (with eight steps the  $20 \times 20$  march diverged); the voxel schemes have many unknowns and a product done by fast Fourier transform. In the timings of these runs, taken on one machine with other work in progress and therefore only indicative, the first-moment voxel at six across was solved in about a sixth of the time of the  $20 \times 20$  march. The march has what the comparison does not use: it returns the reflection and transmission for every incident plane wave of the grid at once, and a laterally periodic or laterally extended medium is its natural problem and not a cost. The comparison is at one frequency (`scripts/measure_graded_sphere_march.py`, `scripts/measure_graded_sphere_planes.py`).

**A sphere with a sharp surface.** The homogeneous sphere is the familiar test, and its voxelisation changes with every grid. Table 20 shows its error for grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at  $k_S a = 0.5$ . The error tracks the staircase's volume, which is 5%, 17% and 4% too large on these grids. Rescaled by the ratio of the sphere's volume to the staircase's, it falls below 1% at  $k_S a = 0.5$  and to at most 3% at  $k_S a = 1.0$ , but it still does not converge: what remains is the staircase's shape. A refinement study of a sharp sphere measures its staircase; the smooth sphere, like the layer, measures the scheme.

**Table 20.** A homogeneous sphere voxelised on grids of  $n_{\text{sub}}$  voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere's volume to the staircase's. It does not converge with refinement, because the staircase changes with the grid.

| $n_{\text{sub}}$ | voxels | $k_S a = 0.5$         |                      | $k_S a = 1.0$         |                      |
|------------------|--------|-----------------------|----------------------|-----------------------|----------------------|
|                  |        | raw                   | volume-rescaled      | raw                   | volume-rescaled      |
| 4                | 32     | $4.6 \times 10^{-2}$  | $1.4 \times 10^{-3}$ | $5.0 \times 10^{-2}$  | $5.4 \times 10^{-3}$ |
| 6                | 136    | $1.94 \times 10^{-1}$ | $7.9 \times 10^{-3}$ | $1.68 \times 10^{-1}$ | $3.0 \times 10^{-2}$ |
| 8                | 280    | $4.2 \times 10^{-2}$  | $2.3 \times 10^{-3}$ | $3.5 \times 10^{-2}$  | $8.9 \times 10^{-3}$ |

## E The three-dimensional voxel solver

The three-dimensional results of §§10 and 11 use the Legendre voxels of §8 on a cubic grid, with the following construction (`cubic_scattering/graded_voxel`).

**The voxel.** Each cell carries the nine-component state of §2, displacement and engineering strain, with the strain an *independent* unknown and not the gradient of the displacement: a strain derived from a cell's projected displacement jumps across the faces between cells and leaves a spurious source there (§8). The state is expanded in the Legendre products of degree at most  $p$  in the cell's local coordinates  $\xi \in [-1, 1]^3$ , and the contrast operator  $\Delta$  in those of degree at most  $r$ , its coefficients the  $L^2$  projection of the medium;  $p = r = 1$  gives the first-moment voxel of 36 weights. The source  $\Delta\psi$  is then a polynomial of degree  $p + r$ , expanded in monomials  $m_c$ . The Lippmann–Schwinger equation is tested with each basis function over each cell (Galerkin), and cells  $m$  and  $n$  couple through the blocks

$$\mathbf{K}_{ac}(\mathbf{R}) = \int_{V_m} \int_{V_n} L_a(\xi) \mathbf{P}(\mathbf{x} - \mathbf{x}') m_c(\xi') d^3x d^3x', \quad \mathbf{R} = \mathbf{x}_m - \mathbf{x}_n, \quad (54)$$

with  $\mathbf{P}$  the  $9 \times 9$  point propagator of §2.

**One integral in place of two.** With  $\mathbf{s} = \mathbf{u} - \mathbf{u}'$  the difference of the two local positions, Eq. (54) becomes  $\mathbf{K}_{ac}(\mathbf{R}) = \int W_{ac}(\mathbf{s}) \mathbf{P}(\mathbf{R} + \mathbf{s}) d^3s$  over  $\mathbf{s} \in [-2h, 2h]^3$ , where  $W_{ac}$  is a product over the three axes of one-dimensional cross-correlations of the test and source polynomials: continuous, polynomial on  $[-2h, 0]$  and  $[0, 2h]$ , with kinks at 0 and  $\pm 2h$ . For cells that do not touch, the kernel is smooth on each of the eight pieces, and a three-dimensional Gauss rule integrates it, with 14, 10 or 8 points per axis at a Chebyshev distance of 2, 3 or 4, and 5 or more cells; this agrees with six-dimensional quadrature of Eq. (54) to  $10^{-9}$  one cell beyond contact.

**The self cell and its 26 neighbours.** The static part of  $\mathbf{P}$  is a sum of derivatives of  $r^{-1}$  and  $r$ , distributions at the origin. Derivatives are moved onto  $W$  until the kernel is at most as singular as  $1/r$ , and the kinks of  $W$  then carry plane  $\delta$  functions; every piece is a box or a rectangle with its corners on the breakpoints, so the singular point is either a vertex of a piece, integrated with Duffy's pyramids, or at least  $2h$  away. The  $\delta$  function of the inclusion problem is carried by those plane  $\delta$  functions and is not added by hand. The construction reproduces the sum rule  $\sum_p \int_V \int_V \partial_p \partial_p |\mathbf{x} - \mathbf{x}'|^{-1} d^3x d^3x' = -4\pi V$ , the biharmonic rule for  $r$  and the cube's Coulomb self-energy to  $10^{-11}$ , and every block equals the sum of the 64 blocks between the two cells' half-size sub-cells, with the polynomials re-expressed on them, to  $2 \times 10^{-15}$ . The dynamic remainder of  $\mathbf{P}$  is at most weakly singular and is integrated by the Gauss rule of the pieces.

**Symmetry, and the solve.** A block depends only on the offset  $\mathbf{o}$  between the two cells. The background is isotropic and the cube is mapped to itself by its 48 signed permutations  $\mathbf{Q}$ , under which  $\mathbf{P}(\mathbf{Q}\mathbf{r}) = \mathbf{S}\mathbf{P}(\mathbf{r})\mathbf{S}^T$ , with  $\mathbf{S} = \text{diag}(\mathbf{Q}, \mathbf{R}_6(\mathbf{Q}))$  and  $\mathbf{R}_6$  the representation of  $\mathbf{e} \mapsto \mathbf{Q}\mathbf{e}\mathbf{Q}^T$  on Voigt vectors. With the test and source polynomials transforming by matrices  $\mathbf{T}$  and  $\mathbf{U}$ ,  $\mathbf{K}_{ac}(\mathbf{Q}\mathbf{o}) = \sum_{b,d} T_{ab} U_{cd} \mathbf{S} \mathbf{K}_{bd}(\mathbf{o}) \mathbf{S}^T$ , so one block is computed for each orbit of offsets and the rest follow exactly. The coupling is then a convolution over the grid, applied by the fast Fourier transform on a  $(2n-1)^3$  circular embedding, and the system is solved by GMRES preconditioned with each cell's own block. Where both can be run, the solution equals that of a dense solve to  $10^{-10}$  relative (`tests/test_graded_voxel_fft.py`). Every cell that overlaps the body is

kept, each with the  $L^2$  projection of its contrast by a six-point Gauss rule per axis; keeping only the cells whose centre lies inside the body drops the contrast of the partly filled cells, and the mean-only voxel then converges at order one instead of two. The far field is read at  $5 \times 10^8$  radii, where the near-field terms that the asymptotic form omits are negligible.

**The first-moment voxel on the graded sphere.** The voxel with  $p = r = 1$  is applied to the graded sphere of §10. To isolate the discretisation error from the rescattering the grid cannot yet resolve, the contrast is first scaled by  $10^{-6}$ , and the exact sphere is solved at the same scaled contrast, so that no Born approximation enters the reference: the relative error is then that of the scheme's first-order term, in which the field is the incident one carried across each cell. With 4, 6 and 8 voxels across, 2304 to 14688 unknowns, the error is  $7.5 \times 10^{-5}$ ,  $5.0 \times 10^{-6}$  and  $1.6 \times 10^{-6}$  at  $k_S a = 0.5$ , and  $2.3 \times 10^{-4}$ ,  $9.1 \times 10^{-5}$  and  $2.8 \times 10^{-5}$  at  $k_S a = 1$ . The apparent orders between 6 and 8 voxels are 4.01 and 4.13; between 4 and 6 they are not yet asymptotic, a voxel at  $n = 4$  being as wide as the graded shell (`scripts/pilot_graded_voxel_sphere.py--weak`). At the full contrast the fast solve is carried to 16 voxels across, where the error is  $4.9 \times 10^{-6}$  at  $k_S a = 0.5$  and  $5.9 \times 10^{-6}$  at  $k_S a = 1$ , and the apparent order approaches four at a rate set by the number of voxels across the graded shell.

## F The uniform-strain closure against the exact sphere

**Against the exact sphere.** These statements are tested against the exact elastic Mie solution for a sphere, with no Eshelby formula in between. The concentration of Mie order  $n$  is  $E_n = a_n/a_n^{\text{Born}}$  (orders 0, 1, 2 are the dilatational, density and shear channels), extrapolated to  $ka \rightarrow 0$ . The formulation is applied to a ball of radius  $a$ , with its moment  $\mathbf{M}$  evaluated by the surface route of Eq. (26) on  $|\mathbf{r}| = a$ . Its dilatational and shear channels equal Mie's to  $2 \times 10^{-10}$ , and the density channel is 1. The cube's closed forms meet the same numbers. Its dilatational channel is Mie's  $E_0$  to  $2 \times 10^{-10}$ , and its isotropic shear average, Eq. (32), is the depolarisation read from Mie's  $E_2$  to  $10^{-8}$ . The same moment taken as a principal value, over the ball with a small sphere excluded around the origin, has zero depolarisation in every channel. The whole static response of the sphere is the  $\delta$  term.

At finite frequency the closure uses the dynamic ball moments, which are in closed form. It departs from Mie at second order in  $ka$  and first order in the contrast, and the departure is closed-form too: the ball's moments are elementary, and Mie's low-frequency series with the contrast kept exact gives  $E_n$  to relative order  $(ka)^3$ . To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta \lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta \rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta \mu}{\mu}, & n = 2, \end{cases} \quad (55)$$

with no  $O(ka)$  term and no  $O((ka)^3)$  term. The imaginary  $(ka)^3$  terms, the radiation damping,

agree exactly in all three channels. So the closure's only error at this order is the real  $(ka)^2$  term, the uniform-field ansatz's own. That is the order the hierarchy assigns to the omitted second gradients (§6.3), and it has the same origin, the variation of the field across the scatterer, as the collocation error of §7. Eq. (55) is derived symbolically (`SphereClosureDynamic.w1`). It agrees with the coefficient extracted from the numerical Mie solve and an independent implementation of the closure to  $5 \times 10^{-4}$ , which is the size of the next order in the contrast.

## G The scattering series of the layer, and the projection-error law

*The scattering series.* Multiply the whole contrast,  $\Delta\lambda$ ,  $\Delta\mu$  and  $\Delta\rho$  together, by a number  $s$ , and let  $u(s)$  be the scattered displacement at an observer at depth  $z_o$  outside the layer. It vanishes at  $s = 0$  and is analytic in  $s$  there, and its expansion is the Born series,

$$u(s) = sT_1 + s^2T_2 + s^3T_3 + \dots, \quad (56)$$

so that  $T_n$  is the part of the field at the full contrast,  $s = 1$ , that has been scattered  $n$  times.

*The first two terms of the exact layer.* They are integrals over the layer. With the incident state  $w^0 = (u^0, \varepsilon^0)$ , the contrast  $\Delta_1 = \text{diag}(\omega^2\Delta\rho, \Delta M)$  with profile  $f(z)$ , and the kernel  $\mathbf{K}$  of Eq. (38), whose first row  $(g, g')(z_o - z)$  gives the displacement at the observer,

$$T_1 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^0(z) dz, \quad (57)$$

$$w^1(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^0(z') dz', \quad T_2 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^1(z) dz. \quad (58)$$

Here  $w^1$  is the state scattered once, inside the layer; its strain includes the local term of the kernel,  $-f(z) \Delta M \varepsilon^0(z)/M_p$ . Each further term repeats the step,  $w^n(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^{n-1}(z') dz'$ . For a plane force at  $z_s$  above the layer,  $w^0(z) = c e^{ik(z-z_s)}(1, ik)$  with  $c = i/(2M_p k)$ , every integrand is an exponential times the profile. Writing  $a = \omega^2\Delta\rho$  and  $b = \Delta M$ , and carrying out the products of Eqs. (57) and (58), the terms seen in reflection ( $z_o < 0$ ) and in transmission ( $z_o > D$ ) are, for any profile,

$$T_1^R = c^2 e^{-ik(z_o+z_s)}(a + bk^2) \int_0^D f(z) e^{2ikz} dz, \quad T_1^T = c^2 e^{ik(z_o-z_s)}(a - bk^2) \int_0^D f(z) dz, \quad (59)$$

$$\begin{aligned}
T_2^R &= c^2 e^{-ik(z_o+z_s)} \left[ c(a^2 - b^2 k^4) \int_0^D \int_0^D f(z) f(z') e^{ik(z+z'+|z-z'|)} dz' dz - \frac{b^2 k^2}{M_p} \int_0^D f(z)^2 e^{2ikz} dz \right], \\
T_2^T &= c^2 e^{ik(z_o-z_s)} \left[ c(a - bk^2)^2 \int_0^D \int_0^z f(z) f(z') dz' dz + c(a + bk^2)^2 \int_0^D \int_z^D f(z) f(z') e^{2ik(z'-z)} dz' dz \right. \\
&\quad \left. + \frac{b^2 k^2}{M_p} \int_0^D f(z)^2 dz \right].
\end{aligned} \tag{60}$$

The first-order reflection is the Fourier transform of the profile at the wavenumber  $2k$ , and the first-order transmission its integral. In each second-order term the last integral is the local term of the kernel, in which both scatterings are at one point and the profile appears squared; the others are the two scatterings at different depths. In transmission the wave scattered twice forwards carries the contrast of velocity twice,  $(a - bk^2)^2$ , and the wave scattered backwards and then forwards again the contrast of impedance twice,  $(a + bk^2)^2$ . For the uniform layer ( $f = 1$ ) the integrals give

$$T_1^R = c^2 e^{-ik(z_o+z_s)} (a + bk^2) \frac{e^{2ikD} - 1}{2ik}, \quad T_1^T = c^2 e^{ik(z_o-z_s)} (a - bk^2) D, \tag{61}$$

$$\begin{aligned}
T_2^R &= \frac{c^2 e^{-ik(z_o+z_s)}}{4k^3 M_p} \left[ -i(a^2 + b^2 k^4) + e^{2ikD} (a^2(i + 2kD) + b^2 k^4(i - 2kD)) \right], \\
T_2^T &= \frac{ic^2 e^{ik(z_o-z_s)}}{8k^3 M_p} \left[ a^2(1 - e^{2ikD} + 2kD(i + kD)) - 2abk^2(e^{2ikD} - 1 + 2kD(kD - i)) \right. \\
&\quad \left. + b^2 k^4(1 - e^{2ikD} + 2kD(kD - 3i)) \right].
\end{aligned} \tag{62}$$

The reflection sees the contrast of impedance,  $a + bk^2 = k^2 M_p (\Delta\rho/\rho + \Delta M/M_p)$ , and the transmission that of velocity,  $a - bk^2$ . As  $kD \rightarrow 0$ ,  $T_2^R \rightarrow -c^2 e^{-ik(z_o+z_s)} b^2 k^2 D/M_p$ : the local term alone, which is the starting point of the argument at the end of this appendix. For the smooth profile the integrals of Eqs. (59) and (60) are evaluated in closed form symbolically, and by quadrature they return the same values to  $10^{-16}$ . Equations (61) and (62) equal, identically, the first two Taylor coefficients in  $s$  of the exact reflection and transmission of the layer of §3 (ContinuumLimit\_BornTerms.wl). At  $\omega = 300 \text{ rad s}^{-1}$ ,  $|T_2|/|T_1|$  is 0.039 in reflection and 0.17 in transmission for the uniform layer.

*The first two terms of the scheme.* Write Eq. (38) as  $(M - K)c = b$ , with  $M$  the diagonal matrix of the  $\int_i \phi_a^2 dz$ ,  $K$  the matrix of the coupling blocks times  $\Delta_1$ , and  $b$  the moments of the incident field, and let  $r$  be the row that gives the scattered displacement at the observer from the coefficients,  $u = rc$ . Scaling the contrast by  $s$  scales  $K$  and  $r$  by  $s$ , so  $u(s) = sr(M - sK)^{-1}b$  and

$$T_1^{\text{scheme}} = r M^{-1} b, \quad T_2^{\text{scheme}} = r M^{-1} K M^{-1} b. \tag{63}$$

$M^{-1}b$  is the projection of the incident field onto each cell's functions: the first term needs no solve. Splitting  $K$  into its blocks within a cell and its blocks between cells splits  $T_2^{\text{scheme}}$  into the part scattered twice in one cell and the part scattered in two; the same split of the inner integral of Eq. (58) splits the exact  $T_2$ .

*A check by differences.* The terms can also be read from solutions at four scaled contrasts,  $s = \pm\epsilon$  and  $\pm 2\epsilon$ , with no knowledge of the formulas above. The odd and even parts of  $u$  in  $s$  separate the odd orders from the even, and the two step sizes remove the next term of each (the five-point central differences for  $u'(0)$  and  $\frac{1}{2}u''(0)$ ):

$$\begin{aligned} T_1 &= \frac{8[u(\epsilon) - u(-\epsilon)] - [u(2\epsilon) - u(-2\epsilon)]}{12\epsilon} + 4\epsilon^4 T_5 + \dots, \\ T_2 &= \frac{16[u(\epsilon) + u(-\epsilon)] - [u(2\epsilon) + u(-2\epsilon)]}{24\epsilon^2} + 4\epsilon^4 T_6 + \dots \end{aligned} \quad (64)$$

A negative  $s$  is a medium slower and lighter than the background by the same fractions, as legitimate a medium as the other. With  $\epsilon = 0.01$  the remainders  $4\epsilon^4 T_5/T_1$  and  $4\epsilon^4 T_6/T_2$  are below  $2 \times 10^{-12}$  for this layer, and the formulas applied to the exact layer return Eqs. (61) and (62) less exactly those remainders, to  $10^{-17}$ . The measurements of §9 use the differences for the scheme and the exact layer alike, each in its own arithmetic, and they agree with Eqs. (57), (58) and (63).

*Why the terms must be separated.* The law of Eq. (44) is a statement about  $T_2$ . At the full contrast the terms of third and higher order are 7% of the second and carry errors of their own, and if every order above the first is taken together the ratio of Eq. (44) reads 0.90 to 0.94 where the second-order term alone gives 1.00 to 1.01. The shortfall is itself the single site's. In the long-wave limit the kernel is local, the exact strain is  $\epsilon_0/(1 - cf)$  with  $c = -\Delta M/M_p$ , and the scheme's is the Galerkin solution of  $\epsilon = \epsilon_0 + cf\epsilon$  in each cell on that cell's polynomials; the relative error of the term of order  $n$  is  $(\int f^n dz - \int (\Pi f)^n dz) / \int f^n dz$ , 2.455 times  $\mathcal{E}_q$  for the third term of this profile, which has the opposite sign to the second. At  $\omega = 30 \text{ rad s}^{-1}$  the whole nonlinear part has 0.896 times  $\mathcal{E}_q$ , and the cell-by-cell Galerkin solution reproduces it to four digits with  $c$  from the contrast (`scripts/measure_layer_t3_low.py`).

Equation (44) is exact in the long-wave limit, for a reason that shows what each basis is for. The strain-strain entry of the kernel,  $g'' = -k^2 g - \delta(z)/M_p$ , is local as  $kD \rightarrow 0$ , and the incident strain  $\epsilon_0$  is then uniform across the layer, so the strain scattered once is  $-(\Delta M/M_p) f(z) \epsilon_0$ : it is the profile itself. A cell holds the contrast as  $\Pi_r f$  and the field to degree  $p$ , so it holds the once-scattered strain as  $\Pi_p(\Pi_r f) = \Pi_q f$  with  $q = \min(p, r)$ . The second-order term pairs the contrast with that strain:

$$T_2 \propto \int f \cdot f dz = \|f\|^2, \quad T_2^{\text{scheme}} \propto \int (\Pi_r f)(\Pi_q f) dz = \|\Pi_q f\|^2, \quad (65)$$

the last because  $\Pi_q f$  lies in the range of  $\Pi_r$ . The relative error is  $1 - \|\Pi_q f\|^2 / \|f\|^2 = \mathcal{E}_q$ . Because the kernel is local in this limit, the second scattering takes place in the cell of the first:  $T_2$  is the second-order term of the single-site  $T$ -matrix, and Eq. (44) is a statement about the single site. The  $T$ -matrix sums the rescattering within a cell exactly for the cell as it is represented, a contrast  $\Pi_r f$  and a field of degree  $p$ ; its error is that the cell is not the medium. The split is measured directly (`scripts/measure_layer_bases.py`): at  $\omega = 300 \text{ rad s}^{-1}$  on 4, 8 and 16 cells,  $T_2$  of the scheme and of the exact layer are each divided into the part whose two scatterings lie in one cell and the part between different cells. For all nine pairs the error of the same-cell part is 0.998 to

1.013 times  $|T_2| \mathcal{E}_{\min(p,r)}$ , and that of the part between cells at most 0.002 times it. The coupling between cells contributes only the departure from the limit. Measured, the ratio of the two sides of Eq. (44) for  $p = r = 1$  on 8 cells departs from one by  $1.7 \times 10^{-2}$ ,  $4.3 \times 10^{-3}$ ,  $4.8 \times 10^{-4}$  and  $4 \times 10^{-5}$  at  $kD = 0.24$ , 0.12, 0.04 and 0.012: as  $(kD)^2$  (`scripts/measure_layer_bases.py`). The pairing in Eq. (65) is verified symbolically for all nine pairs, and the projection errors are computed exactly (`ContinuumLimit_LayerDefect.wl`); they fall by  $4^{q+1}$  for each halving of the cells.

## H Relation to other methods

**Relation to the coupled-dipole method.** The same partition appears in the coupled-dipole method as the choice of polarisability. The Clausius–Mossotti relation, the lattice dispersion relation [11], the polarisability taken from the exact static solution [12] and the integrated Green tensor [13] all adjust the single site so that the lattice reproduces the continuum. For space-filling cubes the adjustment is neither needed nor a free parameter: the tiling identity fixes the local field exactly, the self term of the collocation closure is the one that makes it cancel, and what is left is a closed-form error rather than an empirical one. That the method is a collocation of the volume integral equation was shown for the electromagnetic case by Lakhtakia [9]; Eq. (35) is the same statement for the elastic lattice with the self term evaluated over the finite cube, and it is what makes the error a midpoint-rule error. The filtered coupled dipoles of Piller and Martin [14] and Yurkin et al. [15] band-limit the source to the grid; the source-cell average does the complementary thing exactly, its form factor vanishing on the reciprocal lattice, and the spectral construction of Eq. (49) is the periodic coupled-dipole method of Chaumet et al. [16] with the Galerkin double average in place of the point dipole. The second-order convergence reported for the DDA by Yurkin et al. [3] corresponds to the uniform-field voxel’s, and the opposite-signed constants of Eq. (36) explain why a polarisability correction can improve one observable only at the expense of another, as the package’s own form-factor corrections do in reflection and transmission. Going beyond second order requires a richer internal field, which is the point of higher-order bases for volume integral equations [2, 70]; the first-moment voxel is the lowest such basis, and in its spectral form the Galerkin double average makes the lattice sum converge without Ewald splitting.

**Relation to the spectral element method.** The voxel of degree  $p$  is close kin to the spectral element [71], which was taken up for seismic wave propagation in the 1990s, is set out for three-dimensional models by Komatitsch and Tromp [72], and is now a standard tool for it. Both divide the medium into hexahedra, represent the field in each by polynomials of high degree, and determine the coefficients by a Galerkin condition on a weak form. The spectral element uses Lagrange interpolants through the Gauss–Lobatto–Legendre points of the element; the voxel uses the Legendre polynomials themselves. Four of the present results have their counterparts there.

1. *A diagonal mass matrix.* The spectral element obtains one by integrating at the Gauss–Lobatto–Legendre points, at which each interpolant vanishes but one [72]. That rule is not exact for the product of two interpolants, so that the diagonal form is reached, in

their words, by a process of subintegration, as in the finite-element methods that lump the mass. Here the Gram matrix  $M$  of Eq. (63) is diagonal with no approximation, because the Legendre functions are orthogonal on the cell; this is why the first-order term needs no solve. The two routes serve different ends: theirs an explicit march in time, ours the terms of the scattering series.

2. *The doubled order.* A scattered amplitude is a functional of the field, and a Galerkin method gives a functional to twice the order of the field itself. The order  $2p + 2$  is this doubling: the leading error of Eq. (39) is the product of the part of the incident wave that the cell's polynomials miss and the part of the outgoing wave that they miss, each of order  $h^{p+1}$ .
3. *The medium to the degree of the field.* In a spectral element the density and the elastic moduli are evaluated at the element's own grid points and may vary from one to the next, so that a fully heterogeneous medium is held to the degree of the field [72]. The rule  $\min(2p + 2, 2r + 2)$  says why that is the right choice, and what a coarser representation of the medium would cost.
4. *Elements that honour the interfaces.* A spectral-element mesh is designed to honour the free surface and the main geological boundaries of the model [72]. The stratified layer of §9 is the same requirement seen from the integral side: with voxel faces on the interfaces the projection error vanishes, and a jump inside a voxel costs the order.

The two methods differ as a differential formulation differs from an integral one, and they suit different problems. The spectral element discretises the weak form of the equations of motion in the time domain. Its coupling is local and its matrices are sparse; it treats heterogeneity, anisotropy and topography of any form throughout a three-dimensional model; and the price is that the whole model is meshed, that the outgoing waves are absorbed at an artificial boundary, and that over a long simulation the accuracy is governed by the time scheme [72]. The voxel scheme discretises the volume integral equation in the frequency domain. Only the departure from a stratified background is meshed, the field of each voxel is independent of its neighbours', and propagation through the background, the radiation condition included, is carried exactly by the background's Green's tensor; the price is a coupling between every pair of voxels, and a background restricted to one whose Green's tensor is known. It differs also from the boundary integral methods, which are restricted to a finite number of homogeneous regions [72]: a volume integral equation admits a medium that varies within the scatterer. The voxel scheme is therefore a candidate where the heterogeneity is localised in a layered medium, and not a replacement for the spectral element in a model that is heterogeneous throughout. A spectral-element mesh is designed by a rule of resolution, about five grid points to the shortest wavelength at polynomial degrees between five and ten [72]. The error analysis of this paper is the counterpart of that rule for the integral formulation: it gives the size and the degree of the voxels for a required accuracy in closed form, and it adds a term, the projection error of the medium.

## I The cost of higher order, and what the error analysis offers a fast solver

**The cost of fourth order.** A first-moment voxel carries four times as many unknowns as a uniform one (36 against 9 in three dimensions). The closed forms show what that buys. To reach an error  $\varepsilon$  in reflection, the uniform voxel needs  $kd = \sqrt{8\varepsilon}$  and the first-moment voxel  $kd = (720\varepsilon)^{1/4}$ ; at  $\varepsilon = 10^{-6}$  that is  $kd = 2.8 \times 10^{-3}$  against 0.16, voxels 58 times larger in each direction. In a three-dimensional model the voxel count falls by  $58^3 \approx 2 \times 10^5$  and the number of unknowns by about  $5 \times 10^4$  (these are the Born-order constants; at the contrast studied the first-moment voxel's measured errors lie within 5% of them, while the uniform voxel's transmission constant grows with the contrast, so the comparison is if anything conservative). The stratified model of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-moment voxels for the same accuracy. Higher moments continue the trend in one dimension: by Eq. (40) the second-moment voxel reaches  $\varepsilon = 10^{-6}$  at  $kd = 0.68$  and the third-moment voxel at  $kd = 1.5$ , where a voxel is already a quarter of a wavelength. In three dimensions, with the Legendre products of total degree at most  $p$ , a voxel carries 10 ( $p = 2$ ) or 20 ( $p = 3$ ) weights per state component, against 4 for the first-moment voxel. The voxel of degree 2 converges at sixth order in the layer at oblique incidence (Table 14); that of degree 3 has not been built in three dimensions. The gain is largest precisely where it matters: when the model varies on scales that are coarse compared with the accuracy the wavefield must have.

**What the error analysis offers a fast solver.** A voxel scheme is dense, and for a large medium it is solved iteratively with a fast product. The adaptive integral method [73] and the precorrected fast Fourier transform [74] are the standard forms: the sources are projected onto a regular grid, in the first by matching their multipole moments and in the second by matching the potentials they produce at a set of test points, the interactions between distant cells are evaluated on that grid by fast Fourier transforms, and those between near cells are computed directly and corrected. Such a method has three accuracies to set, that of the discretisation, that of the projection, and the radius of the near zone, and they are usually set by trial. The results of this paper set them beforehand, in five ways.

1. *The target is known.* The discretisation error is known before the solve, in order and in size: the first-order factor of a cell, Eq. (39) and its form for any pair of wavenumbers (§9), and the projection error of the medium, Eq. (44). A fast product need be no more accurate than that, and it is known when it is accurate enough.
2. *The two errors do not mix.* The error of the scheme is the single site's: the coupling between cells, averaged over the cells, is exact (§5). An approximation of the far coupling to a set tolerance therefore adds its own error to the discretisation's and does not alter it, so the two can be budgeted separately.
3. *For cubes on a lattice there is nothing to project.* The cells are the regular grid. The coupling depends on the separation of the cells alone, so the product is a convolution, block by block in the polynomial degrees, and the fast Fourier transform applies to it exactly, as

in Georgakis et al. [2] for the linear voxel. What the projection onto the grid approximates in general is here an identity.

4. *For cells of several sizes the moments are already in hand.* Where the cells differ in size or degree and are mapped to a uniform auxiliary grid, the quantities to be matched are the polynomial moments of a cell, which are the unknowns of the scheme; and the first-order factor says how many of them matter at a given  $kd$ .
5. *A stratified background splits as the method requires.* The Green's tensor of a plane-stratified background is the whole-space tensor, which is invariant under translation in three dimensions and carries all the singularity, plus a reverberation, Eq. (50), which is smooth, is invariant under lateral translation, and separates in depth into one-dimensional moments. The first is the convolution above; the second is a fast Fourier transform laterally and a product in depth, with no near-zone correction except where a voxel touches an interface. A splitting of the kernel into a term that carries the singularity and a nonsingular remainder is used to a related end by Sharma and Triverio [75], for surface integral equations in homogeneous and in layered backgrounds: there the singular term is independent of frequency, so that the direct near-zone integrals are computed once for a whole sweep in frequency. And because the orders and the error law are unchanged by the background (§9), the error budget of the whole space carries over to it.

No such solver is built here. These are consequences of the analysis, and they are to be tested.

## J The evidence for each result, and where to reproduce it

Table 21 lists each result with the evidence for it. Every numerical value is reproducible from the Mathematica notebooks `Mathematica/ContinuumLimit_X.wl`, for X in Reference, Thin-Layer, SpecularSums, StaticLocal, AveragedSums, CubeT, Chain, ErrorConstant, FourthOrder, FourthOrderError, Reduction3D, Oblique, IncidentS, Heterogeneous, SecondMoment, Graded-Contrast, LayerDefect and Figure (data for the figure), and their reference dumps `scripts/continuum_limit_*.py`; the moment integrals are `Mathematica/CubeMomentCore.wl` (gate `CubeMomentCoreTest.wl`), `CubeA22Block.wl`, `CubeMomentArchiveCheck.wl` and `CubeShearSplit.wl`, with the Python half of the last in `scripts/gate_cube_shear_split.py`; the stratified reference against Kennett's recursion is `scripts/gate_heterogeneous_reference_vs_kennett.py`; the voxelised spheres are `scripts/pilot_sphere_voxel_vs_mie.py` (sharp) and `scripts/pilot_graded_voxel_sphere.py` (graded, with the mean-only voxel of `cubic_scattering/graded_voxel`), the graded sphere's exact solution `Mathematica/ContinuumLimit_GradedSphere.wl` with the library `Mathematica/TakeuchiSaito.wl` and `scripts/crosscheck_graded_sphere.py`; the test against exact Mie is `scripts/gate_sphere_closure_vs_mie.py`, with the closed forms in `Mathematica/SphereClosureDynamic.wl`; the exact kernel is `cubic_scattering/cell_averaged_lattice.py`; the independent implementation of the first-moment voxel is `scripts/crosscheck_first_moment_voxel.py`, and of the voxel of any degree `scripts/crosscheck_second_moment_voxel.py`, and of the graded cell `scripts/crosscheck_graded_contrast.py`; the projection-error law and its split are measured by `scripts/measure_layer_bases.py`,

in a stratified background by `scripts/measure_layer_stratified_background.py`, `scripts/measure_oblique_stratified_background.py` and `scripts/measure_oblique_two_term.py`, and the voxel of degree 2 at oblique incidence by `scripts/measure_oblique_degree.py`; the convergence figure is drawn by `scripts/plot_convergence_orders.py`; the near field of the sphere is measured by `scripts/measure_graded_sphere_near_field.py`; the moments of every grade are in `Mathematica/CubeMomentHigherGrades.wl`, `Mathematica/CubeMomentStore.wl` and `Mathematica/CubeMomentCanonical.wl`, with their independent evaluation in `scripts/crosscheck_cube_moments_ball_shell.py`; the hierarchy on the layer is `Mathematica/ContinuumLimit_GradientHierarchy.wl` and `scripts/measure_layer_taylor_hierarchy.py`, and in three dimensions `scripts/measure_ball_gradient_hierarchy.py`, `scripts/measure_cube_gradient_hierarchy.py` and `scripts/measure_graded_sphere_gradient_hierarchy_fft.py`. All are available at <https://github.com/tmnestor/MultipleScatteringCalculations>.

**Table 21.** The results of the paper, and the evidence for each.

| Result                                                    | Evidence                                                                                                                  |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Continuous layer, exact and to $O(D^k)$                   | reference to $2.5 \times 10^{-8}$ ; series order $k$                                                                      |
| Thin-layer series, Eq. (7)                                | each coefficient symbolically, to $D^4$                                                                                   |
| Point-kernel local term survives $d \rightarrow 0$        | 11× the continuum term at $m = 1$                                                                                         |
| Averaged coupling between planes = continuum× sinc        | $2.8 \times 10^{-16}$ , 81 entries                                                                                        |
| Tiling identity, Eq. (13)                                 | 21/21 components to $4 \times 10^{-9}$                                                                                    |
| Exact kernel, Eq. (14)                                    | replaces a $9.4 \times 10^{-4}$ static error                                                                              |
| Moments as distributions, Eq. (26)                        | sum rule (27) exact; M, Q equal an independent archive, 81/81                                                             |
| Channels (29)–(31)                                        | a production code by another route, $2 \times 10^{-15}$                                                                   |
| Cube vs sphere, Eq. (32)                                  | symbolic; and numerically at five $\nu$ , $4 \times 10^{-16}$                                                             |
| Ball closure = exact Mie, static                          | $2 \times 10^{-10}$ ; cube’s bulk and shear average $10^{-8}$                                                             |
| Ball closure vs Mie, dynamic, Eq. (55)                    | symbolic; numerical Mie $5 \times 10^{-4}$ ; $(ka)^3$ terms exact                                                         |
| Self term cancels, Eq. (35)                               | $4 \times 10^{-16}$                                                                                                       |
| Package $T$ = closure statically                          | $2 \times 10^{-5}$ at $k_S h = 0.01$                                                                                      |
| Second-order convergence                                  | order 2.000, three frequencies                                                                                            |
| Closed-form error, Eq. (36)                               | $n = 1-16$ , to $10^{-4}$ of the error                                                                                    |
| First-moment voxel, Eq. (38)                              | order 4.00, three frequencies                                                                                             |
| First-moment voxel, oblique, Eq. (49)                     | order 4, P and S, $20^\circ$ and $40^\circ$                                                                               |
| Incident SV and SH, all schemes                           | orders 2 and 4; constants with $k_S$ to $4 \times 10^{-4}$                                                                |
| First-moment error $-(kd)^4/720$ , Eq. (39)               | $n = 1-8$ , to $10^{-4}$ of the error                                                                                     |
| Voxel of degree $p$ : order $2p + 2$                      | 6.00 ( $p = 2$ , three frequencies), 8.00 ( $p = 3$ )                                                                     |
| Constant $c_p$ , Eq. (40)                                 | symbolic, $p = 0-5$ , twice; chain to $4 \times 10^{-7}$                                                                  |
| One plane of degree $p$ = thin-layer series to $D^{2p+2}$ | exact series, $p = 0-3$ , two layers; $D^{3.00}$ , $D^{5.00}$ , $D^{7.00}$                                                |
| Independent implementation, any degree                    | within round-off; orders 4.02, 6.02                                                                                       |
| Independent implementation of Eq. (49)                    | $1.1 \times 10^{-15}$ , 40 cases: P, SV, SH, stratified                                                                   |
| Random stratified layer, §9                               | orders 2 and 4, P and S, normal and $20^\circ$                                                                            |
| Stratified reference against Kennett’s recursion          | $10^{-14}$                                                                                                                |
| Contrast varying within a cell, Eq. (42)                  | 8 cases, linear and smooth profiles; closed forms tested one by one to $2 \times 10^{-16}$                                |
| Projection-error law, Eq. (44)                            | nine pairs $(p, r)$ to 1.3%; departure as $(kD)^2$ ; pairing symbolic, nine pairs                                         |
| Stratified background, Eq. (45), normal incidence         | orders 2, 4, 6; first-order term to $5 \times 10^{-16}$ ; law to $10^{-4}$ at $kD = 0.012$                                |
| Stratified background at $20^\circ$ , Eq. (50), Table 15  | orders 2, 4, 6 in 36 cases; background reflection against Kennett to $5 \times 10^{-15}$                                  |
| First-order term at $20^\circ$ , Eq. (52)                 | 36-pair prediction to $2 \times 10^{-15}$ , 12 cases, $p = 0, 1, 2$                                                       |
| Projection-error law at $20^\circ$ , Table 16             | $p = 0, 1, 2$ : to 6% at $k_S D = 0.2$ , 0.6% at 0.067; graded reference by impedance march and by propagator, $10^{-12}$ |

## Acknowledgments

This work began thirty years ago in the author’s doctoral research at the Research School of Earth Sciences of the Australian National University, Canberra [21].

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

**Software development.** Software development followed a human-directed, AI-assisted workflow. The author designed the software originally, but only directed and validated recent improvements, which build on the author’s original Fortran implementation (the [PhD\\_fortran\\_code](#) folder of the repository given under Data availability) and an extensive library of Mathematica notebooks, developed and extended since 1992, that carry out and verify the symbolic derivations; an agentic coding assistant (Anthropic Claude, Opus 5.5) operated under human-in-the-loop supervision to implement specified changes and diagnostic tests. The design, the choice of numerical methods, and the resolution of the principal technical difficulties were the author’s. All AI-generated changes were subject to code review by the author and validated by automated tests and independent physical checks such as reciprocity, optical theorem, Hamiltonian symmetry, analytic limits and convergence.

**Manuscript preparation.** During the preparation of this work, the author used Claude (Anthropic) in order to search for relevant open-access research, to write the TikZ code of the diagrams and to help redraft the paper into more coherent sections for readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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